Contents

Abstract

The "Wow! Signal," detected on August 15, 1977, by the Ohio State University Big Ear Radio Observatory, has remained one of the most enigmatic phenomena in modern astronomy for nearly half a century. This study presents an alternative interpretation of the signal's characteristic "6EQUJ5" sequence, proposing that it may represent not merely an intensity profile, but rather a deterministic encoding system indicative of specific orbital dynamics.

Drawing upon principles of serendipity and Occam's razor, a dual-hypothesis framework was developed that interprets the "6EQ=5UJ" formulation through two complementary approaches: a multi-planetary alignment model (E=Earth, U=Uranus, J=Jupiter) and a single-variable (E=Earth) interpretation. Both models converge on the hypothesis that the signal encodes a low-energy transfer network within the Solar System—an Interplanetary Transport Network (ITN).

Through systematic data mining and statistical analysis of the NASA JPL Horizons database, employing P-value calculations, Z-scores, Fisher's combined probability test, and DBSCAN clustering algorithms, a remarkable celestial configuration unique to the 1977 epoch was identified. The analysis isolated 32532 Thereus as the terminal destination, while 55701 Ukalegon and 84011 Jean-Claude emerged as critical waypoints designated as "Primary Gateway" and "Secondary Gateway," respectively.

The most significant finding emerged from network resilience stress tests conducted using Dijkstra's algorithm with a weighted cost function based on orbital elements (semi-major axis, eccentricity, inclination, Tisserand parameter). Results demonstrated that the physically shortest route (Route 6, ~18.00 AU Euclidean distance) proves energetically suboptimal. Conversely, Route 5 (Earth → 3763 Qianxuesen → 15025 Uwontario → 55701 Ukalegon → 32532 Thereus), despite being the physically longest path at 26.29 AU, exhibits the highest efficiency coefficient (EFF: 8.87) due to orbital resonances and Tisserand parameter compatibility. This validates what is termed the "Distance-Energy Paradox" in the literature.

In conclusion, this work presents compelling statistical and astrodynamics evidence suggesting that the Wow! Signal may encode a dual-layered navigational instruction set, specifying both a destination ("where"—32532 Thereus) and an optimal transfer route ("how"—the Ukalegon Corridor). The findings reveal the existence of dynamical tunnels within the chaotic asteroid belt that enable low-energy transfers independent of physical distance metrics.

Keywords: Wow! Signal, SETI, Interplanetary Transport Network (ITN), Orbital Mechanics, 32532 Thereus, 55701 Ukalegon, Technosignature, Data Mining

Section 1. Introduction

1.1. Background: Search for Extraterrestrial Intelligence (SETI)

For millennia, humans have wondered whether our species and home planet are unique [1]. With modern advances in astronomical understanding and technology, searches are underway to scan the far reaches of space in an organized fashion for evidence of other life [1]. Current searches for evidence of extraterrestrial (ET) life are accomplished in a number of distinctly different ways. The various searches can be viewed in three general categories: (1) 'SETI' searches for messages from extraterrestrial civilizations, (2) exploration for extrasolar or habitable planets, and (3) searches and research within the solar system (e.g., planetary missions, meteorites, cosmochemistry) [1]. Astrobiological searches for ET life encompass a broad spectrum of scientific research efforts. In general, this multidisciplinary field seeks evidence of life (not necessarily life itself), searching everywhere we can explore, using diverse scientific methods [1].

The modern era of the Search for Extraterrestrial Intelligence (SETI) began around 1960 with the realization that humanity was capable of producing signals that humanity was capable of detecting at interstellar distances [2]. In 1959, Cocconi and Morrison showed that radio transmissions would very soon meet this standard [2], and in 1960 Frank Drake began the first SETI Program, Project Ozma, on the radio [2]. The pioneer project in listening to the possible radio signals from the galaxy (called Project OZMA, with reference to the princess of the mythical kingdom of Oz) commenced in 1960. It was focused on the direction towards Epsilon Eridani, Tau Ceti, Omicron-2 Eridani, Epsilon Indi, Alpha Centauri, 70 Ophiuchi and 61 Cygni. However, this search was abandoned after two months of negative results [3]. Drake observed two nearby stars with an 85-foot radio telescope, but after 150 hours of listening, all he heard was a steady crackle of radio noise [4].

Since that time, a variety of ideas and approaches have been suggested and conducted [2]. The SETI project which was initiated in October 1992 (to celebrate the quincentenary of the landing of Columbus in the 'New World') represented a significant milestone [3]. In 1992, the US NASA High Resolution Microwave Survey began on the 500th Anniversary of the discovery of America by Columbus [5]. Most SETI searches are done at radio wavelengths, focusing on the "water hole" region near 1.5 GHz, a clean window through the universe where noise from the galaxy and the big bang is minimized [4]. Some researchers also speculated that alien civilizations might use powerful lasers to communicate at optical wavelengths, leading to several optical SETI experiments in progress [4]. More recently, the concept of techno signatures has emerged as an alternative strategy to look for signs of a civilization's energy use or artifacts they might create [4].

Despite such an enormous effort, SETI has not succeeded [4]. SETI is a "needle in a haystack" situation. There are billions of stars in the galaxy, billions of radio frequencies to search for, and infinite time that could be devoted to the experiment [4]. So far, SETI has sampled a glass of water compared to the water in all of Earth's oceans [4]. The failure of successful radio contacts with extraterrestrial civilizations could be explained in many ways, including the absence of technological civilizations in the galaxy advanced beyond our own, their inability to engage in interstellar colonization, or that they are not attempting overt contact with us [3]. It is also possible that their communication signals, if present, are not coded in a simple pattern [3]. In spite of sophisticated methods to detect signals that might have been sent by technologically highly developed life forms elsewhere, such signals have never been received (or, at least, they have not been recognized as such) [6].

However, one remarkable exception stands out in the history of SETI. In 1977, the Big Ear radio telescope at Ohio State detected what became known as the "Wow! Signal"—the strongest and most intriguing candidate for an extraterrestrial transmission ever recorded [2, 4]. The signal's source has remained a mystery for nearly five decades, with various explanations proposed including natural astrophysical events (pulsars, fast radio bursts), specific star systems in the Sagittarius constellation, terrestrial interference, and even a comet's hydrogen cloud theory—yet none have provided conclusive evidence. This study aims to address this enduring mystery by proposing a novel approach: applying an unconventional, intuition-based methodology to identify potential source candidates for the Wow! Signal.

1.2. Definition and Characteristics of the Wow! Signal

On August 15, 1977, at 22:16 EST (19:21 LST), in the course of a routine survey of the "water hole" with the Ohio State University Radio Telescope, a unique narrow-band signal was recorded that would become one of the most intriguing mysteries in the search for extraterrestrial intelligence [7]. A few days after the detection, Dr. Jerry R. Ehman began his routine review of the computer printout from the multi-day run. Several pages into the printout, he was astonished to see the string of numbers and characters "6EQUJ5" in channel 2. He immediately recognized this as the pattern expected from a narrowband radio source of small angular diameter in the sky. In the red pen he was using, he immediately circled those six characters and wrote the notation "Wow!" in the left margin of the computer printout opposite them [8]. This moment would mark the most dramatic event in SETI history.

The signal was detected at a frequency of 1420.3556 MHz, remarkably close to the hydrogen spin-flip transition frequency (1420.40575177 MHz)—the "magic frequency" suggested by Cocconi and Morrison for interstellar communication [7, 9, 10]. This frequency falls within the internationally protected 1400-1427 MHz band reserved for HI observations, in which "all emissions are prohibited" [7]. The signal was exceptionally strong, reaching a peak of 30 sigmas—30 times the average noise level per channel [7, 11]. Most significantly, the signal bandwidth was less than 10 kHz [7, 11, 12]. Because natural emission lines in centimeter-wave astronomy usually have frequency bandwidths on the order of 100–1000 kHz due to Doppler shifts along the line of sight, the narrow <10 kHz bandwidth of the Wow signal suggests that it was generated by technology, i.e., an artificial transmitter [9, 11].

The signal lasted for 72 seconds, which was the maximum amount of time that the Big Ear radio telescope was able to observe any given point in the sky, suggesting that the signal likely lasted longer [13]. The signal-strength sequence "6EQUJ5" represents the following sequence of signal-to-noise ratios: 6 (6 up to 7), E (14 up to 15), Q (26 up to 27), U (30 up to 31), J (19 up to 20), 5 (5 up to 6) [8]. Each data point represents approximately 12 seconds of sidereal time [8]. The signal flux versus time indicated that the Wow source was above the telescope and apparently fixed with respect to the stars, because it matches the expected transit time and signal-to-noise ratio profile for a celestial object passing through the beam of OSU's transit telescope [11]. The intensity pattern detected so closely matched the signature of a transiting celestial source that it provided strong evidence for a signal fixed on the sky rather than terrestrial interference [9].

The signal originated from the direction of the constellation Sagittarius, northwest of the globular star cluster M55, approximately 20 degrees from the center of our galaxy [7, 10]. The coordinates of the signal are RA: 19h25m31s ± 10s (for the positive horn) or 19h28m22s ± 10s (for the negative horn), and DEC: −26°57' ± 20' (J2000 equinox) [13]. However, determining the exact location has proven problematic because the telescope used two separate feed horns to search for radio signals, and the signal was intermittent, appearing in only one horn of the dual-horn feed [7, 10]. Since the digital output printed only magnitude at that time, it is not known whether the signal appeared in the east or west horn, resulting in the right ascension ambiguity [7]. Despite it was detected in just one of the two feed horns of the radio telescope, the data was processed in a way that does not allow us to determine which of the feed horns actually received the signal [13]. The other reason that makes it difficult to determine the exact source is the high uncertainty in declination: 20 arcminutes [13]. These ambiguities have made subsequent searches extremely challenging.

The Wow signal stands as one of the most significant detections in SETI history. John Kraus, director of the OSU Radio Observatory, described it in a memo to Carl Sagan as "highly suggestive of extraterrestrial intelligent origin," writing that "no other signal like it has been observed in all the years of our sky surveys" [11]. As of October 2020, the Wow! Signal remains the strongest candidate SETI signal [13]. The narrow bandwidth, match to the antenna pattern, and high intensity were so suggestive of an interstellar radio signal that they continue to fascinate researchers nearly five decades later [9]. Not only has the "Wow!" signal not been reobserved, but no other signal like it has been detected in all the years of subsequent sky surveys [7]. Thus, due to these restrictions, the origin of the famous "Wow!" signal received in 1977 remains uncertain [14].

1.3. Search for Origin and Previous Studies

The signal location was reobserved over 30 times during the six weeks following its detection and many times in later years, with nothing significant being observed [7]. Ohio State made approximately 100 additional transit scans at the same declination over the next few years but did not detect the emission again [9]. Robert H. Gray of Chicago maintained a continuing fascination with the "Wow!" signal and reobserved its location repeatedly with Prof. Paul Horowitz's Harvard telescope and megachannel receiver, all with negative results [7]. The main problem was that the signal never repeated. Follow-up observations of the area conducted by many observatories during several years never detected another signal. Nonetheless, the fact that the signal never repeated does not necessarily discard that it was produced by extraterrestrial intelligence [13].

In 1987 and 1989, the Harvard/Smithsonian-META system was used to observe the nominal Ohio State positions for 4-hour periods in a search for extremely narrow bandwidth and possibly intermittent radio signals over several 400 kHz segments of the 21 cm band [9, 12]. With 0.05 Hz resolution, the 8.4×10^6 channel META spectrometer could detect ultra-narrowband radio signals to a limit of 4×10^{-24} W m^{-2}, with a 19.5 σ threshold [9]. Six pointing directions were observed for about four hours each [11, 12]. However, the META system would not have detected the Ohio State signal if it had been wider than approximately 10 Hz or if it had not been Doppler corrected to a constant frequency [9]. No ultra-narrowband signals were detected in these observations [9, 12].

In 1995 and 1996, searches for evidence of the Ohio State emission were conducted using the Very Large Array (VLA) [9]. These observations improved on Ohio State's sensitivity by a factor of greater than 100, offering the prospect of detecting weak underlying sources [9]. The VLA observations used channel widths of 6.1 and 12.2 kHz, comparable to Ohio State's receiver, eliminating spectrometer resolution as a variable [9, 11, 15]. The observations covered the large uncertainty in declination and obtained much higher spatial resolution [9]. No narrowband emission resembling the Ohio State Wow was found in searches of the apparent source positions using the VLA, to flux limits as low as 20-22 mJy at the nominal positions, with detection thresholds of 5.9 σ and 6.9 σ [9, 11]. Two continuum sources were detected inside Ohio State's coordinate error boxes but displayed no unusual spectral features, no unusual spectral indices, and no temporal variation [9].

Between 1998 and 1999, the University of Tasmania Hobart 26-meter radio telescope was used to investigate a periodic emission hypothesis [15]. The telescope allowed continuous tracking for nearly 14 hours, the maximum time the target coordinates were above that telescope's horizon [15]. Each of the two nominal positions was tracked during two days for nearly 14 hours each [15]. Additional observations were made in 1999, 15 degrees north and south of the nominal coordinates, to fully cover the Ohio State 40-arcminute declination half-power beamwidth with the Hobart beam [15]. The observations used a bandwidth of 2.5 MHz and 4.9 kHz frequency resolution [11, 15]. No emissions resembling the Wow were detected over this bandwidth to a flux density limit of about 18 Jy, with a detection threshold of 5.9 σ and rms noise of 3 Jy [15]. This result constrains a putative periodic emission to a period greater than 14 hours because the observations were sufficiently sensitive and extended in time to detect emissions with a shorter period at least once [15].

The Allen Telescope Array (ATA) was used to perform interferometric imaging in a search for a signal with characteristics similar to the Wow signal—especially a signal at a position consistent with the original detection [11]. Approximately 100 hours of interferometer observing time was devoted to this study, making this the longest single Wow follow-up survey to date [11]. Images were time-averaged over periods of up to three hours and in selected cases down to the 10-second cadence of the integrations [11]. The analysis bandwidth of 10.5 MHz was more than 20 times that of the original Ohio State 0.5 MHz bandwidth, and the 12.8 kHz frequency resolution was chosen to approximate the 10 kHz bandwidth of the original channels [11]. The ATA permitted simultaneous monitoring of a very wide field of view, 2.5 degrees full width at half maximum (FWHM) or about 5 square degrees [11]. By simultaneously matching or improving on all observational parameters of the original experiment, these observations covered most of the ambiguities of the original signal [11]. No signal closely matching the characteristics of the Wow signal was found within the field of the OSU telescope [11]. One interesting signal was observed with less than 10-second duration, but outside the nominal locales covered by the OSU telescope, and since it was observed only once, it did not meet the standard for a confirmed detection [11].

In 2016, the Center for Planetary Science proposed a hypothesis arguing that a comet and/or its hydrogen cloud were a strong candidate for the source of the "Wow!" Signal [8]. It has been suggested that the signal was produced by hydrogen clouds from Comets 266/P Christensen and P/2008 Y2 (Paris and Davies, 2015) [13]. From November 27, 2016, to February 24, 2017, the Center for Planetary Science conducted 200 observations in the radio spectrum to validate the hypothesis using a 10-meter radio telescope equipped with a spectrometer centered at 1420.25 MHz [16]. The investigation discovered that comet 266/P Christensen emitted a radio signal at 1420.25 MHz, and observations of three randomly selected comets (P/2013 EW90, P/2016 J1-A, and 237P/LINEAR) also detected radio signals at 1420 MHz [8]. However, this hypothesis has been dismissed by the scientific community, and the source of the signal remains unknown [13].

1.4. Why It Remains Unsolved and The Role of This Study

The failure to redetect the Wow signal can be attributed to several fundamental limitations of traditional search methods. The inability to re-observe the signal means it does not comply with the principle of repeatability, which is one of the fundamental pillars of the scientific method [8]. Few attempts have been made to determine the exact location of the Wow! Signal due to the difficulty involved, primarily because the data was processed in a way that does not allow determination of which feed horn actually received the signal, and the high uncertainty in declination of 20 arcminutes [13]. The Big Ear data acquisition system operated only with channels 10 kHz wide, recording data with 12-second integrations, meaning any variation of signal amplitude within the 12-second interval would not have been detected [8]. If the Wow! signal was modulated at a frequency less than 0.00694 Hz (a period longer than 144 seconds) or at a frequency greater than 0.2 Hz (a period shorter than 5 seconds), that modulation would not have been detected [8]. Ohio State's 100 follow-up transit observations would not significantly constrain the possibility of a source with a period of more than several hours because they sampled sources for only 72 seconds twice each terrestrial day; for sources that are themselves periodic, periodic sampling makes a second confirming detection extremely difficult because in many cases the source will be out of phase with the observations for many subsequent transits [15]. All previous searches used telescopes with approximately 0.5-degree fields of view, which covers only a fraction of the 40-arcminute declination half-power OSU beam during an observation, and only one of the two OSU ambiguities in right ascension [11]. The observations reported do not rule out the possibility of a source with wide bandwidth, large-frequency drift rate, or quiescent periods longer than 4 hours, and they do not rule out the possibility of a source at a declination outside of the range observed [12].

The possibility of the signal being Earth-originated has been carefully considered and largely ruled out for various reasons. The signal frequency of 1420.3556 MHz was inside the internationally protected 1400-1427 MHz band reserved for HI observations in which "all emissions are prohibited," meaning no transmitter on Earth or in space should have been transmitting in this protected band [7, 8]. The Wow! source intensity pattern received matched almost perfectly the pattern expected from a small-angular-diameter (point) radio source on the "celestial sphere," indicating a source at such a large distance that there is no perceptible motion relative to the background stars [8]. The signal direction was at least 90 degrees from all the planets including Pluto [7]. Airborne or orbital vehicles would not in general remain in the beam for a time comparable to a transiting celestial source; low Earth-orbiting satellites in east-west orbits are ruled out because they would cross the beam (8-degree half-power beamwidth in right ascension) in only a few seconds, assuming a motion of 4 degrees per minute [9]. The Wow signal cannot be ascribed to a random noise fluctuation because a single 30 σ noise peak would not be expected in thousands of years at the Ohio State sampling rate, and because six consecutive peaks over 5 σ in the same channel and mimicking the antenna pattern cannot be due to noise [9]. The Wow frequency was near the 21 cm HI emission line, but there are several reasons to think it was not due to hydrogen: first, the line is usually detected over approximately 100 kHz, yet this emission was not detected in adjacent 10 kHz channels; second, HI flux would not be expected to vary on a timescale of minutes to produce the single-beam detection [9].

Thus, since all the possibilities of terrestrial origin have been either ruled out or seem improbable, and since the possibility of an extraterrestrial origin has not been able to be ruled out, Dr. Jerry Ehman concluded that extraterrestrial intelligence might have sent the signal that we received as the Wow! source [8]. However, as a scientist, he stated that he awaits the reception of additional signals like the Wow! source that can be received and analyzed by many observatories, and that the origin of the Wow! signal is still an open question, noting: "There is simply too little data to draw many conclusions. In other words, I choose not to draw vast conclusions from 'half-vast' data" [8]. For nearly five decades, the source of the Wow! signal has remained unknown despite numerous systematic searches including META, VLA, Hobart, and ATA observations, all of which returned negative results, and despite the comet hypothesis being rejected by the scientific community [9, 11, 12, 13, 15, 16]. Traditional methods and existing technologies have proven insufficient to solve this mystery. The complex nature of this problem suggests that its solution may lie hidden in a surprisingly simple pattern—an insight that has characterized many breakthrough discoveries in science. This study aims to investigate the possible source of the Wow! signal through an alternative, intuition-based approach that diverges from conventional search methodologies.

Section 2. Creative Processes and the Principle of Simplicity in Scientific Discoveries

2.1 Creative Discovery and Simplicity in Scientific Inquiry

Throughout the history of science, many groundbreaking discoveries have emerged not from systematic analysis alone, but through intuitive thinking and creative insight. The creative process of problem-solving has been conceptualized by Wallas (1926) as consisting of four distinct stages: preparation, incubation, illumination, and verification [17]. The incubation period, in particular, represents a crucial phase where the conscious mind steps away from the problem, allowing unconscious processes to continue working on potential solutions [18, 19]. This phenomenon often culminates in what is known as the "Aha!" or "Eureka!" moment—a sudden illumination where the solution appears unexpectedly and with remarkable clarity [20, 21]. The legendary example of Archimedes discovering the principle of buoyancy while bathing exemplifies this sudden insight, where the solution emerged not through deliberate calculation but through an unexpected moment of realization [22, 23]. Such creative leaps have characterized many of the most significant advances in physics, mathematics, and astronomy, from Newton's insights on gravitation to Poincaré's mathematical breakthroughs and Einstein's thought experiments [17, 19].

At the heart of scientific methodology lies a fundamental principle known as Occam's Razor, or the principle of parsimony, which states that among competing hypotheses, the simplest explanation is generally the correct one [24, 25, 26]. This principle has guided scientific inquiry for centuries, reflecting the deep-seated human preference for elegant, economical theories that explain complex phenomena with minimal assumptions [27, 28]. In the search for natural laws, simplicity is not merely an aesthetic preference but a methodological imperative—the universe often reveals its secrets through surprisingly simple mathematical relationships hidden beneath apparent complexity [17]. As noted in the literature, "simplicity is the goal of science" [28], and researchers across disciplines have consistently found that parsimonious models not only facilitate understanding but often prove more robust and generalizable than their more complex counterparts. This principle holds particular relevance for enigmatic problems like the Wow! signal, where the search for underlying patterns may benefit from focusing on simple, fundamental relationships rather than elaborate theoretical frameworks.

The role of serendipity—the fortunate discovery of valuable findings while searching for something else—has been well documented in the history of science [29, 30, 31]. As Louis Pasteur famously observed, "chance favors the prepared mind" [32], highlighting that serendipitous discoveries require not merely luck but also the sagacity to recognize and pursue unexpected observations. Many pivotal scientific breakthroughs, from the discovery of penicillin to the identification of cosmic microwave background radiation, have resulted from this combination of accident and insight [31]. These historical examples demonstrate that complex scientific problems can sometimes yield to simple, intuitive approaches when researchers remain open to unexpected patterns and relationships. As we move forward to examine specific historical cases, we will see how the triumph of simple thinking has repeatedly led to transformative discoveries across diverse scientific domains.

2.2 Historical Evidence: Triumphs of Simple Thinking

The history of science is replete with discoveries that emerged not from systematic planning but through unexpected observations recognized by prepared minds. Perhaps the most celebrated example is Alexander Fleming's discovery of penicillin in 1928, when culture plates left on a laboratory bench became contaminated with mold, and Fleming noticed that "around a large colony of a contaminating mould the staphylococcus colonies became transparent and were obviously undergoing lysis" [33]. Remarkably, Eugene Semmer had observed the same curious effects of Penicillium notatum nearly 60 years earlier yet failed to recognize its significance—his methods and assumptions constrained his thinking [34]. This contrast exemplifies Pasteur's famous dictum that "chance favors the prepared mind" [32]. Similarly, Wilhelm Conrad Röntgen's discovery of X-rays on that fateful Friday, November 8, 1895, occurred entirely by accident when, working in a completely darkened laboratory with a cardboard-covered tube, he observed by chance a barium platinocyanide-coated screen fluorescing nearby [35]. The key experiment came when, holding a small object in the beam with his fingers, Röntgen recognized the image of his bones on the screen—the first X-ray image ever, revolutionizing medicine fundamentally [35]. Percy Spencer's serendipitous discovery of microwave cooking power, watching popcorn pop in front of an open waveguide, transformed food preparation technology [36]. The pharmaceutical industry has witnessed similar unexpected breakthroughs: sildenafil was initially developed for angina pectoris, but when clinical trial volunteers reported penile erections as a side effect, researchers recognized a new therapeutic application, leading to one of the most commercially successful drugs in history [37]. Even the discovery of saccharin emerged from an accidental observation when Constantin Fahlberg, after overboiling a beaker in June 1878, noticed at dinner that his thoroughly washed hands tasted sweet prompting him to rush back to the laboratory and systematically taste all beakers and vials until identifying the source [38]. The discovery of polytetrafluoroethylene (PTFE, or Teflon) has been described variously as "an example of serendipity, a lucky accident, and a flash of genius—perhaps all three were involved" [39], revolutionizing the plastics industry and enabling applications not otherwise possible.

Beyond fortuitous accidents, the principle of simplicity has proven equally powerful in driving scientific breakthroughs. The discovery of graphene, which earned the 2010 Nobel Prize in Physics, exemplifies how remarkably simple methods can yield revolutionary results: atomically thin carbon films were prepared using mechanical exfoliation, essentially a "repeated peeling" or "Scotch tape" method applied to highly oriented pyrolytic graphite [40]. This straightforward approach, requiring no complex equipment or elaborate procedures, produced high-quality, stable films that transformed materials science and opened entirely new research frontiers. George de Mestral's invention of Velcro in 1955 similarly arose from a simple observation during a walk in 1948, when he examined burrs stuck to his clothes and noticed "their spines were tipped with tiny hooks that stick out from the seeds and provide a hook-and-loop construction" [41]—a biomimetic insight that required no sophisticated instrumentation, only careful attention to natural phenomena. Charles Goodyear's 1839 invention of vulcanization, though he "did not recognize what happened in rubber mixed with sulfur and white lead on heating" and the chemical nature remained unexplained for decades, succeeded because fundamentally "it has worked for rubber products"—a pragmatic triumph of empirical observation over theoretical understanding [42]. The development of Post-it Notes, spanning from Spencer Silver's creation of high-tack, low-adhesion glue in 1968 to the eventual product launch in 1980, demonstrates how simple concepts can require extended incubation periods before their full potential is realized [34]. These historical examples validate the methodological principles outlined earlier: complex problems in science often yield to approaches that prioritize simplicity and remain receptive to unexpected patterns. As we turn to our investigation of the Wow! signal, we adopt this same philosophy—seeking elegant mathematical relationships that may have been overlooked in favor of more elaborate theoretical frameworks, guided by the conviction that nature's deepest secrets often hide in plain sight, waiting for minds prepared to recognize them.

Section 3. Methodology

3.1 Temporal Analysis and Formula-Based Object Identification

Celestial objects conforming to the 6EQ=5UJ formula within a ±1% error margin were systematically catalogued across three distinct temporal frameworks: a single day corresponding to the Wow! signal date of August 15, 1977, a 10-day period, and a 100-day period. The 10-day and 100-day periods were randomly generated through computational methods to assess the formula's consistency across different temporal scales. The distribution density of heliocentric distances, measured in Astronomical Units (AU), for these matched objects was subsequently determined and analyzed to identify spatial patterns.

Point-wise representations of the actual AU value distributions were generated for all objects that satisfied the formula criteria across the three temporal frameworks. This analysis provided a comprehensive view of the spatial distribution patterns of formula-compliant celestial bodies. The frequency of matches for each celestial object and the specific dates on which these matches occurred were systematically documented across all study periods. A comprehensive dataset spanning 111 days was constructed, with matched objects arranged alphabetically and dates ordered chronologically, enabling visualization of temporal patterns in a unified graphical representation.

The percentage distribution of total matches and the recurrence frequency of individual matched objects were quantified and visualized for each study period. This analysis revealed the relative contribution of each celestial object to the overall matching pattern, providing insights into whether specific objects dominated the matches or whether the pattern was distributed across multiple bodies.

3.2 Statistical Randomness and Hypothesis Testing

Comprehensive statistical analyses of observed celestial object matches were conducted across four distinct temporal scales (1 day, 10 days, 100 days, and 111 days) to evaluate the randomness of formula-based correlations. The statistical significance of matches occurring on the specific date of the Wow! signal reception was assessed using P-Value testing, with both null (H_0) and alternative (H_1) hypotheses rigorously evaluated to determine whether the observed pattern could be attributed to chance.

For the randomly selected 10-day dataset, short-term consistency analyses were performed to assess the formula's reliability over brief periods. Z-scores were calculated using proportional Z-tests to quantify the deviation of observed matches from expected values under random conditions. Fisher's Combined Probability Test evaluated the regional selectivity of the formula by comparing probabilities across 10 independent days using chi-square statistics with 2k degrees of freedom. Signal-to-Noise Ratio (SNR) calculations were employed to determine whether results indicated specific objects or broader spatial regions, providing a quantitative measure of the formula's discriminatory power.

To evaluate long-term behavior and continuity, a randomly selected 100-day dataset underwent comprehensive randomness analysis. The H_0 hypothesis was tested using Fisher's Combined Probability Test, with P-values from independent days calculated and combined to assess the overall significance of the observed pattern. Results were fitted to chi-square distributions with appropriate degrees of freedom to determine statistical significance. Expected daily match frequencies were established based on the Wow! signal date analysis, with Poisson distribution calculations determining the probability of observed match totals for each day. Natural logarithms of probability values were computed and summed, with chi-square distributions compared against critical values to test the null hypothesis of random occurrence.

Frequency distribution analysis of the complete 111-day dataset was conducted to characterize the inherent nature of formula-based matches, revealing fundamental patterns in celestial object correlations and providing insight into the consistency of the formula across extended temporal periods.

3.3 Temporal Dependency and Clustering Analysis

Epoch dependency analysis was performed on AU values of all formula-matched objects across the 111-day dataset, examining temporal variations observed in the 1-day, 10-day, and 100-day studies. Kernel Density Estimates (KDE) were calculated for probability density curves, with median value comparisons providing insights into temporal stability and revealing whether the spatial distribution of matched objects remained consistent over time or exhibited systematic variations.

The inherent structure of the comprehensive dataset, comprising AU values and corresponding temporal data, was analyzed using unsupervised machine learning approaches to identify natural patterns without prior assumptions. Natural groupings were identified based on AU proximity and density characteristics, revealing intrinsic clustering patterns that emerged from the data itself. This analysis also identified noise and outlier values within the dataset, distinguishing between systematic patterns and anomalous observations that deviated from the general trend.

3.4 Trend Analysis and Regression Modeling

Comprehensive trend analysis was conducted on AU values of matched celestial objects to identify systematic patterns and assess statistical significance. A data-driven 30-point centered moving average method was employed for trend detection, smoothing short-term fluctuations to reveal underlying long-term patterns in the spatial distribution of matched objects. Outliers were identified using the 2-sigma method with standard deviation calculations, ensuring that extreme values did not unduly influence the trend analysis. Correlation strength was quantified using Spearman's rank correlation coefficient, which assessed the monotonic relationship between temporal sequence and AU values without assuming linearity.

Following trend analysis, a third-degree (cubic) polynomial regression equation was developed to characterize the observed patterns and provide accurate mathematical representation of the dataset. The cubic polynomial form was selected based on its ability to capture the non-linear characteristics evident in the data while avoiding overfitting. Model performance was evaluated using the coefficient of determination (R^2) to assess explanatory power, quantifying the proportion of variance in AU values explained by the temporal progression.

The polynomial regression model's performance was comprehensively evaluated through error distribution analysis supplementing the R^2 coefficient. This approach involved examining the residuals between predicted and observed values to verify that errors were randomly distributed and exhibited no systematic patterns. The error analysis provided robust validation of the model's predictive accuracy and reliability, ensuring that the regression equation appropriately captured the underlying relationship in the data.

3.5 Second Formula Application Approach and Statistical Validation

Celestial objects conforming to the 6EQ=5UJ formula within a ±1.5% error margin were identified for the single day corresponding to the Wow! signal date (August 15, 1977). Heliocentric distances in Astronomical Units were retrieved through multiple authoritative sources including NASA JPL Small-Body Database (SBDB) API, Minor Planet Center (MPC) via web scraping, Le Système Solaire API, and NASA JPL HORIZONS for instantaneous position and distance data. All retrieved data were systematically stored in a dedicated database for subsequent analysis. A total of 20,590 matches satisfying the 6EQ=5UJ formula (where E represents Earth exclusively) were identified. To visualize the distribution of these matches, percentage deviations from perfect formula compliance were extracted from the database. The differences in percentage deviations between the best matches and remaining matches were illustrated using pie charts, providing clear visualization of the match quality distribution across the dataset.

The distinction between the first eight matches and all remaining matches was demonstrated through elbow analysis graphing, revealing a clear inflection point in the match quality distribution. The density of deviation from perfect formula compliance was calculated and visualized for all matches, quantifying how closely each match conformed to the theoretical formula predictions. To assess the statistical significance of this inflection point, two distinct randomness analysis methods were employed. In the Atomic Approach (Individual Probability) model, the eight matches with the smallest percentage deviations from the total 20,590 matches were designated as successful matches. In the Group Approach (Cluster Probability) model, the successful match results were treated as a single cluster event. The universe of matches was conceptually divided into groups of eight (20,590 / 8 = 2,573.5 groups), with the assumption that only one of these groups achieved the target criteria. For both approaches, P-Value testing, null hypothesis (H_0) analysis, and Z-tests (significance scores) were calculated to measure how many standard deviations the results deviated from randomness (meaningless noise).

The statistical differences in percentage deviations from formula compliance for all 20,590 matches were calculated and displayed using log-log histograms. This logarithmic representation revealed the power-law characteristics of the deviation distribution and highlighted the exceptional nature of the top-performing matches relative to the broader population.

3.6 Orbital Characterization and Gateway Selection Framework

The AU distributions and spatial configurations of all celestial objects within each of the first eight distinct matches were systematically mapped and visualized. This analysis revealed the geometric arrangements of these objects in heliocentric space at the time of the Wow! signal, providing insight into their collective spatial organization. Orbital element data for all unique celestial objects across the 20,590 matches were retrieved using the astroquery.jplsbdb library to access NASA JPL Small-Body Database (SBDB) services. The following orbital parameters were extracted and permanently stored in the database: semi-major axis (a), eccentricity (e), inclination (I), and Tisserand parameter (T_J). Using these orbital parameters, comprehensive comparisons were conducted between the selected first eight match groups and all unique objects in the remaining 20,582 matches. The density distributions of semi-major axis (a), eccentricity (e), inclination (i in degrees), Tisserand parameter (T_J), and heliocentric distances (AU) were systematically compared to identify distinguishing orbital characteristics between high-quality and lower-quality matches.

Due to the natural behavior of percentage deviation values in the decimal places of matches conforming to the formula in the database on the Wow! signal date, and because the first eight matches consisting of four-object groupings exhibited distinctive percentage deviation characteristics compared to other matches, a decision was made to organize matches into groups of eight four-object combinations. A total of seven such eight-match groupings were established for analysis. For each unique celestial object in the first eight-match group (the group with the smallest percentage deviations), evaluations were conducted based on Interplanetary Transport Network (ITN) theory. Proximity to Lagrange points, orbital stability characteristics, and Tisserand (T_J) parameter assessments were performed to select primary gateways (main targets) and intermediate gateways for the principal route.

The selection methodology was guided by the conceptual framework that if a hypothetical highway network existed among the identified objects with Earth as the departure point, the most cost-efficient pathways should be prioritized. A theoretical mathematical formula was developed to identify the most efficient transfer routes between objects, inspired by the Southworth-Hawkins D-criterion and Drummond similarity measures, utilizing orbital shape parameters (a, e, i) and energy coefficients (T_J). To address scale differences inherent in orbital mechanics parameters, statistical normalization was applied to all numerical ranges. Following normalization, each term was weighted according to its physical importance and influence within the dynamical system. The cost value output by the theoretical formula is directly proportional to the theoretical velocity change (Δv) required for orbital transfers between two trajectories. The total cost for the route chain was calculated by summing up the minimum transfer costs computed for each individual step.

Total transit costs between each unique object across all matches with percentage deviations less than or equal to 1.5% were calculated, establishing a comprehensive cost control database. Using the a, e, i, and T_J values of all unique objects in the group with the smallest percentage deviations, Dijkstra's algorithm was employed to calculate and identify the most efficient (lowest cost) route from Earth through intermediate gateways to the ultimate target gateway in the highway network.

3.7 Route Optimization and Network Validation

To verify whether the cost of reaching the main gateway using objects selected for the primary route from the eight total match groups (consisting of the four-object matches with the lowest percentage deviations) was genuinely minimal when compared to routes incorporating objects from other match groups, a Network Resiliency Stress Test was implemented. This analysis established the correlation characteristics between the formula predictions and the lowest-cost routes. For this validation, the five most efficient routes were identified using objects from each eight-match group (composed of four-object matches), and their costs were calculated. This procedure was performed for all seven match groups, and cost comparisons were systematically analyzed to confirm route optimality.

The most cost-efficient routes to the intermediate gateway and main gateway were calculated and visualized through comprehensive mapping. Using the cost values from the previously established database containing efficiency calculations for all unique objects across the 20,590 four-object matches, transfer connection pathways between the intermediate gateway and main gateway were identified. The lowest-cost connection pathways were incorporated into the transportation map, providing a complete visualization of the optimal route network.

3.8 Three-Dimensional Reconstruction and Distance Analysis

For all objects on the primary route to the intermediate and main gateways, as well as objects on connection pathways (including the Sun), three-dimensional position vectors and orbital data were retrieved from NASA's JPL HORIZONS service for August 15, 1977. The following orbital elements were extracted and recorded: a (Semi-major axis), e (Eccentricity Distribution), i (Inclination ° Distribution), Omega° (Ω - longitude of ascending node), w° (ω - argument of periapsis), and M° (M - mean anomaly). Using the orbital elements a°, e (ecc), i°, Omega°, and w°, standard orbital mechanics formulas were applied to calculate the complete three-dimensional coordinates of each object's elliptical orbit around the Sun. The three-dimensional position vectors for each object were utilized to determine their locations in orbit on the Wow! signal date. A three-dimensional graph was constructed showing the sequence of objects along the route, illustrating the primary pathway originating from Earth and proceeding to the intermediate gateway and main gateway targets, including lateral connection pathways.

To evaluate alternative main gateway selections and compare the energetic cost efficiency of reaching the primary target object in the first section using these alternatives as departure points, the k=7 limit was removed. Tisserand (T_J) parameter values were extracted for each unique object across all 20,590 matches. Energy-efficient cost calculations were performed for pathways from both the selected alternative main gateways and the previously identified main gateway in the primary route to the ultimate target object in the first section. These costs were systematically compared to validate gateway selection optimality.

Following energy-efficient cost calculations, the actual shortest distances (time-independent minimum approach distances) between all selected main gateways, the intermediate gateway, and all objects constituting the intermediate and primary routes to the ultimate target in the first section were calculated. Using orbital parameters a (Semi-major axis), e (Eccentricity Distribution), i (Inclination ° Distribution), Omega°, and w°, Minimum Orbit Intersection Distances (MOID) were computed using the scipy.optimize.minimize numerical optimization method. This calculation determined the minimum distance that the three-dimensional orbits of these objects could approach each other in space, independent of their temporal positions.

For all objects for which MOID calculations were performed, the actual distances between their positions on their orbits on the Wow! signal date and the location of the target object identified in the first section on the same date were calculated. The Mean Anomaly (M) values for the relevant objects were used in Kepler's equation to calculate Eccentric Anomaly (E) values. Three-dimensional (x, y, z) coordinates were computed using orbital parameter data. Three-dimensional Euclidean distances between all objects were calculated, revealing the characteristic differences between the previously calculated most efficient primary route cost and the shortest geometric distances, thereby distinguishing between energy-optimal and distance-optimal pathways.

To rigorously validate the third hypothesis and ensure global route optimality, the initial statistical segmentation limit (k=7) was temporarily lifted to perform a comprehensive scan of the entire dataset of 20,590 matches. An exhaustive analysis was conducted to identify all unique celestial objects satisfying the specific Tisserand parameter condition ($T_J < 3$), which characterizes strong Jovian influence and potential gateway status. This expanded search yielded additional candidate objects (including 169509 Jeffreyrobbins) that were previously excluded by the grouping constraints. Using the previously established Dijkstra optimization algorithm and the weighted cost database, the lowest-cost trajectories from Earth to the final target (32532 Thereus) were recalculated for each of these alternative gateway candidates, resulting in a distinct set of six primary comparison routes.

A **"Triple Metric Comparative Analysis"** framework was developed to systematically evaluate the performance of these six routes. For the specific orbital configuration on the Wow! signal date (August 15, 1977), three quantitative performance indicators were calculated for each route: 1) **Physical Distance (MT):** The cumulative three-dimensional Euclidean distance of all route segments, measured in Astronomical Units (AU), representing the geometric length of the path, 2)**Total Formula Cost (FT):** The aggregate energetic "cost units" derived from the weighted orbital parameter formula (Equation 5), representing the theoretical velocity change (**Δv)** required for orbital transfers,
3) **Efficiency (EFF):** A derived coefficient calculated as the ratio of Total Formula Cost to Physical Distance (EFF = FT / MT). This final metric, **Efficiency**, was utilized to quantify the

"energy expenditure required per Astronomical Unit." This analytical step was crucial to mathematically distinguish between the "geometrically shortest path" (Euclidean optimum) and the "dynamically most fluid path" (Energetic optimum). By correlating the Efficiency (EFF) scores with the original statistical anomaly groupings (k=1), the methodology provided a quantitative basis to verify whether the statistically rarest matches corresponded to the most physically efficient transport manifolds within the Solar System.

Section 4. Results

4.1. Hypothesis 1 Results

In Hypothesis 1, assuming the Wow signal is an encrypted message, the formula 6EQ=5UJ was solved with a ±1% margin of error for E:Earth, U: Uranus, and J:Jupiter, leaving Q as unknown (referring to Q=Quaestio, question, decomposition in Latin), suggesting it points to a specific celestial body .

To evaluate the population structure of celestial bodies matching the formula (6EQ = 5UJ) detected on different days, the results were analyzed across three different time windows (1d, 10d, and 100d). Increasing the number of days analyzed significantly affected the number of detections. In the single-day application where the Wow! signal was received (N=1), only one match was found. A randomly selected 10-day study resulted in 23 matches (N=23), and a randomly selected 100-day study resulted in 165 matches (N=165). In the 10-day analysis, the vast majority of the 23 points captured by the formula clustered in the high AU regime between 12.75 and 13.74 AU. In the 100-day study, the AU values of the detected celestial bodies were not homogeneously distributed. Using the 100d data window yielded more numerous and diverse results (including different AU regimes) compared to the 10d window. The density distribution characteristic of AU in the 100d study (Fig. S1) strongly implies that AU values have a time-dependent characteristic.

(Fig. S2) illustrates the distribution of Heliospheric Distance (AU) values for each of the detected celestial bodies (189 total) across 1d, 10d, and 100d studies, as shown in (Fig. S1), using a stripplot/jitter plot method (Fig. S2). The Bimodal AU Distribution Characteristic observed in (Fig. S1) is more clearly visible. In the 100-day study, a second, denser and more crowded cluster region (Upper-cluster) was observed, extending from AU 12.6 to 13.7, compared to the AU 11.7 - 12.4 sub-cluster. These two regimes are separated by a clear gap where almost no detections were found between AU 12.4 and 12.6. This gap indicates that the AU values of celestial bodies detected as a result of the formula matching do not show a continuous distribution, but rather consist of two distinct populations, exhibiting a bimodal characteristic structure.

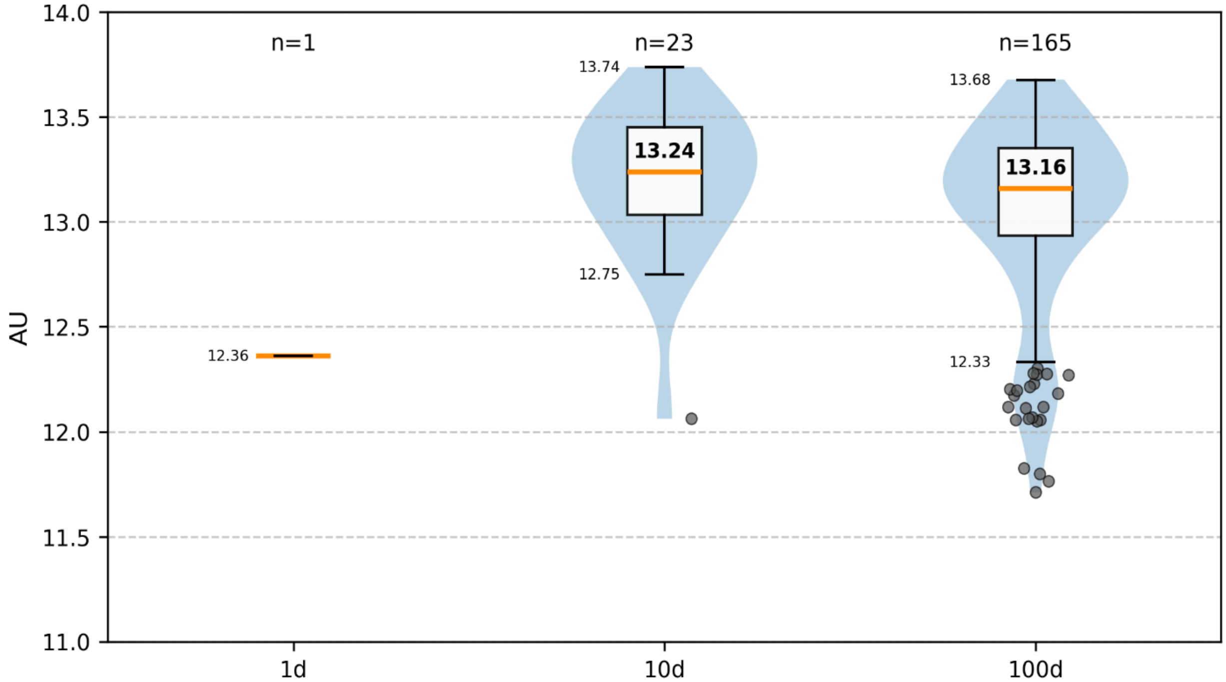

Fig. S1 AU distribution by analysis windows.

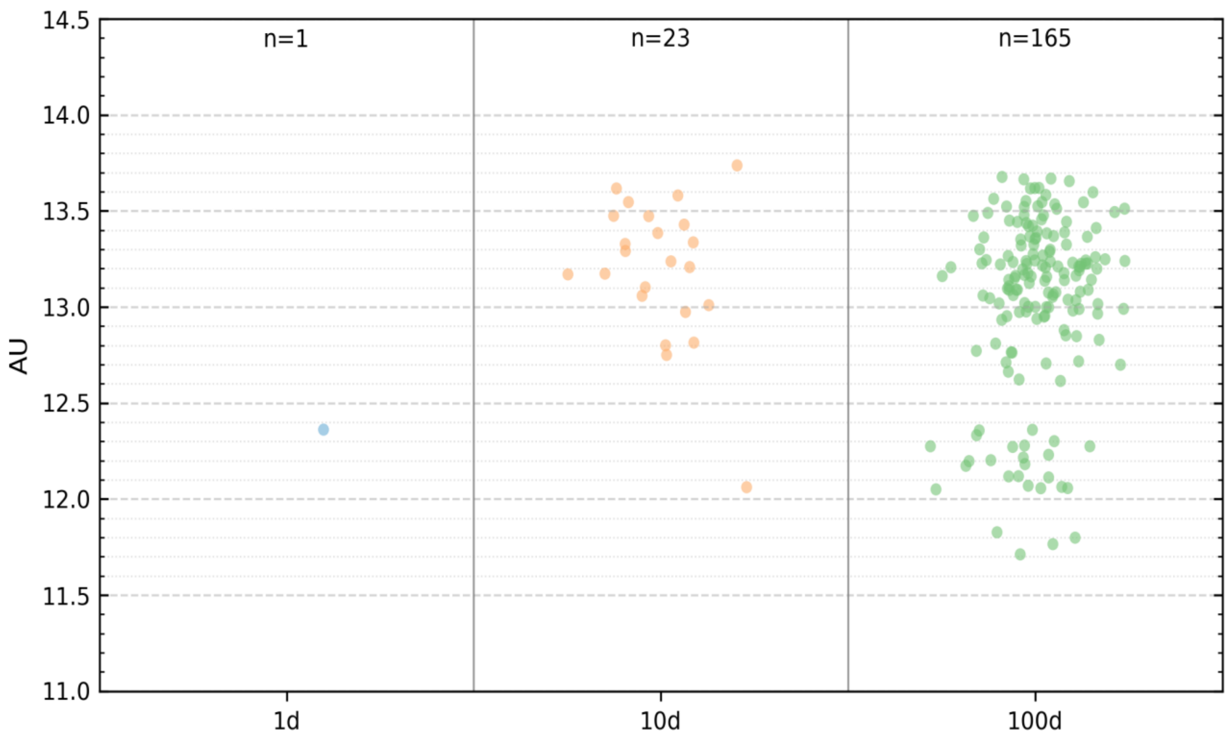

Fig. S2 Scatter of AU values by analysis windows.

(Fig. S3) Temporal presence of objects (10d window) with 1d detail inset, shows the detection dates of the 23 matches found in our 10-day study and the Wow signal date, using our formula, on the chronological axis (X-axis).

The X-axis in (Fig. S3) represents the 10 days analyzed, and the Y-axis represents the 22 unique celestial bodies detected on these days. The inset 'Presence Matrix 1d' in the upper left corner of the main graph shows that 1 single detection was made in our 1-day study on the Wow date, and this sole detection was 32532 Thereus. In our 10-day study, for example, no matches were observed on 1965-03-15 and 1973-11-29. Most 10d matches clustered in post-2000 dates, especially 2005-02-28, 2010-12-05, and 2022-07-22. The only celestial body that matched more than once in the 10d study was 10199 Chariklo. Chariklo's detection in 1999 and 2005 provided the first hint of a pattern in the 10-day study window.

(Fig. S4) Global detection matrix colored by Heliocentric Distance (AU), divides the detection times and frequencies of 149 different celestial bodies (total N=189 detections) detected at 111 unique times, cumulative from 1-day, 10-day, and 100-day datasets between 1960 and 2025, into a 2x2 panel (A, B, C, D), where each detection square represents the Heliospheric Distance (AU) at that moment with a color scale. Blue/purple tones indicate low AU values (~11.7 AU), while red/orange tones indicate high AU values (~13.7 AU). In the main matrix, Object Code numbers are given on the Y-axis; objects are sorted alphabetically without regard to any specific property, and dates on the X-axis are sorted from oldest to newest. The corresponding celestial body names for object and day codes, and the days used in the study, are listed in (Table S1) Day Codes (D1-D111) and (Table S2) Object Codes (O1-O64), (O65-O128), (O129-O149). Detections shown in the Early Period (Panel A) are mostly concentrated in blue/green tones. This indicates that early period detections between 1960 and 1988 belong to the low AU regime (~12.0 AU).

Detections shown in the Late Period (Panel C and D) gradually shifted to orange/red tones. This indicates that post-1988 late period detections predominantly occurred in the high AU regime (~13.5 AU). The average AU value of celestial bodies captured by the formula has significantly and observably increased over time.

The percentage of repetitions for matches found in 1d, 10d, and 100d studies, relative to the total number of matches, is shown in (Fig. S5) Distribution of object repetition frequencies across analysis window. The entire pie chart for the 1-day study belonged to 32532 Thereus (100%). 32532 Thereus was not observed in the 10-day study. The pie chart for the 10-day study shows that the majority of detections are single-time matches (22 unique objects, total N=23 matches). The number of single time matches constitutes 91.3% of all detections. The celestial body 10199 Chariklo repeated 2 times in the 10-day

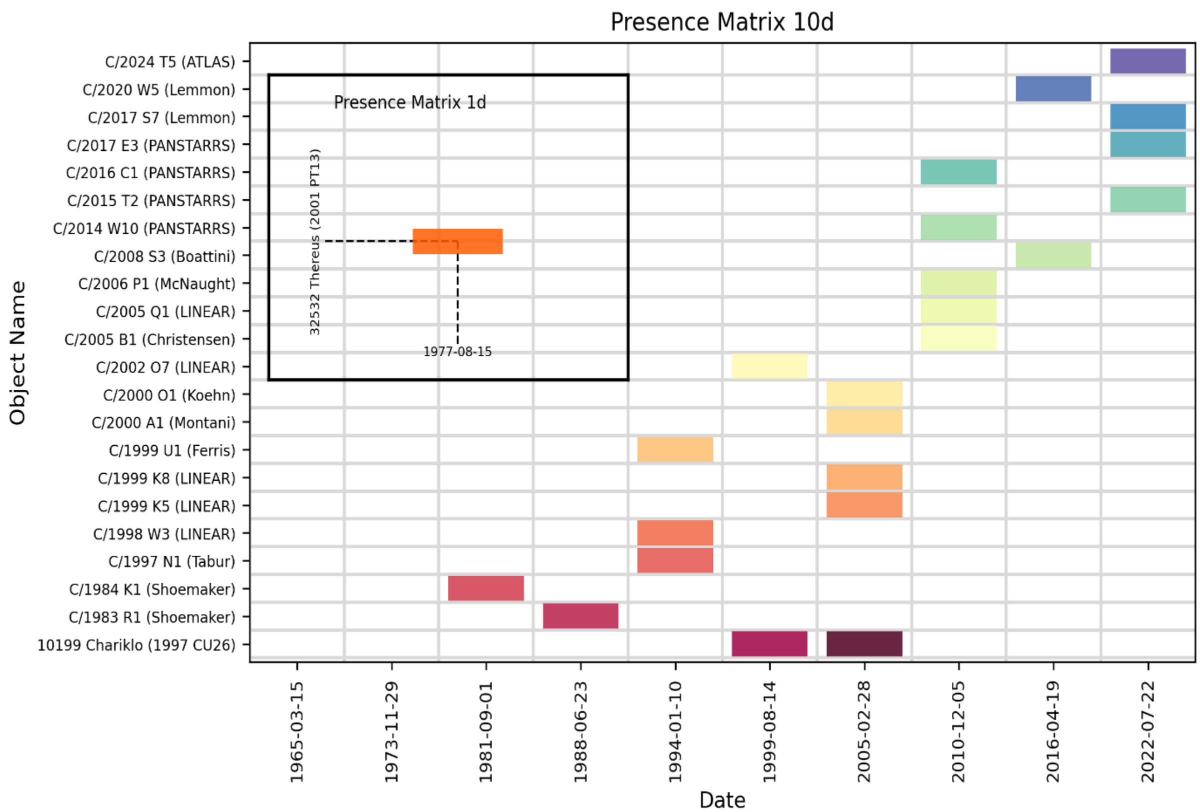

Fig. S3 Temporal presence of objects (10d window) with 1d detail inset.

study (8.7% of all detections), but no conclusion can be drawn that it represents a specific matching pattern. In our 100d study, a total of 165 detections were found, and a total of 131 unique objects were found. The pie chart for the 100-day study shows that the majority of detected matches are still low-frequency detections that occurred 1 time (68.5%) and 2 times (15.8%). The percentage rates for detections repeated 3 or more times and for all detections are: 10199 Chariklo (11 repetitions, 6.7%), 330836 Orius (5 repetitions, 3.0%), 346889 Rhiphonos (4 repetitions, 2.4%), 60558 Echeclus (3 repetitions, 1.8%), 121725 Aphidas (3 repetitions, 1.8%).

In our 1d, 10d, and 100d study windows, the AU distribution characteristics of the celestial bodies matched by the formula (Fig. S1) AU Distribution by Analysis Windows, (Fig. S2) Scatter of AU values by analysis windows, the days on which celestial bodies matched (Fig. S3) Temporal presence of objects (10d window) with 1d detail inset, (Fig. S4) Global detection matrix colored by Heliocentric Distance (AU), and on which days and how frequently celestial bodies were found (Fig. S5) Distribution of object repetition frequencies across analysis windows are shown.

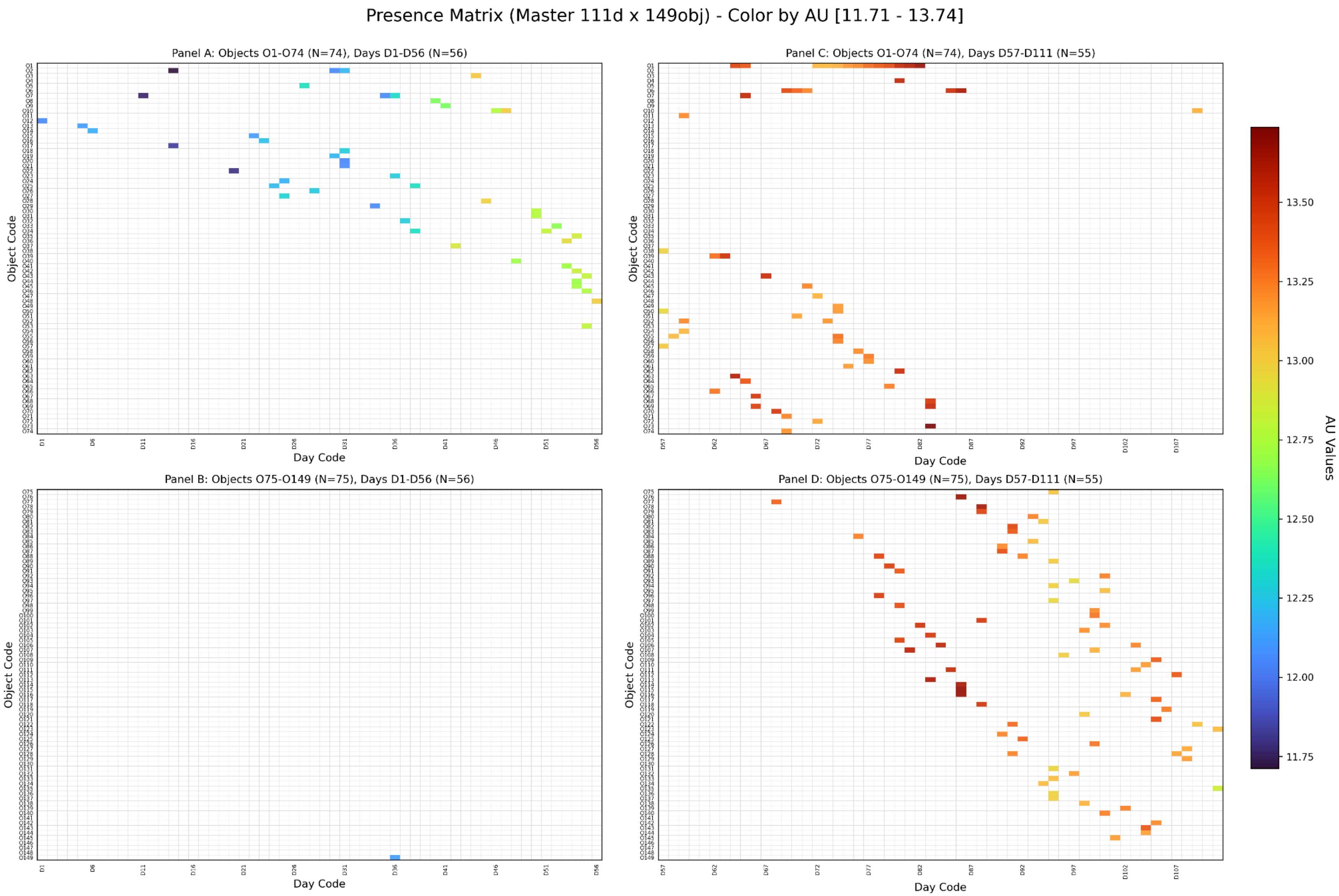

Fig. S4 Global detection matrix colored by Heliocentric Distance (AU).

Table S1. Day codes (D_1-D_{111})

#	Day Code	Match Date	#	Day Code	Match Date	#	Day Code	Match Date
1	D_{1}	1960-07-11	**39**	D_{39}	1984-03-11	**77**	D_{77}	2007-05-30
2	D_{2}	1960-11-28	**40**	D_{40}	1985-08-11	**78**	D_{78}	2007-10-25
3	D_{3}	1961-05-20	**41**	D_{41}	1985-11-06	**79**	D_{79}	2008-01-31
4	D_{4}	1962-08-14	**42**	D_{42}	1986-12-28	**80**	D_{80}	2008-12-13
5	D_{5}	1962-12-03	**43**	D_{43}	1987-02-26	**81**	D_{81}	2009-03-22
6	D_{6}	1963-09-18	**44**	D_{44}	1987-07-15	**82**	D_{82}	2009-08-27
7	D_{7}	1964-02-25	**45**	D_{45}	1988-06-23	**83**	D_{83}	2010-12-05
8	D_{8}	1964-07-03	**46**	D_{46}	1989-01-08	**84**	D_{84}	2011-02-14
9	D_{9}	1965-03-15	**47**	D_{47}	1989-05-19	**85**	D_{85}	2011-09-01
10	D_{10}	1966-01-12	**48**	D_{48}	1990-09-13	**86**	D_{86}	2011-11-30
11	D_{11}	1966-10-30	**49**	D_{49}	1990-12-01	**87**	D_{87}	2012-07-18
12	D_{12}	1967-08-08	**50**	D_{50}	1991-03-05	**88**	D_{88}	2012-11-09
13	D_{13}	1968-04-14	**51**	D_{51}	1992-02-11	**89**	D_{89}	2013-05-11
14	D_{14}	1968-09-05	**52**	D_{52}	1992-06-20	**90**	D_{90}	2014-02-09
15	D_{15}	1969-11-02	**53**	D_{53}	1993-08-09	**91**	D_{91}	2014-07-23
16	D_{16}	1970-01-27	**54**	D_{54}	1993-10-02	**92**	D_{92}	2015-06-27
17	D_{17}	1970-08-22	**55**	D_{55}	1994-01-10	**93**	D_{93}	2015-09-04
18	D_{18}	1971-06-09	**56**	D_{56}	1995-04-22	**94**	D_{94}	2016-04-19
19	D_{19}	1972-01-07	**57**	D_{57}	1995-07-28	**95**	D_{95}	2017-03-14
20	D_{20}	1972-03-21	**58**	D_{58}	1996-02-02	**96**	D_{96}	2017-06-11
21	D_{21}	1973-11-29	**59**	D_{59}	1996-08-30	**97**	D_{97}	2018-02-20
22	D_{22}	1974-05-16	**60**	D_{60}	1997-06-15	**98**	D_{98}	2018-10-18
23	D_{23}	1974-12-11	**61**	D_{61}	1997-11-17	**99**	D_{99}	2019-07-04
24	D_{24}	1975-02-04	**62**	D_{62}	1998-03-29	**100**	D_{100}	2019-12-25
25	D_{25}	1976-03-18	**63**	D_{63}	1998-05-03	**101**	D_{101}	2020-03-09
26	D_{26}	1976-07-29	**64**	D_{64}	1999-08-14	**102**	D_{102}	2020-11-21
27	D_{27}	1977-08-15	**65**	D_{65}	2000-01-23	**103**	D_{103}	2021-04-26
28	D_{28}	1978-01-19	**66**	D_{66}	2000-06-07	**104**	D_{104}	2021-08-16
29	D_{29}	1978-08-02	**67**	D_{67}	2001-07-07	**105**	D_{105}	2022-07-22
30	D_{30}	1979-04-05	**68**	D_{68}	2001-10-15	**106**	D_{106}	2023-01-04
31	D_{31}	1979-06-15	**69**	D_{69}	2002-09-19	**107**	D_{107}	2023-06-03
32	D_{32}	1980-04-29	**70**	D_{70}	2002-12-08	**108**	D_{108}	2024-05-29
33	D_{33}	1980-10-12	**71**	D_{71}	2003-03-12	**109**	D_{109}	2024-09-10
34	D_{34}	1981-09-01	**72**	D_{72}	2004-04-16	**110**	D_{110}	2025-01-17
35	D_{35}	1982-02-17	**73**	D_{73}	2004-06-18	**111**	D_{111}	2025-09-15
36	D_{36}	1982-07-01	**74**	D_{74}	2005-02-28			
37	D_{37}	1983-05-24	**75**	D_{75}	2006-04-09			
38	D_{38}	1984-01-20	**76**	D_{76}	2006-09-24			

Table S2. Object codes (O_1-O_{64})

#	Object Code	Object Name	#	Object Code	Object Name
1	O_1	10199 Chariklo (1997 CU26)	**33**	O_{33}	C/1987 W3 (Jensen-Shoemaker)
2	O_2	121725 Aphidas (1999 XX143)	**34**	O_{34}	C/1988 C1 (Maury-Phinney)
3	O_3	2060 Chiron (1977 UB)	**35**	O_{35}	C/1988 L1 (Shoemaker-Holt-Rodriquez)
4	O_4	31824 Elatus (1999 UG5)	**36**	O_{36}	O36: C/1989 Q1 (Okazaki-Levy-Rudenko)
5	O_5	32532 Thereus (2001 PT13)	**37**	O_{37}	O37: C/1990 K1 (Levy)
6	O_6	330836 Orius (2009 HW77)	**38**	O_{38}	O38: C/1990 M1 (McNaught-Hughes)
7	O_7	346889 Rhiphonos (2009 QV38)	**39**	O_{39}	O39: C/1993 Q1 (Mueller)
8	O_8	5145 Pholus (1992 AD)	**40**	O_{40}	O40: C/1994 N1 (Nakamura-Nishimura-Machholz)
9	O_9	5335 Damocles (1991 DA)	**41**	O_{41}	O41: C/1997 D1 (Mueller)
10	O_{10}	60558 Echeclus (2000 EC98)	**42**	O_{42}	O42: C/1997 J2 (Meunier-Dupouy)
11	O_{11}	7066 Nessus (1993 HA2)	**43**	O_{43}	O43: C/1997 N1 (Tabur)
12	O_{12}	C/1956 R1 (Arend-Roland)	**44**	O_{44}	O44: C/1998 M3 (Larsen)
13	O_{13}	C/1959 Q2 (Alcock)	**45**	O_{45}	O45: C/1998 U1 (LINEAR)
14	O_{14}	C/1966 T1 (Rudnicki)	**46**	O_{46}	O46: C/1998 W3 (LINEAR)
15	O_{15}	C/1970 N1 (Abe)	**47**	O_{47}	O47: C/1999 H3 (LINEAR)
16	O_{16}	C/1971 E1 (Toba)	**48**	O_{48}	O48: C/1999 J2 (Skiff)
17	O_{17}	C/1972 L1 (Sandage)	**49**	O_{49}	O49: C/1999 K5 (LINEAR)
18	O_{18}	C/1975 V1-A (West)	**50**	O_{50}	O50: C/1999 K8 (LINEAR)
19	O_{19}	C/1975 V2 (Bradfield)	**51**	O_{51}	C/1999 S2 (McNaught-Watson)
20	O_{20}	C/1975 X1 (Sato)	**52**	O_{52}	C/1999 S4 (LINEAR)
21	O_{21}	C/1976 D2 (Schuster)	**53**	O_{53}	C/1999 U1 (Ferris)
22	O_{22}	C/1976 U1 (Lovas)	**54**	O_{54}	C/1999 Y1 (LINEAR)
23	O_{23}	C/1978 H1 (Meier)	**55**	O_{55}	C/2000 A1 (Montani)
24	O_{24}	C/1979 M1 (Bradfield)	**56**	O_{56}	C/2000 O1 (Koehn)
25	O_{25}	C/1979 M3 (Torres)	**57**	O_{57}	C/2000 U5 (LINEAR)
26	O_{26}	C/1980 E1 (Bowell)	**58**	O_{58}	C/2001 G1 (LONEOS)
27	O_{27}	C/1980 L1 (Torres)	**59**	O_{59}	C/2001 RX14 (LINEAR)
28	O_{28}	C/1983 R1 (Shoemaker)	**60**	O_{60}	C/2002 A3 (LINEAR)
29	O_{29}	C/1984 K1 (Shoemaker)	**61**	O_{61}	C/2002 E2 (Snyder-Murakami)
30	O_{30}	C/1986 P1-A (Wilson)	**62**	O_{62}	C/2002 J5 (LINEAR)
31	O_{31}	C/1986 P1-B (Wilson)	**63**	O_{63}	C/2002 O7 (LINEAR)
32	O_{32}	C/1987 A1 (Levy)	**64**	O_{64}	C/2003 A2 (Gleason)

Table S2. Object codes (O_{65}-O_{128})

#	Object Code	Object Name	#	Object Code	Object Name
65	O_{65}	C/2003 G1 (LINEAR)	**97**	O_{97}	C/2013 L2 (Catalina)
66	O_{66}	C/2003 S3 (LINEAR)	**98**	O_{98}	C/2013 S1 (Catalina)
67	O_{67}	C/2004 X3 (LINEAR)	**99**	O_{99}	C/2013 V2 (Borisov)
68	O_{68}	C/2005 B1 (Christensen)	**100**	O_{100}	C/2014 AA52 (Catalina)
69	O_{69}	C/2005 Q1 (LINEAR)	**101**	O_{101}	C/2014 B1 (Schwartz)
70	O_{70}	C/2006 E1 (McNaught)	**102**	O_{102}	C/2014 L5 (Lemmon)
71	O_{71}	C/2006 K3 (McNaught)	**103**	O_{103}	C/2014 S1 (PANSTARRS)
72	O_{72}	C/2006 OF2 (Broughton)	**104**	O_{104}	C/2014 W10 (PANSTARRS)
73	O_{73}	C/2006 P1 (McNaught)	**105**	O_{105}	C/2014 W3 (PANSTARRS)
74	O_{74}	C/2006 S2 (LINEAR)	**106**	O_{106}	C/2015 D3 (PANSTARRS)
75	O_{75}	C/2006 S3 (LONEOS)	**107**	O_{107}	C/2015 J1 (PANSTARRS)
76	O_{76}	C/2006 YC (Catalina-Christensen)	**108**	O_{108}	C/2015 K7 (COIAS)
77	O_{77}	C/2007 JA21 (LINEAR)	**109**	O_{109}	C/2015 T2 (PANSTARRS)
78	O_{78}	C/2007 W3 (LINEAR)	**110**	O_{110}	C/2015 V2 (Johnson)
79	O_{79}	C/2008 A1 (McNaught)	**111**	O_{111}	C/2015 X7 (ATLAS)
80	O_{80}	C/2008 FK75 (Lemmon-Siding Spring)	**112**	O_{112}	C/2015 XY1 (Lemmon)
81	O_{81}	C/2008 S3 (Boattini)	**113**	O_{113}	C/2016 C1 (PANSTARRS)
82	O_{82}	C/2009 K5 (McNaught)	**114**	O_{114}	C/2016 E2 (Kowalski)
83	O_{83}	C/2009 O4 (Hill)	**115**	O_{115}	C/2016 J2 (Denneau)
84	O_{84}	C/2009 R1 (McNaught)	**116**	O_{116}	C/2016 K1 (LINEAR)
85	O_{85}	C/2009 UG89 (Lemmon)	**117**	O_{117}	C/2017 E3 (PANSTARRS)
86	O_{86}	C/2010 F4 (Machholz)	**118**	O_{118}	C/2017 F1 (Lemmon)
87	O_{87}	C/2010 U3 (Boattini)	**119**	O_{119}	C/2017 F2 (PANSTARRS)
88	O_{88}	C/2010 X1 (Elenin)	**120**	O_{120}	C/2017 K2 (PANSTARRS)
89	O_{89}	C/2011 F1 (LINEAR)	**121**	O_{121}	C/2017 S7 (Lemmon)
90	O_{90}	C/2011 UF305 (LINEAR)	**122**	O_{122}	C/2017 U7 (PANSTARRS)
91	O_{91}	C/2012 E2 (SWAN)	**123**	O_{123}	C/2017 Y2 (PANSTARRS)
92	O_{92}	C/2012 F3 (PANSTARRS)	**124**	O_{124}	C/2018 C2 (Lemmon)
93	O_{93}	C/2013 B2 (Catalina)	**125**	O_{125}	C/2018 W2 (Africano)
94	O_{94}	C/2013 G2 (McNaught)	**126**	O_{126}	C/2019 E3 (ATLAS)
95	O_{95}	C/2013 G9 (Tenagra)	**127**	O_{127}	C/2019 K1 (ATLAS)
96	O_{96}	C/2013 J5 (Boattini)	**128**	O_{128}	C/2019 M3 (ATLAS)

Table S2. Object codes (O_{129}-O_{149})

#	Day Code	Object Name	#	Day Code	Object Name
129	O_{129}	C/2019 M4 (TESS)	**140**	O_{140}	C/2024 N4 (Sarneczky)
130	O_{130}	C/2019 Q3 (PANSTARRS)	**141**	O_{141}	C/2024 O1 (PANSTARRS)
131	O_{131}	C/2020 N1 (PANSTARRS)	**142**	O_{142}	C/2024 T5 (ATLAS)
132	O_{132}	C/2020 R7 (ATLAS)	**143**	O_{143}	C/2025 B2 (Borisov)
133	O_{133}	C/2020 T4 (PANSTARRS)	**144**	O_{144}	C/2025 F2 (SWAN)
134	O_{134}	C/2020 W5 (Lemmon)	**145**	O_{145}	C/2025 N2 (ATLAS)
135	O_{135}	C/2021 A1 (Leonard)	**146**	O_{146}	C/2ZREİ E2 (ATLAS)
136	O_{136}	C/2021 K3 (Catalina)	**147**	O_{147}	C/2ZREİ K1 (Leonard)
137	O_{137}	C/2021 S1 (ATLAS)	**148**	O_{148}	C/2ZREİ O1 (ATLAS)
138	O_{138}	C/2021 X1 (Maury-Attard)	**149**	O_{149}	Pioneer 11
139	O_{139}	C/2024 N1 (PANSTARRS)			

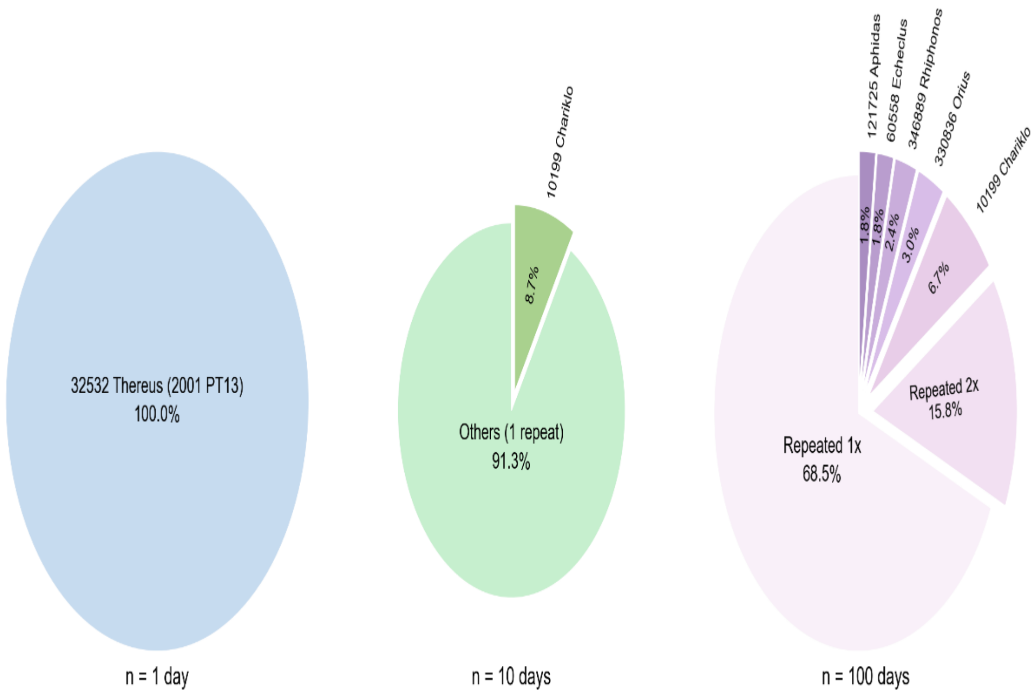

Fig. S5 Distribution of object repetition frequencies across analysis windows.

A comprehensive statistical framework (Sections 1.1-1.4) was implemented across four distinct time scales to ascertain whether the observed celestial matches constitute a random coincidence or indicate a specific, hidden systematic pattern. In the statistical assessment of randomness, a strict significance threshold of $\alpha = 0.001$ (0.1%) was used to maximize the reliability of the results.

4.1.1. Statistical Analysis

4.1.1.1. Single-Day Analysis: Wow! Signal and P-Value Test

The statistical significance of the single match that occurred on the specific day the Wow! signal was received was calculated using the Null Hypothesis H_0 with a pool of N=26576 celestial bodies to compute the P-Value using Eq. (1). The calculated P-Value is 26 times smaller than the determined strict α value.

$$P\text{-value} = \frac{1}{N} = \frac{1}{26{,}576} \approx 0.0000376 < \alpha = 0.001 \quad (1)$$

The probability of this match occurring by chance is statistically negligible. The possibility of randomness was dismissed. This finding, when we apply our formula to the Wow signal date, strongly suggests that the single celestial body match found could be strong statistical evidence of being meaningful and deliberate.

4.1.1.2. 10-Day Analysis: Regional Selectivity and Z-Test

The statistical significance of our applied formula was analyzed for a 10-day study (total trial count 265,760, with 23 total matches) based on H_0 and H_1 hypothesis values. The Z-Score, Fisher's Combined Probability Test with 10 independent days' probabilities, Chi-square χ^2 statistical test statistic, and Signal/Noise Ratio (SNR) were calculated (using Eq. (8)).

$$\hat{\boldsymbol{p}} = \frac{x}{N} = \frac{23}{265{,}760} \approx 0.000086544 \quad (2)$$

$$\boldsymbol{Z} = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}\; Z = \frac{-0.49913456}{0.00097} \approx -515.37 \quad (3)$$

For the day the Wow signal was received, a single celestial body match ($\lambda=1$) expectation was assumed. For each individual day in the 10-day study: the probability of finding k matches was calculated using the Poisson Probability Mass Function (PMF) using Eq. (4) , p-Values using the Right Tail Test Formula (Eq. (5)), the contribution of the χ^2 statistical test statistic using Fisher's Combined Probability Test (Eq. (6)), and the total Fisher's χ^2 Statistic for the 10-day study was calculated using Eq. (7) and was given in (Table S3). The Signal/Noise ratio was calculated by using Eq. (8) to evaluate whether our formula targeted the 32532 Thereus object detected in our single-day study.

$$\boldsymbol{P}(\boldsymbol{k} \mid \boldsymbol{\lambda}) = \frac{\lambda^k \times e^{-\lambda}}{k!} \tag{4}$$

$$\boldsymbol{p_i} = P(X \geq x_i \mid \lambda = 1) = 1 - P(X \leq x_i - 1 \mid \lambda = 1) \tag{5}$$

$$-2 \times \ln(pi) \tag{6}$$

$$\chi^2 = -2 \times \sum_{i=1}^{10} \ln(p_i) \tag{7}$$

$$\text{Message Clarity} = \frac{S}{N} = \frac{\text{Thereus Detection Number}}{\text{Others' detection number)}} \tag{8}$$

Fisher's χ^2 Statistic (λ=1), χ^2 was calculated as 42.5714. The critical χ^2 value, using a degree of freedom df=2k = 20, and a significance threshold of $\alpha = 0.001$, was found to be 45.3147 using the Inverse CDF Method and compared with the calculated Fisher's χ^2 Statistic (Table S4.) The p-value of 0.00008654, being dramatically and immeasurably far from the $p_0 = 0.5$ expectation, and the Z-score value of -515.37 being exactly 515.37 standard deviations away from the randomness expectation, rejects randomness from the Z-test perspective. Although the $\alpha = 0.001$ Fisher's χ^2 Statistic result, $\chi^2 = 42.5714$, is very close to the critical value required for a very strict threshold like α=0.001, it falls short. For $\alpha = 0.001$: the H_0 hypothesis cannot be rejected, for $\alpha = 0.01$ and $\alpha = 0.05$: the H_0 hypothesis is rejected. Although the 10-day data study is not sufficient for a very strict threshold like α=0.001, it is shown that it provides significant and strong evidence at the α=0.01 level, indicating the necessity of expanding the data set.

This also revealed the necessity of expanding the data set. 32532 Thereus was not detected at all in the 10-day period, and the S/N ratio was calculated as 0. Due to $Z \approx -163$, $p \approx 0.000865$, $\chi^2 = 42.5714$ (very close to the critical degree), it was shown that the pattern of formula output matches is not random, and due to S/N = 0, it can be said that the formula detects orbital characteristics or a broader physical principle like a specific region of space.

Table S3. Fisher's χ^2 static values for the 10d analysis window

Day	Date	Match Count	p_i	$-2\times\ln(p_i)$
1	1981-09-01	1	0.632121	0.9174
2	1965-03-15	0	1.000000	0.0000
3	1973-11-29	0	1.000000	0.0000
4	1988-06-23	1	0.632121	0.9174
5	1994-01-10	3	0.080301	5.0439
6	1999-08-14	2	0.264241	2.6618
7	2005-02-28	5	0.003660	11.2207
8	2010-12-05	5	0.003660	11.2207
9	2016-04-19	2	0.264241	2.6618
10	2022-07-22	4	0.018988	7.9279

Table S4. χ^2 statistics across significance levels (10d)

Significance level	Critical χ^2 values	Control	Outcome
$\alpha = 0.05$	31.4104	**$\chi^2 > \chi^2$ (critical)**	✓
$\alpha = 0.01$	37.5662	**$\chi^2 > \chi^2$ (critical)**	✓
$\alpha = 0.001$	45.3147	**$\chi^2 < \chi^2$ (critical)**	X

4.1.1.3. 100-Day Analysis: Regional Selectivity and Z-Test

The statistical significance of our applied formula was analyzed for a 100-day study (total trial count 2,657,600, with 165 total matches) based on Ho and H_1 hypothesis values. The Z-Score, Fisher's Combined Probability Test with 100 independent days' probabilities, Chi-square χ^2 statistical test statistic, and Signal/Noise Ratio (SNR) were calculated.

$$\boldsymbol{\hat{p}} = \frac{x}{N} = \frac{165}{2{,}657{,}600} \approx 0.000062086 \tag{2}$$

$$\boldsymbol{Z} = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}\; Z = \frac{-0.499937914}{0.000306708} \approx -1630.01 \qquad (p_0 = 0.5) \tag{3}$$

For the day the Wow signal was received, a single celestial body match (λ=1) expectation was assumed. For each individual day in the 100-day study: the probability of finding k matches was calculated using the Poisson Probability Mass Function (PMF), p-Values using the Right Tail Test Formula, the contribution of the χ^2 statistical test statistic using Fisher's Combined Probability Test, and the total Fisher's χ^2 Statistic for the 100-day study was calculated (Table S5) .

Table S5. Fisher's χ^2 static values for the 100d analysis window

Day	Date	Match Count	p_i	$-2\times\ln(p_i)$
1	7/11/1960	1	0,632121	0,9174
2	11/28/1960	0	1	0
3	5/20/1961	0	1	0
4	8/14/1962	0	1	0
5	12/3/1962	1	0,632121	0,9174
6	9/18/1963	1	0,632121	0,9174
7	2/25/1964	0	1	0
8	7/3/1964	0	1	0
9	1/12/1966	0	1	0
10	10/30/1966	1	0,632121	0,9174
11	8/8/1967	0	1	0
12	4/14/1968	0	1	0
13	9/5/1968	2	0,264241	2,6618
14	11/2/1969	0	1	0
15	1/27/1970	0	1	0
16	8/22/1970	0	1	0
17	6/9/1971	0	1	0
18	1/7/1972	0	1	0
19	3/21/1972	1	0,632121	0,9174
20	5/16/1974	1	0,632121	0,9174
21	12/11/1974	1	0,632121	0,9174
22	2/4/1975	1	0,632121	0,9174
23	3/18/1976	2	0,264241	2,6618
24	7/29/1976	0	1	0
25	1/19/1978	1	0,632121	0,9174
26	8/2/1978	0	1	0
27	4/5/1979	2	0,264241	2,6618
28	6/15/1979	4	0,018988	7,9279
29	4/29/1980	0	1	0
30	10/12/1980	0	1	0
31	2/17/1982	1	0,632121	0,9174
32	7/1/1982	3	0,080301	5,0439
33	5/24/1983	1	0,632121	0,9174
34	1/20/1984	2	0,264241	2,6618

35	3/1/1984	0	1	0
36	8/11/1985	1	0,632121	0,9174
37	11/6/1985	1	0,632121	0,9174
38	12/28/1986	1	0,632121	0,9174
39	2/26/1987	0	1	0
40	7/15/1987	1	0,632121	0,9174
41	1/8/1989	1	0,632121	0,9174
42	5/19/1989	1	0,632121	0,9174
43	9/13/1990	1	0,632121	0,9174
44	12/1/1990	0	1	0
45	3/5/1991	2	0,264241	2,6618
46	2/11/1992	1	0,632121	0,9174
47	6/20/1992	1	0,632121	0,9174
48	8/9/1993	2	0,264241	2,6618
49	10/2/1993	4	0,018988	7,9279
50	4/22/1995	1	0,632121	0,9174
51	7/28/1995	3	0,080301	5,0439
52	2/2/1996	1	0,632121	0,9174
53	8/30/1996	3	0,080301	5,0439
54	6/15/1997	0	1	0
55	11/17/1997	0	1	0
56	3/29/1998	2	0,264241	2,6618
57	5/3/1998	1	0,632121	0,9174
58	1/23/2000	3	0,080301	5,0439
59	6/7/2000	2	0,264241	2,6618
60	7/7/2001	1	0,632121	0,9174
61	10/15/2001	2	0,264241	2,6618
62	9/19/2002	3	0,080301	5,0439
63	12/8/2002	2	0,264241	2,6618
64	3/12/2003	2	0,264241	2,6618
65	4/16/2004	3	0,080301	5,0439
66	6/18/2004	2	0,264241	2,6618
67	4/9/2006	2	0,264241	2,6618
68	9/24/2006	3	0,080301	5,0439
69	5/30/2007	3	0,080301	5,0439
70	10/25/2007	3	0,080301	5,0439
71	1/31/2008	3	0,080301	5,0439
72	12/13/2008	6	0,000594	14,8566
73	3/22/2009	2	0,264241	2,6618
74	8/27/2009	2	0,264241	2,6618
75	2/14/2011	1	0,632121	0,9174
76	9/1/2011	2	0,264241	2,6618

77	11/30/2011	5	0,00366	11,2207
78	7/8/2012	0	1	0
79	11/9/2012	4	0,018988	7,9279
80	5/1/2013	1	0,632121	0,9174
81	2/9/2014	3	0,080301	5,0439
82	7/23/2014	4	0,018988	7,9279
83	6/27/2015	2	0,264241	2,6618
84	9/4/2015	2	0,264241	2,6618
85	3/14/2017	10	0	32,0198
86	6/11/2017	1	0,632121	0,9174
87	2/20/2018	3	0,080301	5,0439
88	10/18/2018	3	0,080301	5,0439
89	7/4/2019	4	0,018988	7,9279
90	12/25/2019	5	0,00366	11,2207
91	3/9/2020	1	0,632121	0,9174
92	11/21/2020	2	0,264241	2,6618
93	4/26/2021	2	0,264241	2,6618
94	8/16/2021	3	0,080301	5,0439
95	1/4/2023	1	0,632121	0,9174
96	6/3/2023	2	0,264241	2,6618
97	5/29/2024	2	0,264241	2,6618
98	9/10/2024	2	0,264241	2,6618
99	1/17/2025	0	1	0
100	9/15/2025	2	0,264241	2,6618

When we took Fisher's χ^2 Statistic (λ=1), χ^2 was calculated as 270.0597. The critical χ^2 value, using a degrees of freedom df=2k = 200, and a significance threshold of α = 0.001, was found to be 278.5 using the Inverse CDF Method and compared with the calculated Fisher's χ^2 Statistic (Table S6) .

The p-value of 0.000062086, being dramatically and immeasurably far from the p_0 = 0.5 expectation, and the Z-score value of -1630.01 being exactly 1630.01 standard deviations away from the randomness expectation, rejects randomness from the Z-test perspective. Although the α = 0.001 Fisher's χ^2 Statistic result, χ^2 = 270.0597, is very close to the critical value required for a very strict threshold like α=0.001, it falls short. For α = 0.001: the H_0 hypothesis cannot be rejected, for α = 0.01 and α = 0.05: the H_0 hypothesis is rejected. Although the 100-day data study is not sufficient for a very strict threshold like α=0.001, it is shown that it provides significant and strong evidence at the α=0.01 level, indicating the necessity of expanding the data set. 32532 Thereus was not detected at all in the 100-day period, and the S/N ratio was calculated as 0. Due to Z ≈ -1630.01, p ≈ 0.000062086,

χ^2 = 270.0597 (very close to the critical degree), it was shown that the pattern of formula output matches is not random, and due to S/N = 0, it can be said that the formula detects orbital characteristics or a broader physical principle like a specific region of space .

Table S6. χ^2 statistics across significance levels (100d)

Significance level	Critical χ^2 value	Control	Outcome
$\alpha = 0.05$	233.9	**$\chi^2 > \chi^2$ (critical)**	✓
$\alpha = 0.01$	248.4	**$\chi^2 > \chi^2$ (critical)**	✓
$\alpha = 0.001$	278.5	**$\chi^2 < \chi^2$ (critical)**	X

4.1.2. Frequency Distribution Analysis: 111-Day Period (189 Total Detections)

In the large dataset formed by combining 111 randomly selected unique days between 1960-07-11 and 2025-09-15 across 1d, 10d, and 100d studies, the frequencies and repetition count of celestial bodies matching our formula were calculated (Table S7).

In the 111-day dataset, a total of 189 matches were observed, and a total of 149 unique celestial bodies were observed. The top 5 most frequently matched celestial bodies in the 111-day dataset and their respective S/R ratios are: 10199 Chariklo (1997 CU26): 13 matches, ≈ 7.39%; 330836 Orius (2009 HW77): 5 matches, 2.72%; 346889 Rhiphonos (2009 QV38): 4 matches, 2.16%; 60558 Echeclus (2000 EC98): 3 matches, 1.61%, 121725 Aphidas (1999 XX143): 3 matches, 1.61%. 17 unique celestial bodies matched 2 times, and 127 unique celestial bodies matched 1 time. Although the S/N ratio of 10199 Chariklo (1997 CU26) is 7.39%, indicating it is the most suitable object for our formula's orbital resonance, its absolute S/N value is small and remains a small proportion compared to the overall signal. The fact that the most frequently captured objects belong to the Centaur classes with chaotic orbital dynamics, and that the leading Chariklo object has a low S/N ratio, indicates that the formula is sensitive not to a specific object's coordinate point, but to the current orbital states of objects, or orbital resonances, or specific orbital classes. Our formula scans a large spatial volume, and based on the matching results, a second sensitivity of the formula has been observed. It also samples objects from the more distant and chaotic regions of the Solar System, which have irregular orbits and are inclined relative to the Solar System plane. Most of the objects that matched once belong to the

Table S7. Distribution of object repetition frequencies

Repetition count	Number of Unique Objects
13	1
5	1
4	1
3	2
2	17
1	127

long-period distant comet class. The matching characteristics of our formula are observed to point to a specific group and class of objects with specific orbital dynamics, rather than revealing a specific hidden signal.

To analyze the finding that the formula is sensitive to specific orbital dynamics rather than a single coordinate, and to answer whether the detected orbital classes exhibit a random distribution or contain a systematic spatial shift, Epoch-based AU Distribution Density data covering the 1960-2025 range was examined.

(Fig. S6) Density distribution of AU values by epoch (Pre- vs. Post-1990), shows how the AU (Astronomical Unit) values of the total 189 detections matching the formula in a large 111-day dataset, consisting of randomly selected days between 1960 and 2025, change their characteristics, comparing Kernel Density Estimation curves and Median values (Fig. S6).

The frequency density of all data (Overall Histogram) is shown as a pale gray bar graph. Two distinct peaks (modes) are clearly visible in the graph. The dataset of 189 total matches in terms of AU density characteristics is divided into two different periods: pre-1990 and post-1990. The Kernel density curve shown with the Blue Curve shows the AU characteristics of 35 matches from before 1990. Detections from this period peak sharply at a median of 12.22 AU. The Orange Curve shows the characteristics of a total of 154 matches from after 1990. Detections from this period have a flatter peak at a median of 13.22 AU. After 1990, the median AU value of detections shifted by exactly 1.0 AU (13.22 AU - 12.22 AU) further away from the Sun. This indicates that the formula's spatial scanning window shifted outwards over time, or that objects matching the formula became more concentrated in that region. Detections made before 1990 are closer to objects in Jupiter's orbit (JFC - Jupiter Family Comets) or inner Centaur orbits. Post-1990 detections are more focused on the center of typical Centaur orbits, extending more towards Neptune.

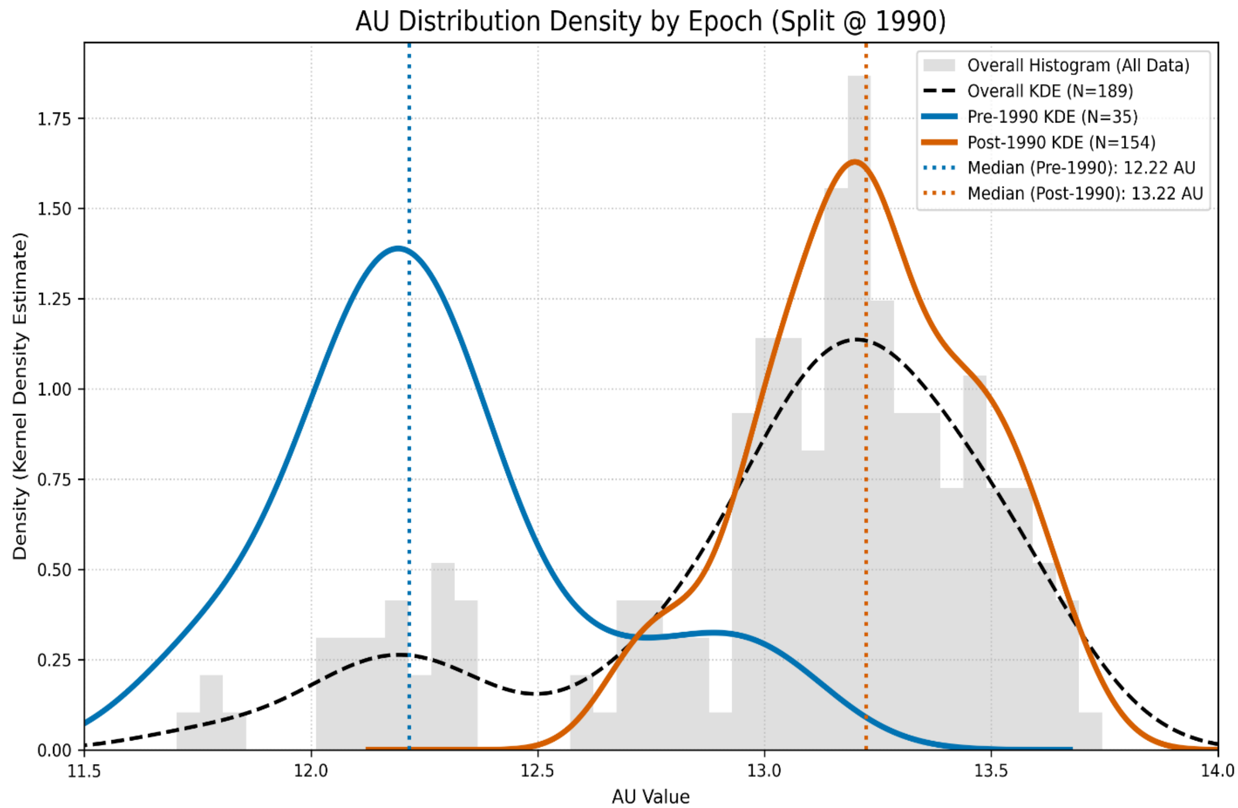

Fig. S6 Density distribution of AU values by epoch (Pre- vs. Post-1990).

In the light of all these results and the bimodal distribution characteristic, it has been concluded that the formula points to objects belonging to a specific spatial class characteristic, and that the spatial distance values in the orbits of these objects change over time, resulting in a shift in the AU values of the matches detected by the formula over time. The main sensitivity of the formula is observed to be to the Centaur class in more dynamic and outer orbits over time.

Using the unsupervised learning technique DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm, the data of 189 objects matched by the formula in our 111-day dataset were analyzed to reveal their natural structures, clusters (density-based groups), and isolated points (noise points) without any prior assumptions, based on time and spatial position. (Day codes for 111 days are in (Table S1), and object codes for 149 unique objects in 189 total matches are in (Table S2)).

By using a Standard Scaler, equal weighting was ensured for the time (day) and AU (distance) axes in the 111-day dataset, eps=0.3 (neighborhood radius), and a minimum cluster size of 5 were chosen.

DBSCAN analysis showed that the 189 detections were not randomly distributed in the Time-AU space, but had a natural structure, separating the 189 detections into two main dense clusters and isolated noise (Fig. S7).

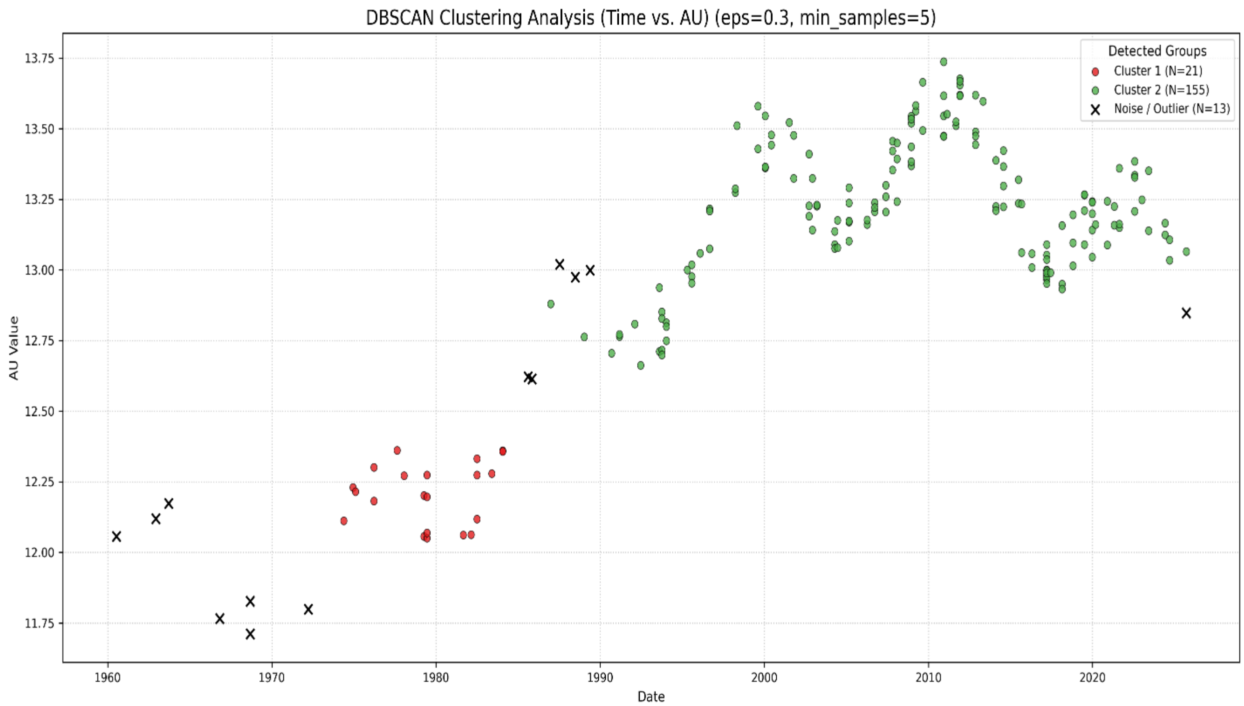

Fig. S7 Spatiotemporal clustering of detections using DBSCAN algorithm

Cluster 1 (Low AU regime), separated by red, showed 21 object matches in the time range of ~1960 – 1985, within ~12.0 – 12.4 AU. Cluster 2 (High AU regime), separated by green, showed 155 object matches in the time range of ~1985 onwards, within ~12.5 – 13.75 AU. The period between 1985-1990 was observed as a natural turning point where the formula's detections shifted from one AU regime to another (from low AU to high AU). DBSCAN analysis identified 13 isolated noise points (these 13 outlier detection points (listed in (Table S8)) are inter-regime detections in dynamically mixed regions) as not fully conforming to the main regime of either the early or late periods, based on the density and neighborhood characteristics of other points in the dataset. This indicates that the formula points to a specific physical resonance law.

Table S8. List of outliers detected by DBSCAN

#	Object Code	Match Date	Object Name	AU Value	Analysis Window
1	O_2	1968-09-05	121725 Aphidas (1999 XX143)	11.7121	100d
2	O_5	1977-08-15	32532 Thereus (2001 PT13)	12.3615	1d
3	O_7	1966-10-30	346889 Rhiphonos (2009 QV38)	11.7654	100d
4	O_{12}	1960-07-11	C/1956 R1 (Arend-Roland)	12.0573	100d
5	O_{13}	1962-12-03	C/1959 Q2 (Alcock)	12.1191	100d
6	O_{14}	1963-09-18	C/1966 T1 (Rudnicki)	12.1733	100d
7	O_{15}	1974-05-16	C/1970 N1 (Abe)	12.1127	100d
8	O_{17}	1968-09-05	C/1972 L1 (Sandage)	11.8270	100d
9	O_{20}	1979-06-15	C/1975 X1 (Sato)	12.0504	100d
10	O_{21}	1979-06-15	C/1976 D2 (Schuster)	12.0694	100d
11	O_{22}	1972-03-21	C/1976 U1 (Lovas)	11.7990	100d
12	O_{24}	1979-04-05	C/1975 V2 (Bradfield)	12.2022	100d
13	O_{26}	1978-01-19	C/1980 E1 (Bowell)	12.2715	100d

The relationship between AU values (Y-axis) and detection times (X-axis) for 189 objects detected between 1960 and 2025 was analyzed using a 30-point centered rolling mean method (Fig. S8). Three detection datasets were represented: objects from 1-day analysis (circles), 10-day analysis (squares), and 100-day analysis (triangles). AU values for detected objects were color-coded according to the AU scale legend displayed in the right panel of the figure. The 100-day study constituted the primary dataset for determining the trend and ±2σ normal range due to its data abundance. Detections from the 10-day study concentrated in the high-AU region after 1990, supporting the observed trend. A single detection from the 1-day study near 1977 was observed during a period of relatively low trend values.

Statistical analysis revealed that the relationship between AU values and detection times was non-random and statistically significant. A moderate positive monotonic relationship was demonstrated, with AU values generally increasing over time (Spearman correlation coefficient $\rho = 0.48$). The probability of this correlation occurring by chance was less than 0.1% ($p < 0.001$). These findings indicate that the formula exhibits sensitivity to a time-dependent, evolving resonance rather than instantaneous specific observations, and possesses a predictable, mathematically modellable relationship.

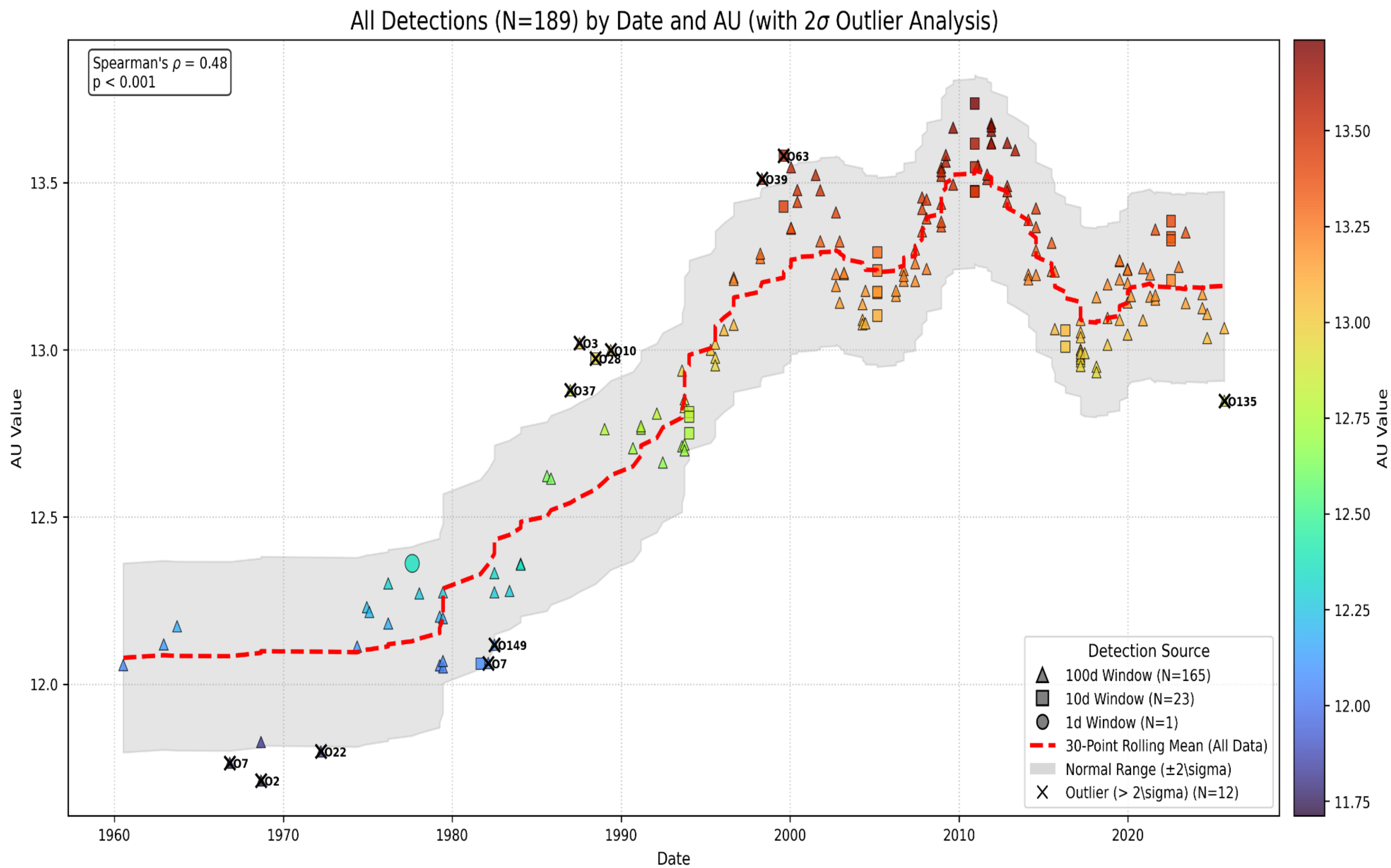

Fig. S8 Temporal Evolution of AU Values with Rolling Mean and Outlier Analysis

The S-shaped trend line, represented by the red rolling mean curve, demonstrated that the trend was not constant over the study period. From the 1960s to 1980, the trend remained relatively stable at approximately 12.1 AU. Between 1980 and the mid-2000s, a sharp increase was observed. The trend peaked at approximately 13.5 AU around 2005–2010, followed by a slight declining tendency.

Within the gray normal range (±2σ) around the trend line, 12 of the 189 total detections (marked with black 'X' symbols in the figure) were classified as statistical anomalies. Objects O2, O7, O22, O135, and O149 (object codes referenced in (Table S2) of 189 matched objects) represented negative outliers below the trend. Objects O3, O10, O28, O37, O39, and O63 constituted positive outliers above the trend. These outliers are listed in (Table S9). The identified outliers possessed AU values inconsistent with the expected trend for their respective time periods and corresponded to detections in the intermediate-regime zone intervals that were not included in the dense clusters demonstrated by DBSCAN analysis.

Table S9. List of Identified Statistical Outliers (2σ) in the Temporal Analysis

#	Object Code	Match Date	Object Name	AU Value (AU)	Study Group
1	O_2	1968-09-05	121725 Aphidas (1999 XX143)	11.7121	100d
2	O_3	1987-07-15	2060 Chiron (1977 UB)	13.0202	100d
3	O_7	1966-10-30	346889 Rhiphonos (2009 QV38)	11.7654	100d
4	O_7	1982-02-17	346889 Rhiphonos (2009 QV38)	12.0633	100d
5	O_{10}	1989-01-08	60558 Echeclus (2000 EC98)	12.7627	100d
6	O_{22}	1972-03-21	C/1976 U1 (Lovas)	11.7990	100d
7	O_{28}	1988-06-23	C/1983 R1 (Shoemaker)	12.9740	10d
8	O_{37}	1986-12-28	C/1990 K1 (Levy)	12.8794	100d
9	O_{39}	1998-05-03	C/1993 Q1 (Mueller)	13.5116	100d
10	O_{63}	1999-08-14	C/2002 O7 (LINEAR)	13.5803	10d
11	O_{135}	2025-09-15	C/2021 A1 (Leonard)	12.8479	100d
12	O_{149}	1982-07-01	Pioneer 11	12.1180	100d

Previous analyses demonstrated that AU values increased over time with a correlation of ρ = 0.48 and separated into two distinct regimes around 1985–1990. The mathematical predictability of the formula's detection pattern and associated AU characteristic trend was tested using third-degree polynomial regression analysis.

(Fig. S9) presents a comparison between the third-degree polynomial model (blue line) that best describes the AU = ƒ (Detection date) relationship and the 30-point rolling mean trend (red line) showing local fluctuations. The polynomial model represents the expected primary signal of the formula's resonance over the 65-year period, while the rolling mean reveals local fluctuations and secondary signals not captured by the model. For instance, the red line exhibits a higher peak than the blue line at approximately 2010, reaching approximately 13.6 AU, indicating that the formula focused on higher AU detections than expected during that brief period.

The polynomial model achieved an R^2 value of 0.746, indicating that 74.6% of AU value variance was correctly predicted based on the time factor. These results demonstrate that AU values do not vary randomly over time but possess a strong, predictable mathematical model.

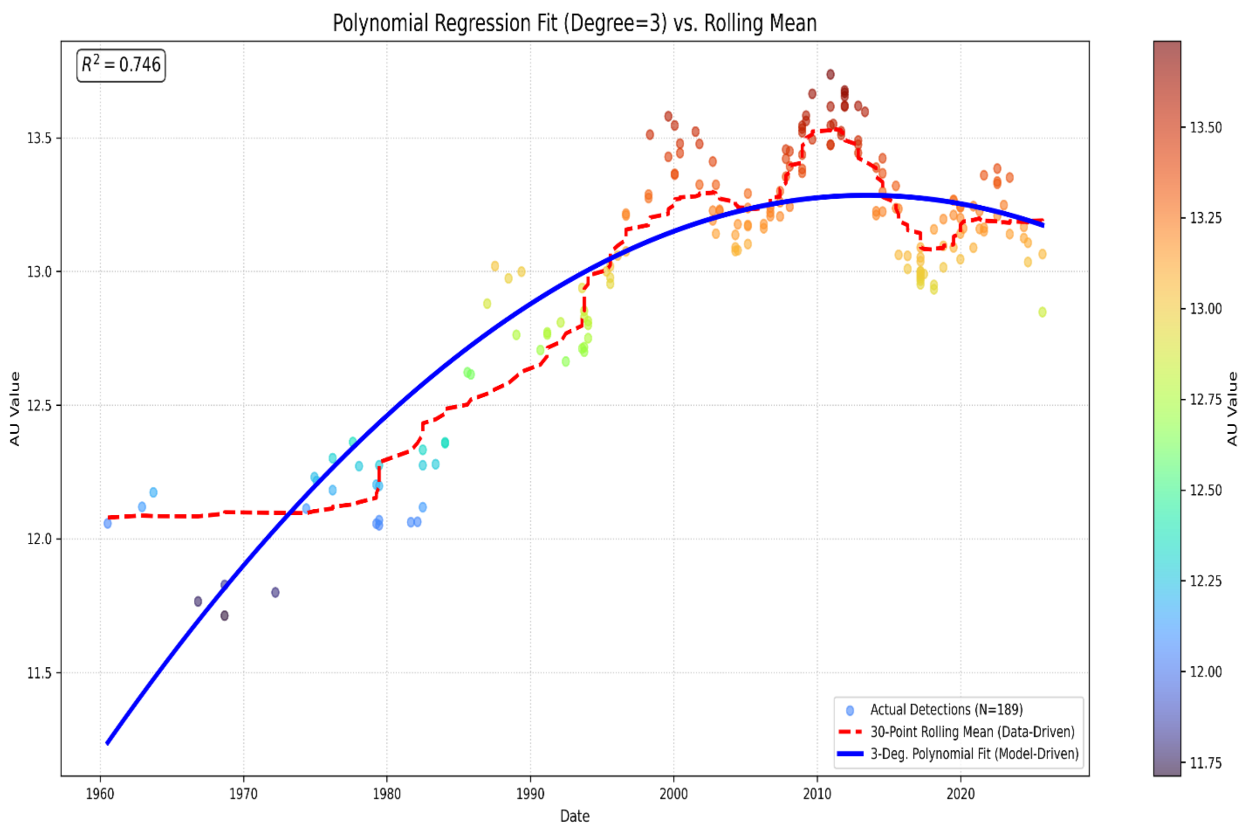

Fig.S9 Polynomial Regression Analysis (Degree=3) of Temporal AU Variations (R^2 = 0.746)

(Fig. S10) displays the distribution of prediction errors (residuals) from the polynomial regression model over time. The model errors were not randomly distributed around the Y = 0 line; instead, systematic, periodic fluctuations (autocorrelation) were observed. The error distribution consisted of four distinct regimes with approximate periods of 15–20 years.

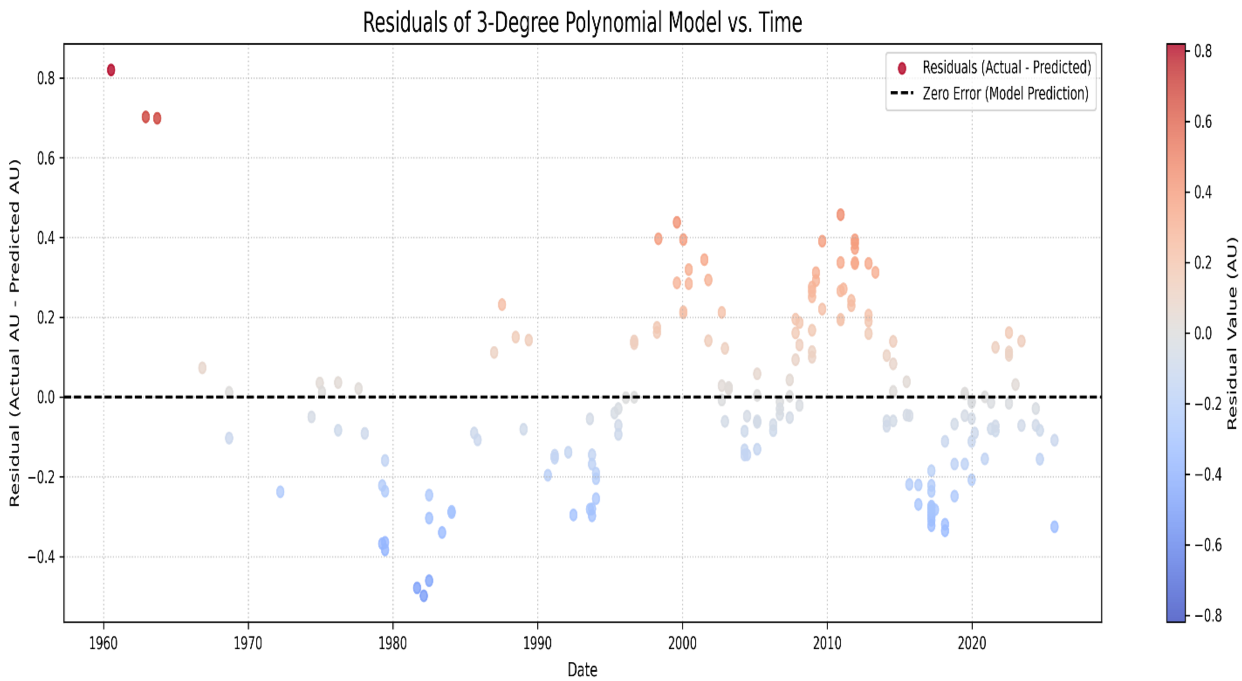

Fig.10 Temporal Distribution of Model Residuals (Actual vs. Predicted)

During the early period (1960–1975, Red Zone), the third-degree polynomial model systematically underestimated AU values, with errors positioned above the Y = 0 line. In the 1975–1995 interval (Blue Zone), the presence of errors below Y = 0 indicated that the model systematically overestimated actual AU values. During 1995–2015 (Red Zone), errors returned above the Y = 0 line. From 2015 to the present (Blue Zone), errors again descended below the Y = 0 line.

Although the polynomial regression model (R^2 = 0.746) successfully captured the main S-curve trend, the residual plot (Fig. S10) demonstrates that the model is not perfect and overlooks more complex structures within the data. This distinct wave pattern in the errors (positive-negative-positive-negative) may indicate the presence of a secondary periodic oscillation embedded within the data. These findings suggest that the formula may be sensitive not only to a global rule but also to this faster, periodic modulation.

To improve the formula's performance, several investigations are recommended: (1) increasing the dataset by incorporating additional daily data; (2) examining the effect on R^2 values and residuals of the polynomial mathematical regression model to be used for predicting AU-time relationships for future dates; and (3) conducting deeper investigations into the existence and origin of the secondary signal.

4.2. Hypothesis 2 (E=Earth) Results

Hypothesis 2 demonstrates that the target region defined in Hypothesis 1 is corroborated by the formula 6EQ=5UJ, applied on August 15, 1977, under the assumption that 'E' represents Earth. For this hypothesis study, first: the percentage deviation distribution data of detection matches from the formula's target value, generated from the massive database using the 6EQ=5UJ formula, applied on 1977-08-15 (Wow signal date), with suggesting E=Earth as a celestial body, is shown in (Fig. S11).

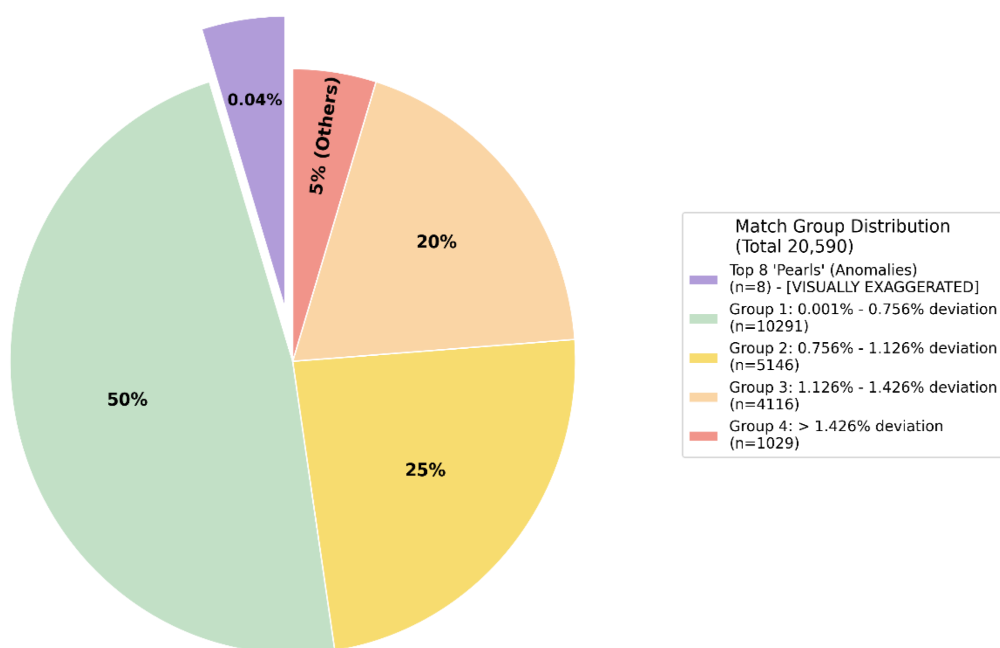

Fig.S11 Distribution of Match Candidates by Deviation Magnitude

20590 groups of matches are divided into 4 main groups and additionally the Top N=8 most isolated (matching groups with the lowest statistical deviation) based on their percentage deviation from the formula's target value. The Top 8 matches (with percentage deviation values between 0.000159% and 0.000778%) account for 0.04% of the total 20590 matches (the percentage deviations of the 8 matches consisting of 4-object matches and their corresponding AU values are shown in (Table S10)). For 20590 matches: Group 1 deviation range 0.001 – 0.756% (50% area of the graph, total 10291 matches), Group 2 0.756-1.126% (25% of the graph and total 5146 matches), Group 3 deviation range 1.126 % 1.426 % (20% of the graph, total 4116 matches), Group 4 matches with percentage deviation greater than 1.426% (5% of the graph, total 1029 matches) are shown .

Table S10. Top 8 High-Precision Candidate Matches (E=Earth)

#	Match Information	percentage deviation
1	364192 Qianruhu (2006 QR1) - 4716 Urey (1989 ULS) - 13441 Janmerlin (2098 P-L) **2.3406 AU** **3.5354 AU** **1.9420 AU**	**%** 0.000159
2	9588 Quesnay (1990 WE2) - 4257 Ubasti (1987 QA) - 17195 Jimrichardson (1999 XQ234) **2.2162 AU** **1.4160 AU** **2.8601 AU**	**%** 0.000170
3	52301 Qumran (1991 RQ2) - 4632 Udagawa (1987 YB) - 58424 Jamesdunlop (1996 DL1) **1.8336 AU** **1.9299 AU** **2.9149 AU**	**%** 0.000192
4	3513 Quqinyue (1965 UZ) - 55701 Ukalegon (1193 T-3) - 5255 Johnsophie (1988 KF) **2.6046 AU** **4.6201 AU** **2.7098 AU**	**%** 0.000194
5	26940 Quintero (1997 GC8) - 19462 Ulissedini (1998 HE20) - 84011 Jean-Claude (2002 OB25) **2.2448 AU** **3.2121 AU** **4.6905 AU**	**%** 0.000541
6	52301 Qumran (1991 RQ2) - 15025 Uwontario (1998 TX28) - 30030 Joycekang (2000 DY61) **1.8336 AU** **3.0757 AU** **2.0907 AU**	**%** 0.000612
7	2255 Qinghai (1977 VK1) - 233880 Urbanpriol (2008 WF93) - 101811 Jakobkaup (1999 JQ6) **2.6688 AU** **3.7098 AU** **1.7225 AU**	**%** 0.000709
8	17438 Quasimodo (1989 SQ4) - 16515 Usman'grad (1990 VN14) - 33889 Jengebo (2000 KZ22) **2.4062 AU** **3.5505 AU** **1.8784 AU**	**%** 0.000778
	Note: On August 15, 1977, Earth's AU value is 1.01279.	

The distribution of deviation values in the total 20590 formula matches was analyzed using the KDE Statistical Elbow test to determine if the data had a natural breaking point (elbow) (Fig. S12). For each match, the percentage deviations from the formula's target value were calculated for 20590 matches, and it was proven that the first 8 candidate matches shown in the previous Conceptual Pie Chart (Fig. S11) are statistically separated from 99.96% of the data by a natural breaking point. The matches closest to the formula's target percentage deviation value were identified.

The upper panel in (Fig. S12) shows that the overall distribution of all 20,590 data points has very little density near zero (perfect fit), and the main density (plateau) is between 0.1% and 1.4%. On the far left of the graph, the density is seen to drop rapidly in an area near zero, indicating that there is very little data that fits the formula almost perfectly. This is preliminary evidence that the candidate objects in the Top 8 matches are absolutely rare.

The lower panel focuses on the most sensitive region of the data (one in a thousand) and examines the first 48 candidates, consisting of 6 groups of 8-object matches. Group 1 Top 8 (Red Lines), on the left of the graph, is tightly clustered very close to zero, isolated from each other. A visibly large gap, a statistical 'elbow' point, exists between the last member of Group 1 and the first member of Group 2 (around 0.00016% and 0.00100% values). This elbow proves that the first 8 candidates in Group 1 are naturally separate and statistically different from all remaining 20,582 data points, forming an anomalous cluster.

A comprehensive statistical framework (Sections 2.1 and 2.2) was implemented to ascertain whether the observed celestial matches constitute a random coincidence or indicate a specific, hidden systematic pattern. In the statistical assessment of randomness, a strict significance threshold of α = 0.001 (0.1%) was used to maximize the reliability of the results. To test the validity of these matches against the null hypothesis (H_0), two distinct analytical approaches were employed: an atomic analysis of individual events (Method A) and a collective group analysis (Method B).

4.2.1. Statistical Analysis

4.2.1.1. Method A: Atomic Approach (8 / 20590)

Since there are N=20590 4-object matches, N=20590 was used. The success criterion is that the percentage deviation of the formula's matches is < 0.0012%. Exactly 8 matches met this criterion. In the analysis, a strict significance threshold of α = 0.001 (0.1%) was used to maximize the reliability of the results. The P-Value was calculated using the Null Hypothesis Ho with a pool of N=20590 celestial bodies by using Eq. (2). The calculated P-Value is 2.57 times smaller than the determined strict α value. The probability of this group of 8 matches coming together by chance is statistically negligible, and the randomness

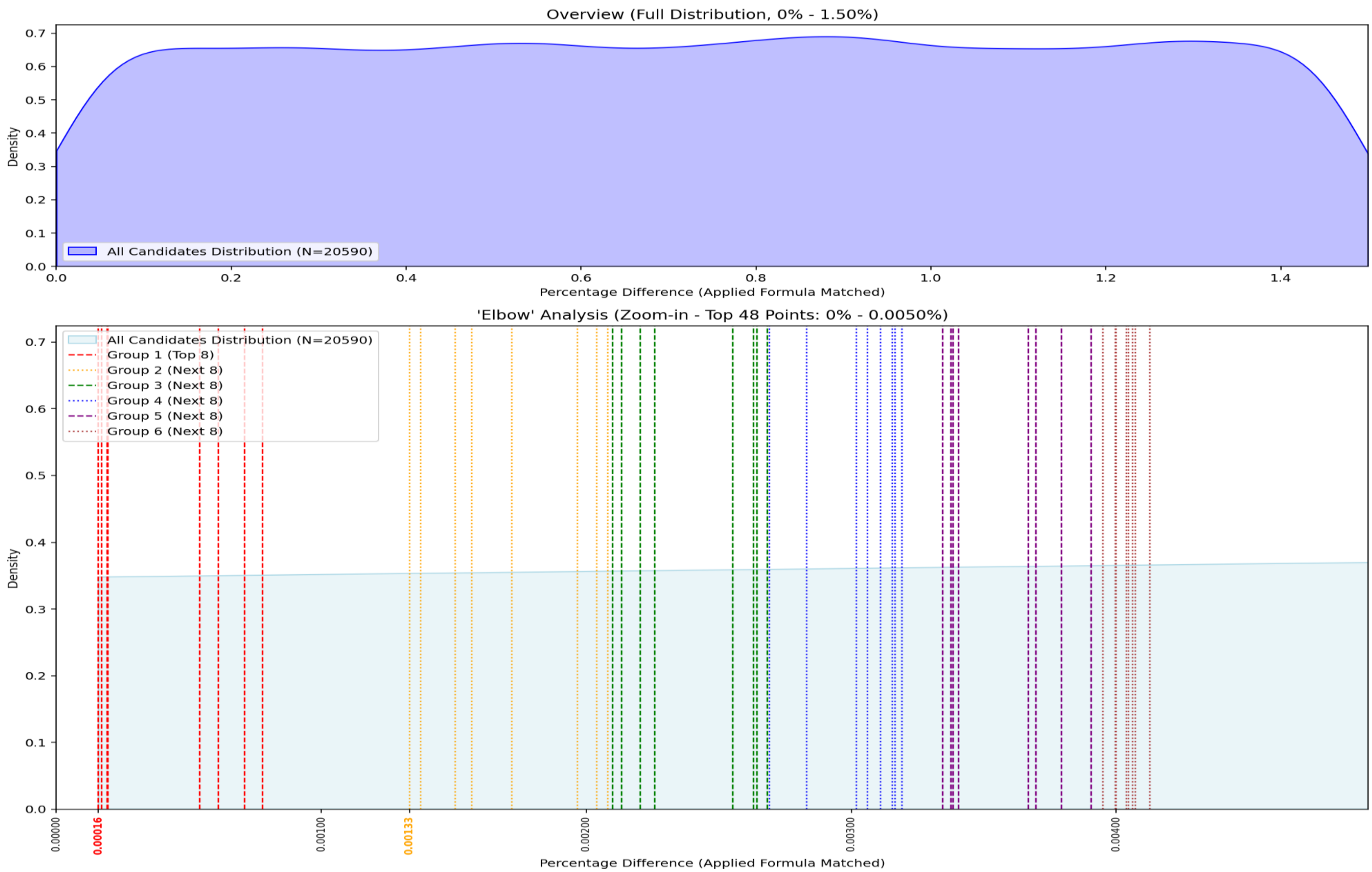

Fig. S12 Kernel Density Estimation (KDE) of Match Deviations: Global Distribution and High-Precision Zoom-in

hypothesis (H_0) has been rejected. The detected matches carry strong statistical evidence of being part of a meaningful and deliberate pattern. The Z-score was calculated using the Null Hypothesis Ho assumption (p_0 = 0.5) by using Eq. (3).

$$\hat{\boldsymbol{p}} = \frac{x}{N} = \frac{8}{20{,}590} \approx 0.0003885 \tag{2}$$

$$\boldsymbol{Z} = \frac{\boldsymbol{\hat{p}-p_0}}{\sqrt{\frac{\boldsymbol{p_0(1-p_0)}}{\boldsymbol{n}}}}\ \boldsymbol{Z} = \frac{\mathbf{-0.4996115}}{\mathbf{0.003484}} \approx \mathbf{-143.38} \qquad (p_0 = 0.5) \tag{3}$$

This score, which is far above the statistical critical thresholds in absolute value (|Z| > 3), proves that the 8 observed matches cannot be explained by random noise or chance factors. The negative value mathematically confirms that the frequency of events is much lower and more selective than the randomness assumption (p_0=0.5); that is, these matches are the product of a rare and specific filtering.

4.2.1.2. Method B: 'Group Approach' (1 / 2573.5)

In Method B, the entire group of 8 4-object matches with minimum deviation was considered as a large group, and assuming there is a total of 2573.5 such 8-object groups, randomness analyses were performed (Eq. (1)).

$$\boldsymbol{P}\textbf{-value} = \frac{1}{N} = \frac{1}{2573.5} \approx 0.0003885 < \alpha = 0.001 \tag{1}$$

In a universe consisting of 2573.5 groups of 8 matches, each consisting of 4 objects, the probability of this best group of 8 matches (each consisting of 4 objects) occurring by chance is statistically negligible, and the randomness hypothesis (H_0) has been rejected. The detected matches carry strong statistical evidence of being part of a meaningful and deliberate pattern. The Z-score was calculated using the Null Hypothesis H_0 assumption by using (Eq. (2)) (p_0 = 0.5).

$$\boldsymbol{Z} = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}\ Z = \frac{-0.499615}{0.009856} \approx -50.69 \qquad (p_0 = 0.5) \tag{2}$$

In the study, a Z-Test performed on N=2573.5 groups of 8 matches, each consisting of 4 objects, yielded a value of ≈ -50.69. This result shows that the formula successfully isolated a meaningful, rare, and deliberate signal in a universe consisting of 2573.5 groups of 8.

It is postulated in Hypothesis 2 that if the 'Wow!' signal contains specific spatial encryption, the application of the formula with the only input E=Earth yields a group of top 8 matches that delineate a hidden Interplanetary Transport Network (ITN) route.

Following the randomness analysis, a log-log deviation histogram analysis was conducted to demonstrate that Hypothesis 2 specifically points to a distinct region.

The logarithmic distribution of percentage deviations for the entire dataset of 20,590 matches is presented in the Log-Log Deviation Histogram (Fig. S13). Log-Log scaling was used to zoom in on the almost perfect fit to the formula (region near zero) and to represent the enormous data volume differences on the Y-axis in an understandable way on a single graph.

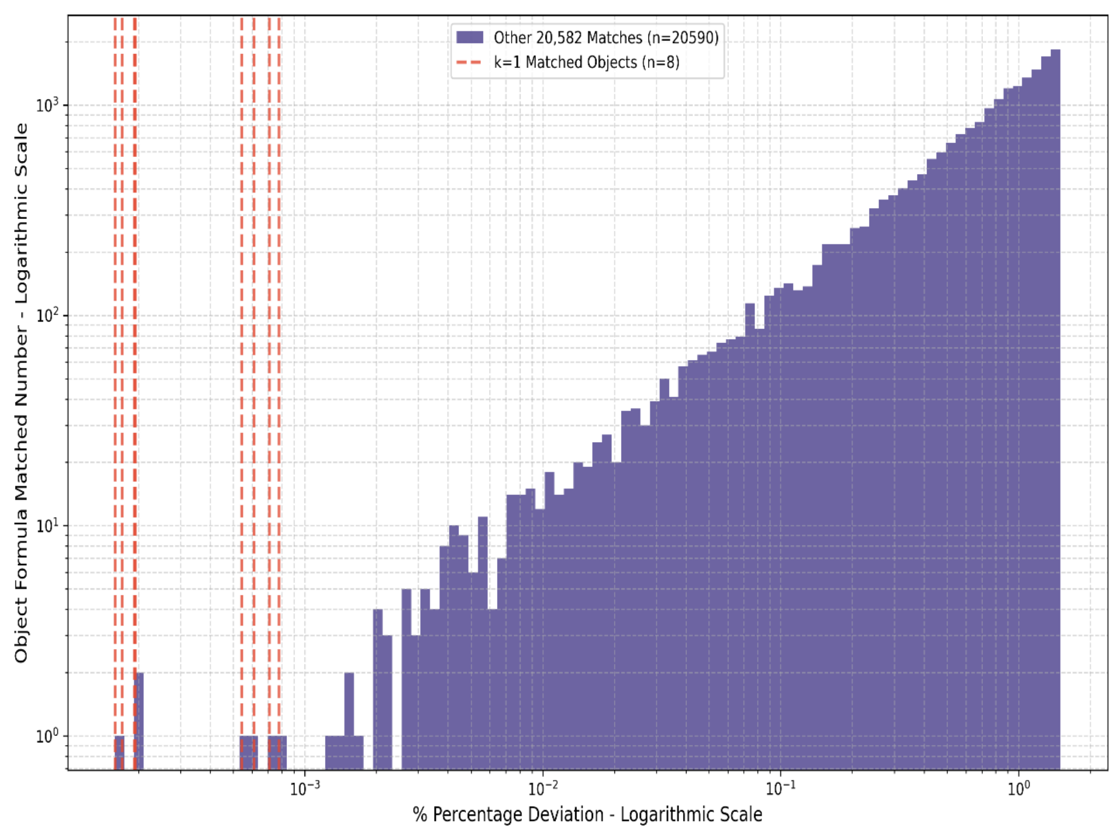

Fig.S13 Log-Log Frequency Distribution of Percentage Deviations Across the Full Dataset

The Log-Log Deviation Histogram (Fig. S13) is the most comprehensive frequency analysis performed on N=20590 formula matches. (Fig. S13) shows that the dataset does not have a continuous distribution, but rather a statistical chasm between two distinct and separate populations. All the Top groups of 8 matches, each consisting of 4 objects (k=1 Matched Objects) (Red Dashed Lines) are clustered to the left of the X-axis at 10^{-3} (0.001%) value, meaning in the region closest to perfect. There is almost no data in the blue histogram in this region. This is the anomalous cluster where the groups of 8 matches, each consisting of 4 objects, shows the best fit to the formula. The second distinct group (Blue Histogram), which constitutes 99.96% of the histogram graph, starts to the right of the 10^{-3} value and extends up to 100 (1%).

There is a huge gap in the X-axis between the Red Lines and the main body of the Blue Histogram. This statistical gap is the elbow point where the Top groups of 8 matches, each consisting of 4 objects candidates, are separated by a natural statistical barrier from the remaining 20,582 matches. This result directly shows the statistical truth behind our Elbow Analysis (Fig. S12) in selecting the Top 8. Selecting more than Top 8 (e.g., 9, 10, or 20 candidates) would scatter this unique anomalous cluster, shift the focus of the analysis and scientifically represent a transition to noise. The Top 8 cluster is a unique anomalous cluster in terms of its percentage fit to the formula. The formula is shown to detect a physical principle and to be able to extract only 8 extremely rare and unique objects that enter maximum resonance.

(Fig. S14) shows the instantaneous distances in AU from the Sun for the Top 8 matches that best fit the formula with the lowest deviation, from Match 1 to Match 8, on the Wow date. The fixed Earth Reference is always at 1.0 AU. This indicates that the analysis is based on heliocentric (Sun-centered) distances and that Earth plays the role of the reference frame.

(Fig. S14), (Fig.S11), (Fig. S12), and (Fig. S13), and the Top 8 objects identified in (Table S10), represent a spatial fingerprint showing where they were physically located in the Solar System at the time of matching.

Each vertical profile: none of the vertical profiles from Match 1 to Match 8 are identical. The distances of the Q, U, and J objects change dramatically from match to match. The majority of matches: Q, U, J objects are concentrated further than 1.5 AU. Each Match ID represents the unique alignment created by the Q, U, J objects together with Earth at that single critical moment in the Solar System to capture the maximum resonance of the formula.

(Fig. S15) compares the density distributions (KDE) of Group A, consisting of the 24 unique objects that form the k=1 Top 8 best matches, with Group B, consisting of the other 20582 4-object match groups, on the same axis. The orbital parameters of Group A and Group B,

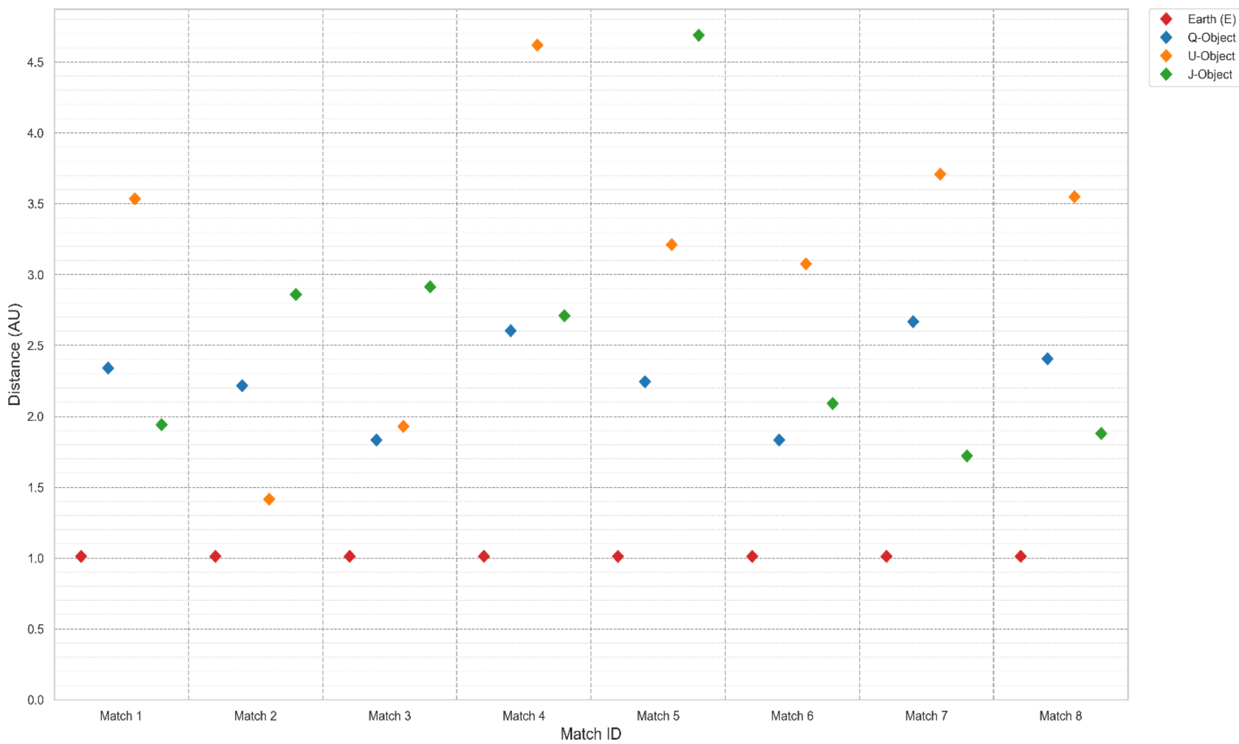

Fig. S14 Heliocentric Distance Composition of the Top 8 Candidate Matches

drawn from NASA JPL SBDB: a (Semi-major axis), e (Eccentricity Distribution), i (Inclination ° Distribution), Tj (Tisserand Distribution), and physical AU density, were compared. Filtering was applied to exclude extreme outliers in the datasets and to focus on the main belt (AU ≤ 10, Inclination ≤ 75 °).

(Fig. S15) shows that the unique objects in the Top 8 matches (Group A) are fundamentally different from the unique objects in the remaining 20,582 matches (Group B) in terms of orbital parameters.

The semi-major axis (a) parameter values of Group A are densely clustered between ≈ 2.0 – 3.3 AU, indicating that the orbital size and energy of Group A are clustered within a limited range. The semi-major axis (a) parameter values of Group B are much more widely distributed, with a broader distribution observed between 1.0 – 5.5 AU. The analysis observed that the formula favors a specific orbital size range. The comparison analysis of e (Eccentricity) (Eccentricity) shows an extraordinarily tight clustering of Group A's e parameter values between 0.0 and 0.02 (the body of the violin plot is in this region). Group B's e parameter values showed a broad and scattered distribution between 0.2 and 0.6. The analysis concluded that the unique objects in the Top 8 matches (Group A) have much more circular orbits.

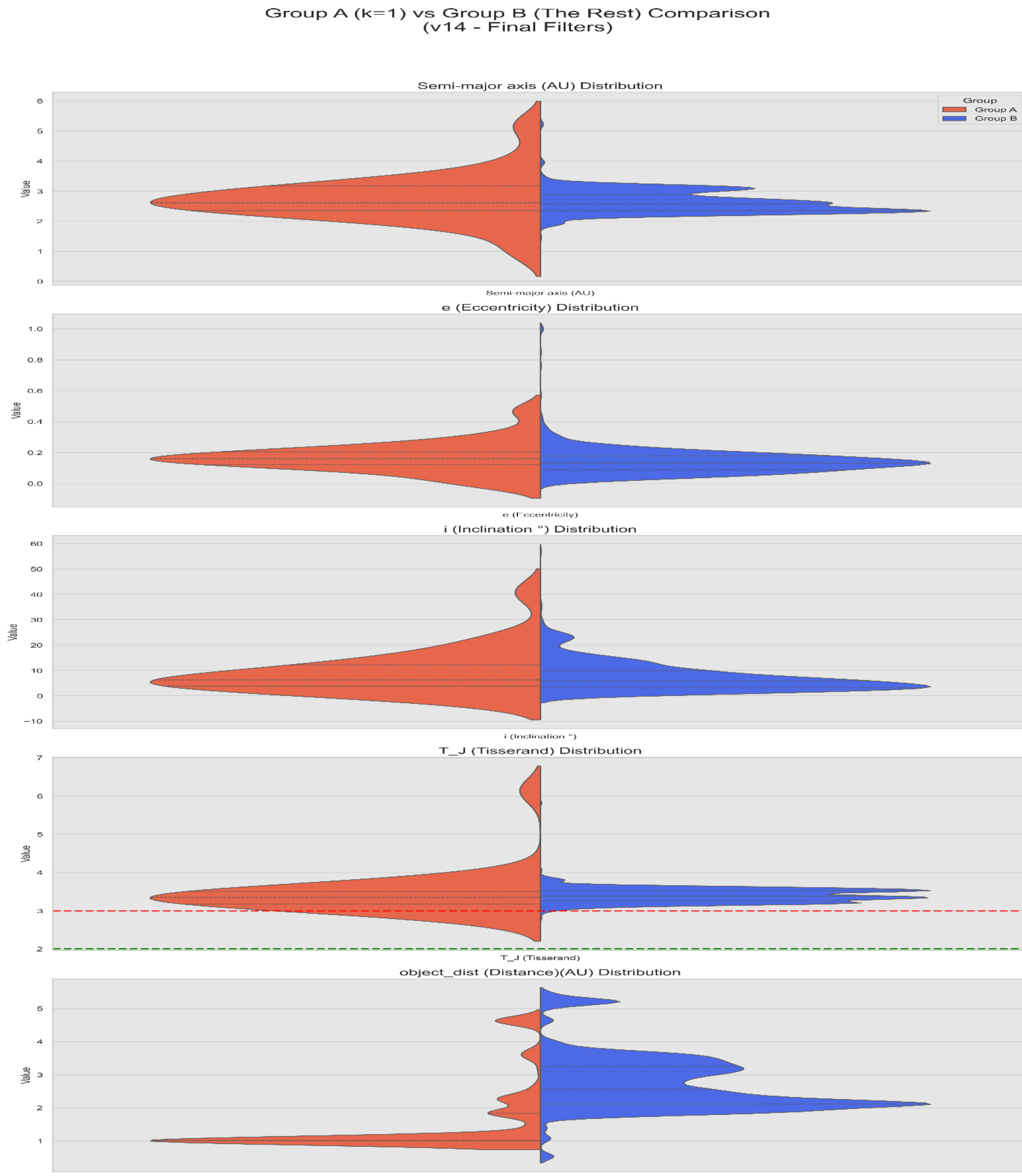

Fig. S15 Comparative Distribution of Orbital and Physical Parameters: Top Candidates (Group A) vs. General Population (Group B)

When i (Inclination °) parameters were compared, Group A's i (Inclination °) parameters were mostly concentrated between 0 ° and 20 °. Group B's i (Inclination °) parameters also started in a similar range but scattered more rapidly. This indicates that the unique objects in Group A are closer to the Ecliptic plane.

The Tj (Tisserand) parameters of Group A and Group B were compared. Group A's Tj parameter values were tightly constrained in a narrow range between 2.8 and 3.7. Group B's Tj parameter values showed a wide distribution between 2.0 and 4.0. The Tisserand parameter analysis showed that Group A objects are concentrated in a dynamically related region with Jupiter (around Tj ≈ 3.0). This result proves that the unique objects in Group A are dynamically related to Jupiter Trojans and Hilda groups, not Inner Main Belt Asteroids.

Comparing the AU values of Group A and Group B, Group A was found to be between ≈ 0.5-5 AU, and Group B had similar distributions between ≈ 0-5.5 AU. Group A has a small peak concentrated around 1.0 AU, while Group B has a more prominent broad peak around 2.0 AU.

(Fig. S15) Comparitive Distribution of Orbital and Physical Parameters analysis proves that the Top 8 candidates selected by Elbow Analysis are not a noise signal randomly indicated by the formula but rather possess a consistent and unique dynamic identity.

The members of Group A are in narrower ranges in terms of orbital plane i and size a compared to Group B, but most importantly, their Eccentricities e are extremely low and Tisserand Parameters (Tj) are extraordinarily clustered around ≈ 3.0. This finding supports the thesis that the formula is sensitive to an astrodynamics law with strong quantitative physical evidence.

The difference in instantaneous distances from the Sun is not very noticeable when regional approximation is not performed, showing a similar distribution. In the previously given sections of the article, the unique objects in the Top 8 matches were identified through regional approximation and comparison analyses of AU% deviations (Fig. S12) and Fig. S13).

Hypothesis 2 postulates that these unique celestial bodies within the Top 8 matches, belonging to a specific object class, correspond to solutions of the 'navigation equation' required for Earth-based entry into the Interplanetary Transport Network (ITN). Consequently, these objects are identified as candidate waypoints along rare and highly efficient trajectories, as further detailed in the subsequent section.

The unique objects in the Top 8 matches, located around the entry ramps of the Interplanetary Transport Network (ITN) (Tj ≈ 3.0), were interpreted as safe 'Tolls' with high orbital stability. It has been proven that the unique objects in the Top 8 matches belong to a statistically different population in terms of orbital dynamics.

To determine the most efficient routes, the solution is derived by comparing the orbital parameters: a (Semi-major axis), e (Eccentricity Distribution), i (Inclination ° Distribution), Tj (Tisserand Distribution) of objects in this 'spatial alignment' configuration that achieved maximum resonance. The formula's solution is a combination of orbital characteristics and spatial alignment.

According to the Interplanetary Transport Network (ITN) hypothesis, the unique celestial bodies identified within the Top 8 matches serve as potential candidates—acting as insertion points, entry ramps, and terminal destinations—for accessing the 'space highway' from Earth. The determination of intermediate and primary routes among these objects was achieved by calculating the orbital physics parameters, specifically regarding geometry and energy, for each respective body.

4.3. ITN Hypothesis Route Target Results

For the Wow date, 1977-08-15, the Tj (Tisserand) parameters are shown for each of the unique objects in our Top 8 Matches (Table S11).

According to the Interplanetary Transport Network (ITN), objects with $Tj > 3$ cannot be controlled directly and strongly by Jupiter. The $Tj \approx 3$ region is the boundary region of Jupiter's sphere of influence, and objects from this region can be captured by the ITN's gravitational river with very little energy, so they will be chosen as entry ramps or intermediate gates on the route. Objects in the $Tj < 3$ region are in strong interaction with Jupiter, and their orbits are controlled by Jupiter and are in a locked resonance with Jupiter (like Trojan Asteroids). These are stable equilibrium points (L_4/L_5 Lagrange points), the main intersections, the main gates themselves.

The hypothesis posits that the distinct objects in the Top 8 matches function analogously to 'toll booths' and stops on a 'superhighway' (ITN) that permits low-energy transfer. The formula serves to solve the 'navigation equation' necessary for Earth-departure onto this path. If the 'Wow!' signal is interpreted as an encrypted transmission delineating a specific route and destination, then stable stop points along this highway must be established using the configuration specified by the message. With this hypothesis, among the candidate stop objects indicated by the formula, stop candidates will be selected from Asteroids with billions of years of stable orbits, instead of centaurs with chaotic orbits, which are unsuitable for a navigation network that needs to remain stable for millions of years.

Table S11. Tisserand Parameters (T_J) of the Unique Constituent Objects within the Top 8 Matches

#	Object Name	Tj Value
1	399 Earth	6.138
2	55701 Ukalegon	**2.850**
3	84011 Jean-Claude	**2.995**
4	2255 Qinghai	3.158
5	4716 Urey	3.161
6	17195 Jimrichardson	3.169
7	233880 Urbanpriol	3.169
8	15025 Uwontario	3.173
9	16515 Usman’grad	3.179
10	19462 Ulissedini	3.275
11	13441 Janmerlin	3.320
12	5255 Johnsophie	3.351
13	9588 Quesnay	3.352
14	26940 Quientero	3.361
15	101811 Jakobkaup	3.383
16	3513 Quqinyue	3.399
17	30030 Joycekong	3.415
18	58424 Jamesdunlop	3.428
19	364192 Qianruhu	3.501
20	52301 Qumran	3.541
21	17438 Quasimodo	3.563
22	33889 Jengebo	3.611
23	4632 Udagawa	3.633
24	4257 Ubasti	3.913

For stable highway route gate selection, the main gate was chosen from objects in the Tj < 3 region and from stable equilibrium points (L_4/L_5 Lagrange points), and the intermediate gate was chosen from objects near the Tj ≈ 3 region, as objects in this region can be captured by the ITN's gravitational river with very little energy. Information about the selected intermediate gate and main gate targets has been provided, and their proximity to Lagrange points has been evaluated (Table S12).

Only 55701 Ukalegon and 84011 Jean-Claude objects among the 24 unique objects in the Top 8 matches given by the formula have Tj parameters (Tj) less than 3. Ukalegon's orbit being very close to Jupiter's average orbital distance (Jupiter ≈ 5.2 AU), its Tisserand Parameter (Tj): 2.850, and its orbit being locked in a 1:1 resonance with Jupiter, being at the L_5 Lagrange point, being a stable equilibrium point, and meeting the specified condition;

Table S12. Selected Main and Secondary Gateways

#	Gateway Type	Object Name	Characteristics
1	**Main**	55701 Ukalegon	Semi-major axis (a): 5.16 AU, Tisserand Parameter (Tj): 2.850, Orbit Type: Jupiter Trojan (1:1 Resonance), Librating around the L5 Lagrange Point, Asteroid
2	**Secondary**	84011 Jean-Claude	Semi-major axis (a): 4.01 AU, Tisserand Parameter (Tj): 2.995, Orbit Type: Outer Main Belt / Hilda Group, 3:2 Resonance, Borderline Tj ≈ 3, Asteroid

being an Asteroid, it was chosen as the Main Gate. 84011 Jean-Claude, with a Tisserand Parameter (Tj): 2.995 and being close to the Tj ≈ 3 region, and being a stable orbit asteroid, was chosen as the Intermediate Gate. The selected route: the main destination from Earth is 55701 Ukalegon and the intermediate stop is 84011 Jean-Claude. The formula employed in this study, which is expounded upon in the following section, determines the most energetically efficient travel.

Cost Formula for Finding the Most Energetically Efficient Route

The purpose of the cost formula is to find the physically most efficient (lowest cost) route from Earth ('399') to target asteroids, especially 55701 Ukalegon and 84011 Jean-Claude. The formula treats all objects as stops and calculates the cost (difficulty) of travel between two stops. The cost formula used is fundamentally inspired by classical formulas (especially Southworth-Hawkins D-Criterion and Drummond Similarity Measures) for measuring dynamic similarity between small Solar System objects (asteroids, comets, centaurs, etc.).

It is hypothesized that the formation of a sequential route chain (A → B → C → D) relies on selecting the connection with the minimum energetic cost at each discrete step. The total energetic travel cost to the identified primary targets is therefore determined by the cumulative summation of these optimized segment costs, as expressed in Eq. (9)

If the orbital elements (semi-major axis, eccentricity, etc.) of two objects are very similar, these objects are assumed to have originated from a common source or to be dynamically similar in orbital space. Orbital shape (Semi-major axis (a), e (Eccentricity), i (Inclination °) (Inclination) and energy (Tj (Tisserand) (Tisserand Parameter)) assumes that traveling between very similar objects requires less energy or Δv (required theoretical velocity change). The energy cost required to transfer between two orbits is assumed to be directly proportional to the Δv (required theoretical velocity change) cost (Travel Energy Cost ≈ Δv

Requirement). The hypothesis operates on the principle of Low Cost = Low Δv = More Efficient Route. Consequently, this approach evaluates not merely the geometric proximity of celestial bodies but also their orbital dynamics. It selects the chain of orbits that are most like each other. This route offers a theoretical suggestion of a series of 'minimum energy orbital jumps.

The formula used to calculate the energetic travel cost between two objects is given Eq. (10). |ΔP| expresses how close the orbital elements (Eq. (11)) of two objects are and is expressed by the absolute difference of Semi-major axis (a), e (Eccentricity), i (Inclination °), and Tj (Tisserand) (Tisserand Parameter) values. The smaller the difference in orbital elements ΔPj of two objects, the lower the dynamic impulse or Δv required to transfer from one orbit to another, and it is directly related to the efficiency of the route. w; are the semi-major axis, eccentricity, inclination, and Tisserand coefficients of the selected object. Since the numerical ranges of the parameters used in orbital mechanics vary greatly, statistical normalization was applied in the formula to eliminate scale differences and calibrate the coefficients. To ensure that all term coefficients contribute equally to the cost calculation, each difference term was scaled according to its typical maximum change range (or standard deviation) (Eq. (12)). After normalization, each term was weighted according to its physical significance (or its effect on the dynamic system). The cost of traveling from one object to another: is the weighted sum of the absolute differences of the P_i parameters and their respective coefficients (Eq. (13)). A large difference in orbital plane (ecliptic) between two objects requires the most expensive maneuver, a plane change maneuver (Δv), so the orbital inclination (i) coefficient was chosen as high as 10.0. Since the change in different types of orbits during travel requires significant Δv, the Eccentricity (e) coefficient was chosen as 5.0. The Tisserand parameter (Tj) coefficient was chosen as 2.0 because it has a moderate effect on cost. The semi-major axis (a) coefficient, which determines the size of the orbit, was chosen as 1.0 because it has the least effect on the travel cost difference (Eq. (14)).

$$\textbf{Total Cost =}\ \mathrm{Cost}\ (\mathrm{A},\mathrm{B}) + \mathrm{Cost}\ (\mathrm{B},\mathrm{C}) + \mathrm{Cost}\ (\mathrm{C},\mathrm{D}) \tag{9}$$

$$\left(\sum_{\boldsymbol{j}} \boldsymbol{w_j} \cdot \left|\boldsymbol{\Delta P_j}\right|\right) \tag{10}$$

$$\boldsymbol{P_i} = \{\ a_i\ ,\ e_i,\ i_i,\ T_{ji}\ \} \tag{11}$$

$$\text{Normalized Difference} = \frac{|\boldsymbol{\Delta P_j}|}{\text{Typical Range}} \tag{12}$$

$$\mathbf{Cost}\ (\mathbf{P1}, \mathbf{P2}) \;=\; w_a \,.\, |\ a_1 - a_2\ | + w_e \,.\, |\ e_1 - e_2\ | + w_i \,.\, |\ i_1 - i_2\ | + w_{Tj} \,.\, |\ Tj_1 - Tj_2\ | \tag{13}$$

$$\mathbf{Cost}\ (\mathbf{P1}, \mathbf{P2}) \;=\; 1 \,.\, |\ a_1 - a_2\ | + 5 \,.\, |\ e_1 - e_2\ | + 10 \,.\, |\ i_1 - i_2\ | + 2 \,.\, |\ Tj_1 - Tj_2\ | \tag{14}$$

The travel costs were calculated between all objects in the Top 8 matches. The detailed departure network matrix showing the given routes and potential transfer costs between routes was stored as data. A weighted graph was created for unique objects and their potential transfers.

From the set of 23 unique objects identified within the Top 8 matches, the five most energetically efficient trajectories—originating from Earth (399), transiting through the intermediate target 84011 Jean-Claude, and terminating at the primary target 55701 Ukalegon—were calculated utilizing Dijkstra's algorithm in conjunction with the specific cost function defined in Eq. (14).

The Dijkstra algorithm calculates the lowest total cost from '399' Earth to all other objects. For each object, it calculates the lowest energetically efficient travel cost required to reach the previous object, thus forming the path itself. In the Dijkstra algorithm, pruning technique was used for the top 5 alternative routes, and for a main target, the Top 1 route. After finding the Top 2 route by removing the first stop from the graph and re-running the algorithm. The Top 5 routes; this cycle was calculated 5 times.

Selection of the database matches was conducted by clustering the percentage deviations into octets, based on the intrinsic decimal value distribution observed on the 'Wow!' date. Consistent with the trends presented in (Fig. S11), (Fig. S12), (Fig. S13), and (Fig. S15), the distinct objects identified within the primary Top 8 cluster are posited to constitute the most efficient navigational route.

Network Resiliency Stress Test was analyzed to see how the Top 5 route costs from Earth (399) to 84011 Jean-Claude (intermediate gate) and 55701 Ukalegon (main gate) changed when 6 additional groups of 8 matches (k=7, total 56 matches from 4-object groups) were added to the Top 8 (k=1) (Fig. S16). The most efficient travel costs calculated for Earth - 84011 Jean-Claude and Earth - 55701 Ukalegon, between k=1-7, and for the Top 5 Route are shown in (Table S13).

(Fig. S16) consists of two panels comparing two main gateways: left Panel: Primary Gateway (55701 Ukalegon), and right Panel: Secondary Gateway (84011 Jean-Claude). The PIP graphs show the values of the k=1 Top 5 route costs for 55701 Ukalegon and 84011 Jean-Claude.

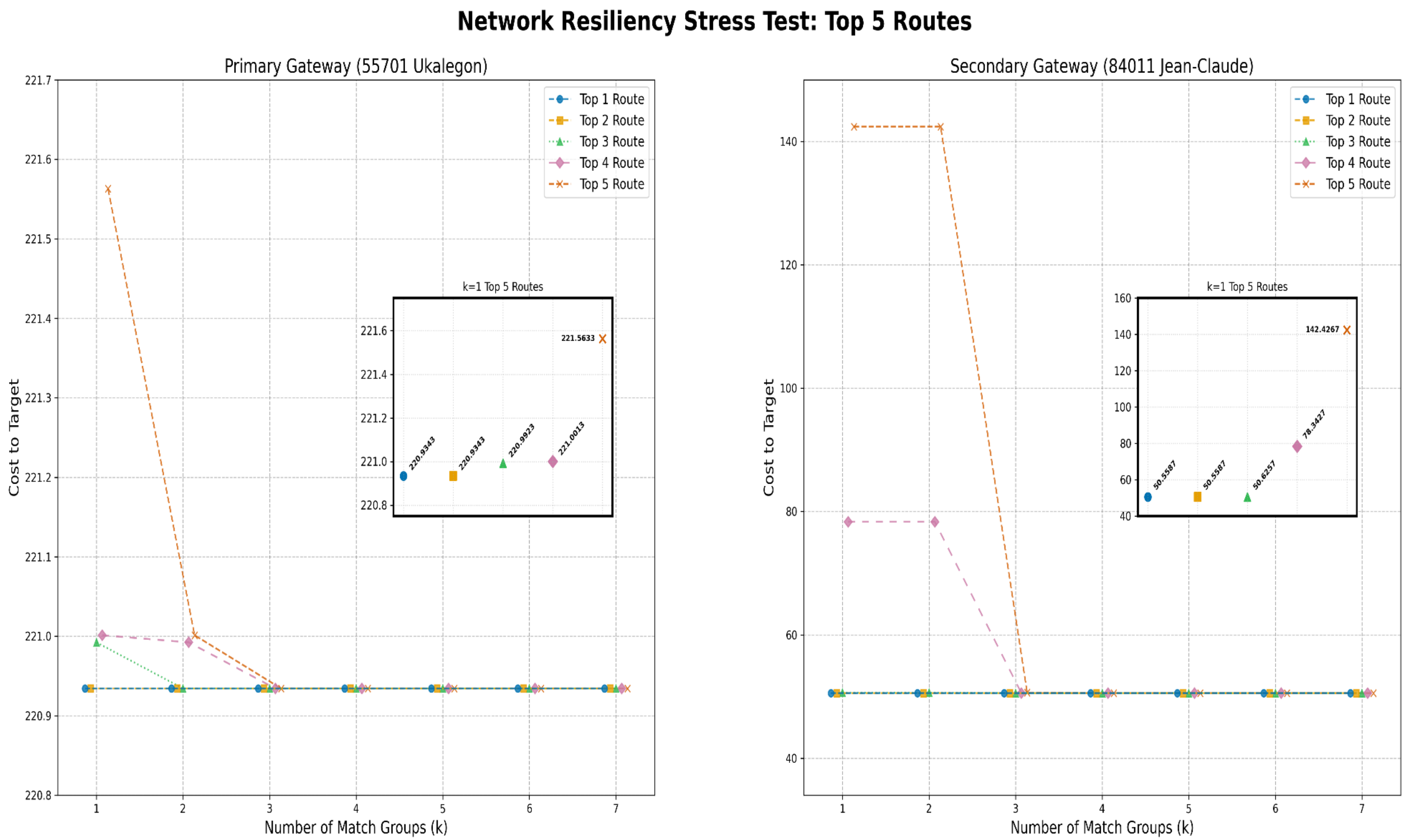

Fig. S16 k=7 Route Stability Stress Test: Evolution of Path Costs under Increasing Group Constraints (k)

In (Fig. S16), the x-axis denotes the stress factor (Number of Match Groups (k)), ranging from highest stress (k=1, least matches), to lowest stress (k=7, most matches). The y-axis plots the 'Cost to Target' required for network success. The curves represent the 'Top 5 Routes,' identifying the best five alternative paths the network can utilize.

(Fig. S16) analysis shows that when k=1 (only one matching group), the network structure is fragile, and cost differences are observed among the Top 5 best routes. For 55701 Ukalegon on the left panel, the Top 1 route has a cost of ~220.93, while the Top 5 route has a cost of ~221.56. The situation is much more dramatic for 84011 Jean-Claude on the right panel. While the best route ('Top 1') costs ~50.55, the 'Top 5 Route' (orange 'x') has a very high cost of ~142.42. These results indicate that, in the k=1 situation, the network can be forced to deviate to very inefficient alternative paths and that it possesses them.

The Network Resiliency Stress test analysis indicates a positive correlation between the number of matching groups (k) and network resilience; as ‘k’ increases (i.e., as more matching groups are added to the network), the network improves and gains resilience. The travel cost network structure of 55701 Ukalegon was observed to converge to the lowest cost path (220.9343) for all routes, including 'Top 5', when k reaches 3. The travel cost network structure of 84011 Jean-Claude was observed to dramatically decrease and converge to the lowest cost path (50.5587) even for 'Top 5' (from 142) and 'Top 4' (from 78) routes when k reaches 4.

The Stress Test results (Fig. S16) demonstrate that beyond the k=4 threshold, the efficiency gains for the Top 5 routes regarding 55701 Ukalegon and 84011 Jean-Claude approach an asymptote, indicating diminishing marginal utility. While a significant optimization of 60% was recorded between k=1 and k=4, the subsequent gain between k=4 and k=7 proved negligible (< 0.01%). This convergence implies that the network rapidly encounters its physical energy limits as defined by orbital mechanics, rendering further data addition ineffective for significant cost reduction. Even in a hypothetical scenario extending to k=1,000 (encompassing a total of 2,573.5 objects in match groups), the impact of any new intermediate transfer object would be restricted to decimal-level precision, thereby leaving the primary strategic findings unchanged.

To test the k=4 travel cost saturation and the resilience of optimum routes calculated in (Fig. S16), and to find the most efficient routes for travel from Earth (399) to the main target 55701 Ukalegon and intermediate target 84011 Jean-Claude, the network resilience was tested by limiting it to k=5, k=7, 56 matching groups, ~224 objects, considering analysis space efficiency and statistical significance .

Table S13. Evolution of Top 5 Route Costs for Primary and Secondary Gateways under Increasing Group Constraints (k=1-7)

1	**399 Earth - 84011 Jean-Claude**					
	#	**Top 1 Route**	**Top 2 Route**	**Top 3 Route**	**Top 4 Route**	**Top 5 Route**
	k = 1	50.5587	50.5587	50.6257	78.3427	142.4267
	k = 2	50.5587	50.5587	50.6257	78.3427	142.4267
	k = 3	50.5587	50.5587	50.5587	50.5587	50.6257
	k = 4	50.5587	50.5587	50.5587	50.5587	50.5587
	k = 5	50.5587	50.5587	50.5587	50.5587	50.5587
	k = 6	50.5587	50.5587	50.5587	50.5587	50.5587
	k = 7	50.5587	50.5587	50.5587	50.5587	50.5587
2	**399 Earth - 55701 Ukalegon**					
	#	**Top 1 Route**	**Top 2 Route**	**Top 3 Route**	**Top 4 Route**	**Top 5 Route**
	k = 1	220.9343	220.9343	220.9923	221.0013	221.5633
	k = 2	220.9343	220.9343	220.9343	220.9923	221.0013
	k = 3	220.9343	220.9343	220.9343	220.9343	220.9343
	k = 4	220.9343	220.9343	220.9343	220.9343	220.9343
	k = 5	220.9343	220.9343	220.9343	220.9343	220.9343
	k = 6	229.9343	229.9343	220.9343	220.9343	220.9343
	k = 7	220.9343	220.9343	220.9343	220.9343	220.9343

The analysis results indicate a two-stage saturation process. The network resilience test performed with k=7 confirmed that the cost optimization shown in (Table S13) and (Fig. S16) reached asymptotic stability at the k=4 level. This finding proves that the network reached its physical energy limits (orbital mechanics constraints) at an early stage. The minimum energy cost for a transfer from Earth to the Jupiter system (or asteroid belt) (Hohmann Transfer Limit) is a physical constant. Therefore, even if the analysis space were expanded to include all 2573.5 matching groups, the cost is not expected to significantly fall below this base value, as physical laws would not change. Study reached the global minimum cost (lowest point) at the k=4 level, and the subsequent process remained at the microscopic optimization level.

Although the cost stabilized at k=4, it was observed that adding k=5 to the analysis space offered 'comfortable' alternative paths with lower inclination differences in terms of orbital geometry (orbital geometry).

Despite the cost stabilizing at k=4, with the addition of k=5 to the analysis space, the selection of nodes forming the route reached its final and stable form at the k=5 level. Although the k=4 dataset optimized the cost, adding the k=5 dataset included 15025 Uwontario (34) (the number in parentheses, 34, indicates that it was the 34th matched object encountered for the first time, and that it is an object in the k=5, 8-object group (within the first 40 matches)) into the route. This object modernized the existing route thanks to its superior orbital geometry (low Δi and Δe difference).

When the analysis space was extended to k=6 and k=7 levels, no change in route topology was observed, proving that the solution reached saturation at the k=5 level and that adding more data sets would not change the main route topology. The algorithm evaluated dozens of new options in the k=6 and k=7 datasets but preferred to maintain the 'Qianxuesen-Uwontario' bridge at k=5. This demonstrates experimental saturation supported by the power of 'Negative Evidence'. The analysis up to k=7 confirms that the solution found at k=5 is not a local minimum, but rather that the system has reached topological saturation.

Despite a total of k=2573.5 matching groups in the database, this study was limited to k=7 (56 groups, ~224 objects) considering computational efficiency and statistical significance. The 224 objects examined adequately represent the orbital parameter distribution (Gaussian distribution) of main belt asteroids statistically. In a possible scenario, even if a new intermediate transfer object were found (e.g., k=1000) in terms of orbital geometry, its effect on the total cost would be at the decimal precision level. According to the principle of efficient use of scientific resources, it is inefficient to perform a search with $O(N^2)$ complexity for a process where the marginal gain approaches zero. The current study exhaustively (brute-force) scanned the first 56 groups with a deterministic approach. In future studies, instead of the Dijkstra algorithm, which has high computational cost, a model trained with the data obtained from the current k=7 analysis using Machine Learning can be used to identify only 'strategically important' objects from the remaining 2500+ groups. This approach could reduce the processing load from $O(N^2)$ to a linear level, allowing the entire database to be scanned.

(Fig. S17) The network graph, using the basic asteroids selected from the k=7 group candidates, shows the lowest cost routes starting from Earth and reaching the main ITN targets 55701 Ukalegon and 84011 Jean-Claude. This cost formula, and the outputs of the Dijkstra Algorithm, are the final evidence visualized. The network map proves that the solution found by the formula is a functional route infrastructure consistent with the principles of the Interplanetary Transport Network.

The most efficient routes were found using the Dijkstra algorithm and are shown in (Fig. S17) The graph shows that the formula creates a reliable and multi-path infrastructure that

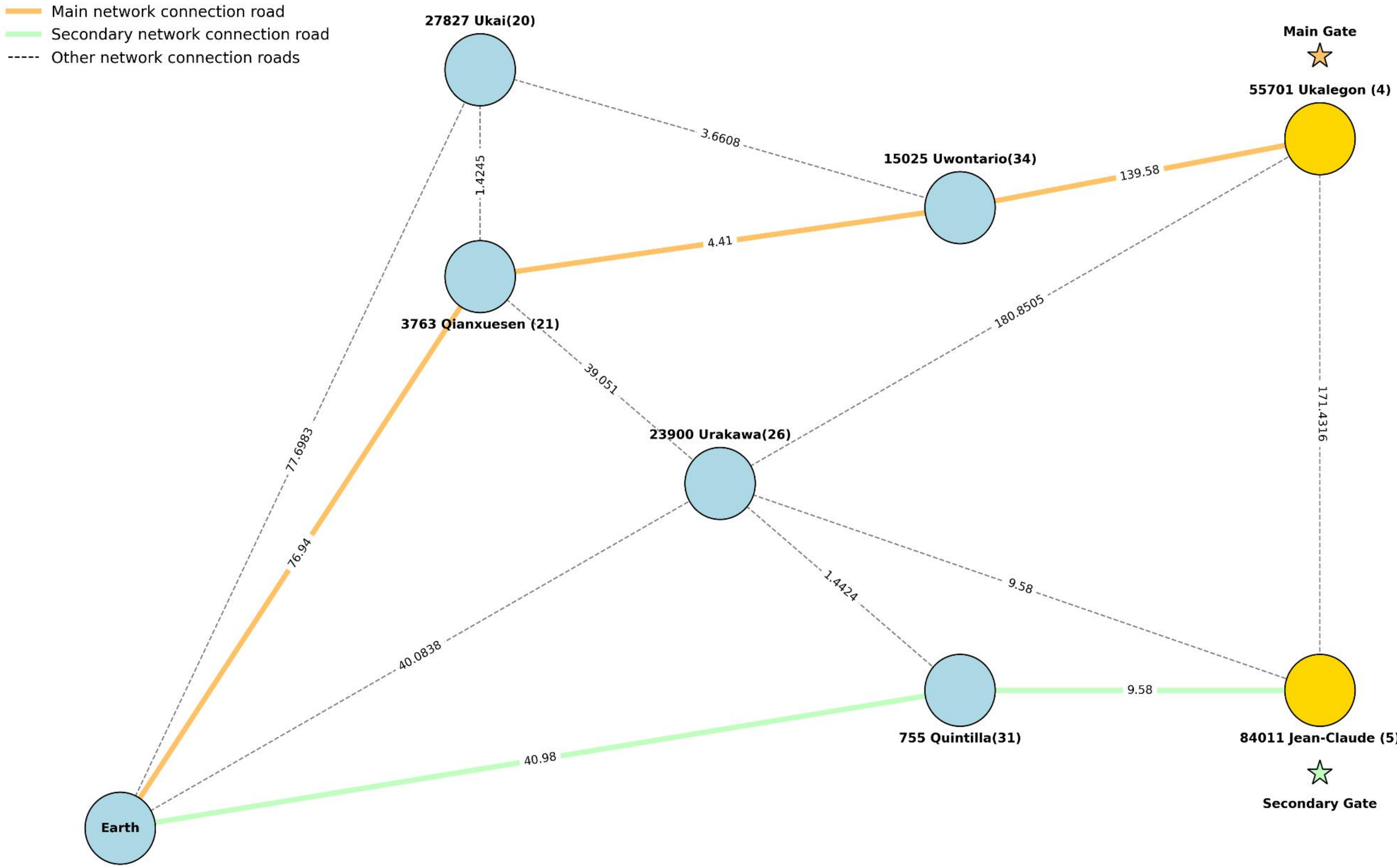

Fig. S17 Network Topology of Optimal Routes Targeting Primary and Secondary Gateways (k=7) perfectly aligns with ITN theory. The main route and intermediate route calculated using the Dijkstra algorithm for the Top 5 routes based on the Network Resiliency Test results for k=7 are: Main route: Earth → 763 Qianxuesen (21) → 15025 Uwontario (34) → 55701 Ukalegon (4) Intermediate route: Earth → 755 Quintilla (31) → 84011 Jean-Claude (5) .

In the main and intermediate routes, the numbers next to the object names in the paragraph indicate the first matched number the formula found. 763 Qianxuesen (21) is first found in the 21st match of the k=3 match group, 15025 Uwontario (34) in the 40th match of the k=5 match group, 55701 Ukalegon (4) in the 4th match of the k=1 match group, 755 Quintilla (31) in the 31st match of the k=4 match group, and 84011 Jean-Claude (5) in the 5th match of the k=1 match group .

55701 Ukalegon, Tj = 2.850, is located at the boundary of Jupiter Trojans, labeled as the main gate, and represents the main intersection of the ITN. 84011 Jean-Claude, with a Tj value of 2.995, is very close to the Tisserand Boundary, labeled as an intermediate gate, and represents the main entry. The main route and intermediate route shown in (Fig. S17) are the most energetically efficient routes from '399' Earth to these gates. The jump cost from '399' Earth to 3763 Qianxuesen (76.94) is high, but the jump from 3763 Qianxuesen to 15025 Uwontario (4.41) is extremely low. This result is consistent with the classical structure of the ITN: entering the network is expensive, but traveling within it is cheap.

The dashed lines shown in (Fig. S17) represent other connection paths, backup paths of the network, and potential routes. This indicates that the system offers multi-path routing and explains the rapid stabilization observed in the network resilience test during cost analysis. Nodes like 23900 Urakawa and 3763 Qianxuesen have many dashed lines and connection points, and these objects are considered main transfer hubs.

The Network Resiliency Stress Test presented in (Fig. S17), conducted using Dijkstra’s algorithm with k=7, corroborates the hypothesis that if the 'Wow!' signal encodes a network map, the deciphered locations must represent distinct matching groups. Analysis of the figure reveals that the permanently designated main gateway, 55701 Ukalegon, and the intermediate gateway, 84011 Jean-Claude, are intrinsic to the k=1 matching group (the primary octet).

The hypothesis stating that the coordinates of the Main and Intermediate Gates are encoded within the initial octet (k=1 matching group) has been substantiated. However, analysis indicates that the optimal trajectory to these fixed nodes allows for refinement as the network expands. For instance, the inclusion of 'Uwontario' from the k=5 subset served to optimize transfer logistics without altering the fundamental identification of the target within the primary group. These findings demonstrate that the proposed route topology exhibits both cost-effectiveness and operational flexibility, while maintaining structural stability.

The 3D instantaneous arrangement of the primary and secondary route components on the 'Wow!' signal date (August 15, 1977) is visualized in (Fig. S18). This configuration is the resultant output of the Network Resiliency Stress Test conducted with Dijkstra's algorithm (k=7). The 3D visualization representing the configuration of the Sun, Earth, and the selected objects (3763 Qianxuesen, 15025 Uwontario, 55701 Ukalegon, 755 Quintilla, and 84011 Jean-Claude) is founded upon precise orbital mechanics data. The 3D visualization of the configuration of the Sun, Earth, 3763 Qianxuesen, 15025 Uwontario, 55701 Ukalegon, 755 Quintilla, and 84011 Jean-Claude is based on orbital mechanics data. The instantaneous 3D position vectors of all these objects on 1977-08-15, and the semi-major axis (a), e (Eccentricity), i (Inclination °), Longitude of the Ascending Node (Ω), Argument of pericenter (ω), Mean Anomaly (M) parameter elements defining the instantaneous orbital path of the objects in 3D space, were retrieved from NASA's JPL Horizons service. The positions of the objects at this date were solved using Kepler's equation with the Mean Anomaly (M) value, and 3D Euclidean distances were calculated.

The orange line in the graph represents the main route, and the numbers on the arrows indicate the stops and step order of the route. The main route from Earth: 763 Qianxuesen (21), 1st stop; 15025 Uwontario (34), 2nd Stop; and the main gate target 55701 Ukalegon (4) is the 3rd stop. The intermediate route from Earth: 755 Quintilla (31), 1st stop; and the intermediate gate target 84011 Jean-Claude (5) is the second and final target. (Fig. S18) only shows the positions of the objects on the main and intermediate routes in their 3D positions.

This 3D (Fig. S18) connects the instantaneous positions of objects with straight lines. However, these straight lines and 3D projection can be misleading about the actual orbital paths of objects. For example, Ukalegon (Main Gate) in the graph appears to be close to the Z-axis. This gives the reader the impression that its route is very short or direct. Therefore, MOID (Minimum Orbit Intersection Distance) calculations were also performed and shared in the following section.

(Fig. S19) shows the potential MOID distances pulled to compare the spatial proximity between each of the objects shown in (Fig. S18) on 1977-08-15, to correct the misleading nature of the straight lines and 3D projection regarding the actual orbital paths of objects: Semi-major axis (a), e (Eccentricity), i (Inclination °), Longitude of the Ascending Node (Ω), Argument of pericenter (ω), and Mean Anomaly (M) parameter values were used. The Mean Anomaly was converted to Eccentric Anomaly, which gives the true angle of the object in its orbit, using Newton-Raphson numerical analysis method through iteration. The distance

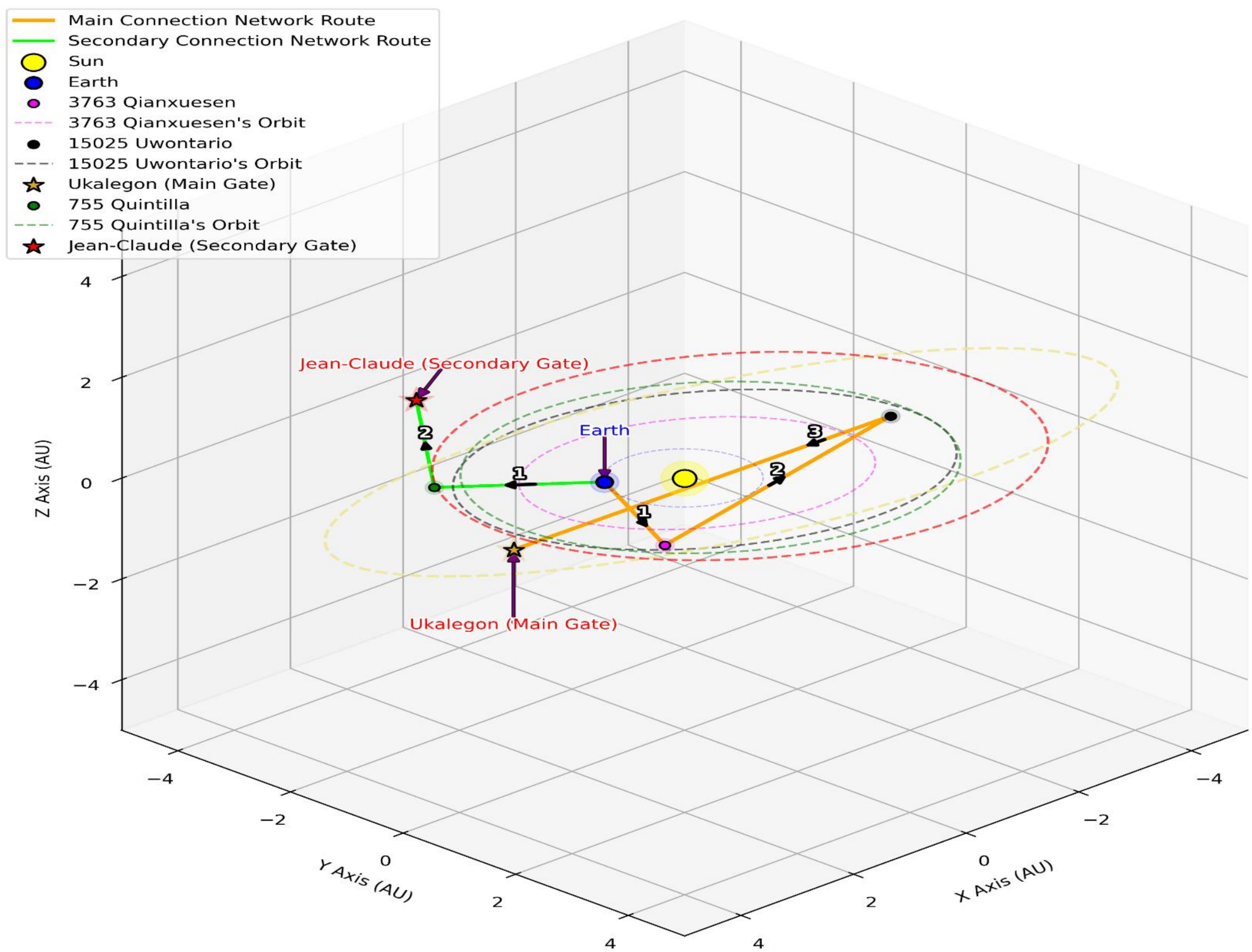

Fig. 18 3D Illustration of the Main and Secondary Connection Gates

and angle of objects from the Sun were determined in their own orbital plane. Using 3 Euler angles, the absolute (x,y,z) coordinates for each object were obtained in AU, and using these coordinates, the Euclidean Distance between two desired objects was obtained (Fig. S19).

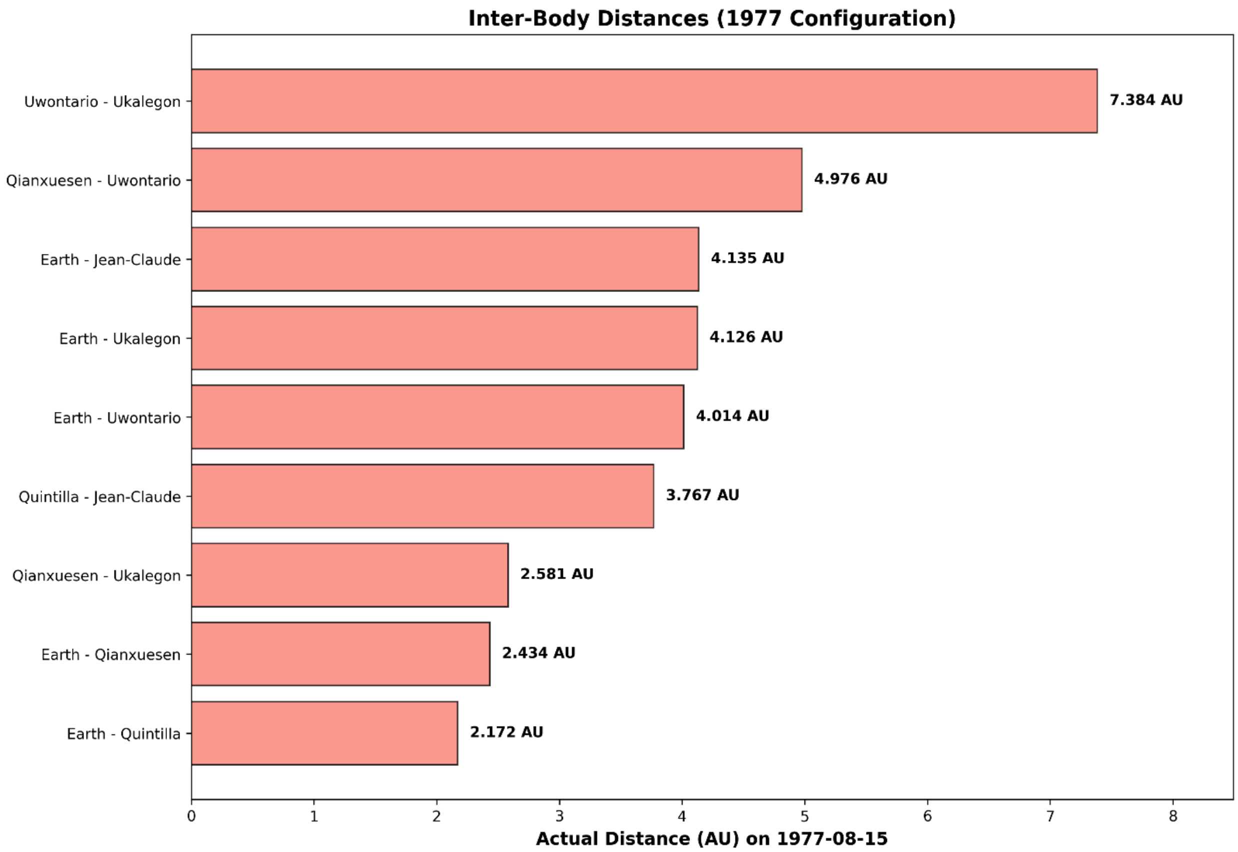

Fig.S19 Calculated Inter-Body Euclidean Distances for the August 15, 1977 Configuration

The findings obtained in (Fig. S19) revealed that transfer efficiency followed a unique energetic topology independent of physical distance for the celestial body alignments on August 15, 1977. Comparing (Fig. S18) and (Fig. S19), a paradox is striking. The direct (linear) distance between Earth and the target object Ukalegon was calculated as 4.126 AU on 1977-08-15, whereas the most efficient route algorithm that minimizes the energy cost used suggests: Earth → 763 Qianxuesen (21) → 15025 Uwontario (34) → 55701 Ukalegon (4). The total physical length (MOID distance) of this selected route (2.434 + 4.976 + 7.384 AU) is approximately 14.8 AU, and the algorithm, based on the principles of Low Energy Transfer Manifolds, chose to physically extend the path by 3.5 times. The analysis of Cost Units (CU) calculated by the algorithm shows that the system optimizes the Delta-V budget (minimum energy) rather than Euclidean geometry (shortest distance).

The step-by-step costs of the route determined by the algorithm: Earth → 763 Qianxuesen (21) → 15025 Uwontario (34) → 55701 Ukalegon (4) are: 1) Earth – Qianxuesen (Cost: 76.94 CU / Distance: 2.434 AU) has been calculated, which is the initial energy 'investment' required to enter the system and exit Earth's gravitational well. 2) Qianxuesen – Uwontario (Cost: 4.41 CU / Distance: 4.976 AU): This is the most critical finding of the study. Despite a significant physical distance of 4.976 AU between the two objects, the cost dropping to a negligible level of 4.41 CU indicates that these two objects entered a perfect 'resonance channel' in terms of their vector velocities and orbital planes in 1977, allowing for an almost free energy transfer. 3) Uwontario – Ukalegon (Cost: 139.58 CU / Distance: 7.384 AU): This is the necessary energy cost for transitioning to the outer belt. Although the physical distance of the route suggested by the formula totals ~14.8 AU, the total energy cost has been reduced to 220.93 CU thanks to gravitational assists and vector alignment. For a direct route: Earth → 55701 Ukalegon (4), despite a physical distance of only 4.126 AU, the calculated total energy cost is 233.1184 CU due to high orbital inclination and velocity vector mismatches. The algorithm, at the cost of extending the path, saved ~12.19 CU in total energy, achieving 220.93 < ~ 233.12, and mathematically confirmed the most efficient route. This situation confirms that the celestial bodies in question were in an exceptional and rare orbital configuration that maximized transfer efficiency at that date.

As postulated in Hypothesis 1, the formula successfully isolated the single object 32532 Thereus on the 'Wow!' signal date (1977-08-15). Following the framework of Hypothesis 2, the model (k=7) selected the primary and intermediate gateways and routes. This selection led to the generation of a transfer network matrix and a comprehensive Network Resiliency Stress Test covering all path variations. The study concluded with the 3D visualization of the spatial configurations for the identified main and intermediate trajectories.

Hypothesis 1, detailed in Section 1, postulates that if the 'Wow!' signal conveys an encrypted message, the decrypted solution for the epoch of August 15, 1977 (corresponding to the 'Q' identifier), reveals the primary target. This procedure identified '32532 Thereus' as the singular matching object. While the randomness analyses in Section 1 demonstrated that the formula systematically selects objects belonging to a specific spatial class, they did not statistically validate '32532 Thereus' as an isolated anomaly. Nevertheless, given its status as the unique solution on the 'Wow!' date, '32532 Thereus' was designated as the primary target. Furthermore, the formula's distinct bias toward a specific spatial characteristic provided the theoretical basis for the formulation of Hypothesis 2.

Under Hypothesis 2, it was assumed that solving the equation with Earth as the sole variable would disclose the critical nodes of an ITN route within the top octet. The objective was to isolate the coordinates for the main and intermediate gateways. Consistent with the cumulative analysis in Section 2, 55701 Ukalegon and 84011 Jean-Claude were identified as the Main Gate and Intermediate Gate, respectively.

Hypothesis 3 synthesizes the preceding postulates, proposing that the decrypted 'Wow!' signal reveals both the primary destination and the topological map of the route network required to reach it. Within this framework, the most energetically efficient trajectories to the terminal target '32532 Thereus' were calculated as follows: Main route: Earth → 763 Qianxuesen (21) → 15025 Uwontario (34) → 55701 Ukalegon (4), and Intermediate route: Earth → 755 Quintilla (31) → 84011 Jean-Claude (5). These trajectories were selected based on optimal energetic travel costs rather than mere geometric proximity (MOID), as established in Hypothesis 2. Consequently, '32532 Thereus' (identified in Hypothesis 1) is defined as the ultimate destination, accessible exclusively via the gateways isolated in Hypothesis 2, with the Main Gate serving as the final staging point.

In the testing phase of Hypothesis 3, the analysis was expanded by removing the k=7 threshold. All 20,590 matches were re-evaluated to detect potential gate candidates defined by Tj < 3 that might have been omitted due to the previous classification. Consequently, four new objects meeting these energetic requirements were identified and are detailed in (Table S14).

Table S14. Newly Identified Celestial Objects Satisfying the $T_J < 3$ Condition from the 20,590 Candidate Pool (Independent of the k=7 Constraint).

	Criterion: $T_J < 3$, 20590 Matches		
	#	**Matched Celestial Object**	**T_J value**
	1	169509 Jeffreyrobbins (2002 CV269)	2.998
	2	25869 Jacoby (2000 JP70)	2.962
	3	306001 Joanllaneras (2009 TD42)	2.945
	4	5254 Ulysses (1986 VG1)	2.81

Utilizing the Dijkstra Algorithm within the Network Resiliency Stress Test framework (k=7), the primary optimal trajectories (Top 1 Routes) were calculated for each gate candidate listed in (Table S14), as well as for the nodes proposed in Hypothesis 2 (55701 Ukalegon and 84011 Jean-Claude). In each simulation, 32532 Thereus was designated as the terminal destination. The resulting data and route metrics are presented in (Table S15).

Table 15. Primary Optimal Trajectories (Top 1 Routes) from Earth to 32532 Thereus via Selected Gate Candidates

	Optimal routes originating from Earth targeting Thereus via $T_J < 3$ gates selected from the 20,590 match pool; values in parentheses denote first time matched group number	
	Route #	**Route**
	1	Earth → 8755 Querquedula (40) → 25869 Jacoby (13) → 32532 Thereus
	2	Earth → 28275 Quoc-Bao (20) → 5254 Ulysses (31) → 32532 Thereus
	3	Earth → 755 Quintilla (26) → 84011 Jean-Claude (5) → 32532 Thereus
	4	Earth → 4372 Quincy (33) → 306001 Joanllaneras (26) → 32532 Thereus
	5	Earth → 3763 Qianxuesen (21) → 15025 Uwontario (6) → 55701 Ukalegon (4) → 32532 Thereus
	6	Earth → 755 Quintilla (26) → 169509 Jeffreyrobbins (438) → 32532 Thereus

For the routes given in (Table S15), comparisons were made for the 1977-08-15 Wow signal date, calculating the total energetic cost values for each route using the formula, actual MOID distance values, and efficiency values (energy required to travel 1 meter), and are shown in (Fig. S20).

As seen in (Fig. S20), Route 6 offers the shortest physical distance at 18.00 AU. Although it appears to be the most advantageous path according to the classical Euclidean geometry approach, it was measured as one of the least efficient routes with an EFF score of 12.29 in the efficiency (EFF) analysis of the proposed model. This indicates that physical proximity does not imply low energy cost in orbital mechanics; on the contrary, inappropriate orbital inclination (i) and eccentricity (e) differences lead to enormous energy costs on a short path.

Routes incorporating the Main and Intermediate Gates, selected from the 'Top 8 Anomaly' cluster identified at the study's outset, demonstrated decisive superiority in efficiency

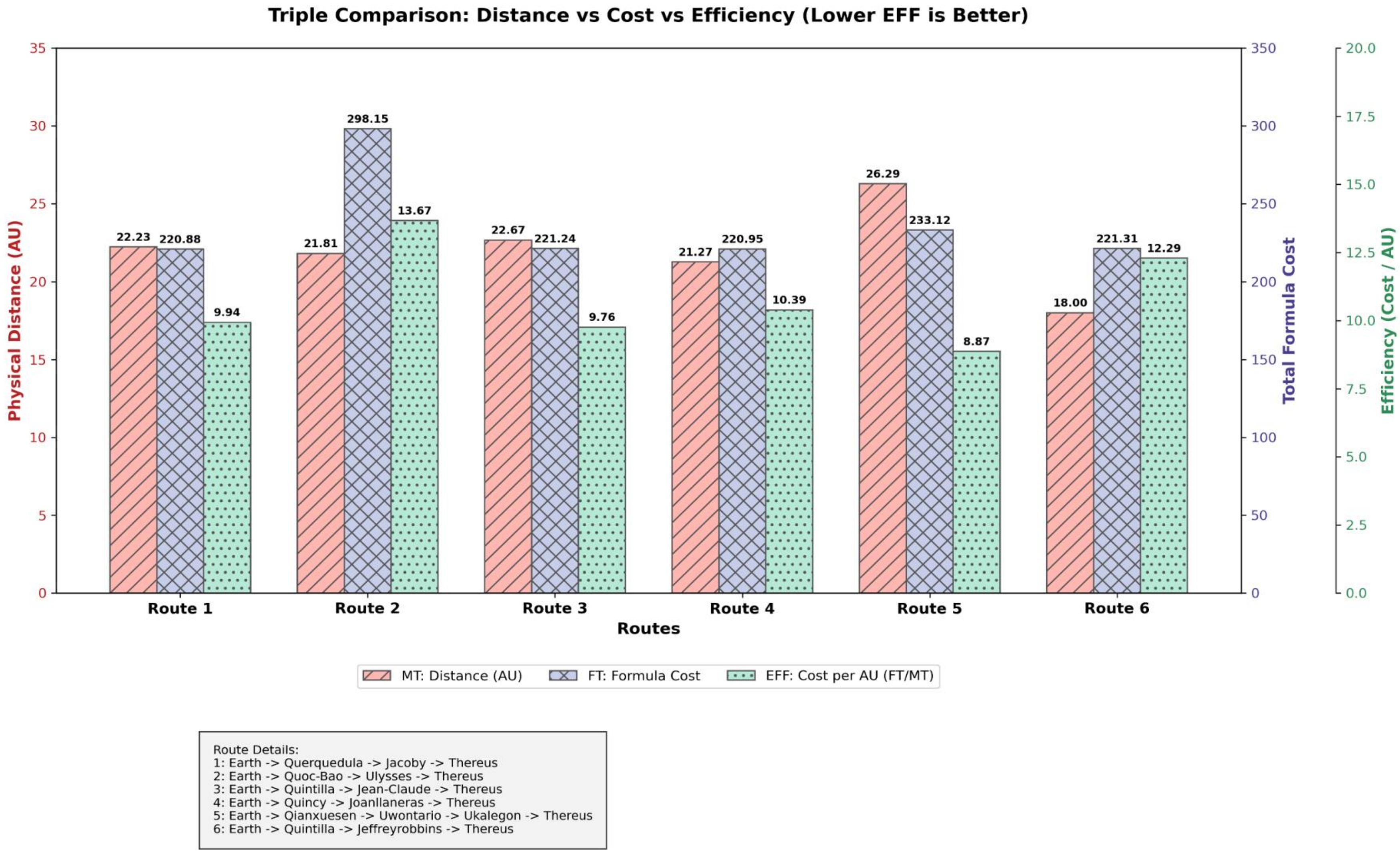

Fig. S20. Comparative Performance Metrics of Optimal Routes: Physical Distance vs. Energetic Cost vs. Efficiency

metrics over competing trajectories. Most notably, Route 5 emerged as the optimal path with an efficiency (EFF) score of 8.87, despite possessing the greatest physical path length of 26.29 AU. This trajectory follows the chain: Earth → 3763 Qianxuesen (21) → 15025 Uwontario (6) → 55701 Ukalegon (4) → 32532 Thereus. This result provides evidence that favorable Tisserand parameters and orbital phasing create a seamless dynamic alignment, thereby minimizing the Formula Cost (FT) despite the increased spatial range. Route 3, with an EFF score of 9.76, was the second most efficient route (Earth → 755 Quintilla (26) → 84011 Jean-Claude (5) → 32532 Thereus).

The calculated energetic travel costs, arranged in ascending order, are Route 1 (220.88), Route 4 (220.95), Route 3 (221.214), Route 6 (221.31), Route 5 (233.12), and Route 2 (298.12). In contrast, sorting by physical MOID distances yields a divergent sequence: Route 6 (18.00), Route 4 (21.27), Route 2 (21.81), Route 1 (22.23), Route 3 (22.67), and Route 5 (26.29). However, upon applying the definitive efficiency evaluations, the hierarchy shifts significantly from optimal to least efficient: Route 5 (8.87), Route 3 (9.76), Route 1 (9.94), Route 4 (10.39), Route 6 (12.29), and Route 2 (13.67).

The second hypothesis of the study is based on the premise that the developed cost formula will find the most energetically efficient path solely based on the compatibility of orbital parameters (a, e, i, Tj), completely independent of physical distance (MOID) data. The results obtained confirm this premise through a complex but consistent 'Cost-Efficiency' relationship.

When the total costs calculated by the formula are examined; Route 1 (220.88), Route 4 (220.95), Route 3 (221.24) and Route 6 (221.31) values are seen to form a 'cluster' that is very close to each other, separated by decimal precision. Route 5, at 233.12 points, is slightly above this cluster. At first glance, the formula does not seem to have chosen Route 5 as 'the best', but when physical distances are considered, the real finding emerges.

Although Route 6 (18.00 AU) represents the geometric minimum and Route 5 (26.29 AU) the maximum in terms of MOID, the calculated energetic costs diverge from classical expectations. The formula yielded a cost of 221.31 for Route 6, effectively equating it with Routes 1 and 3 despite its short physical distance. This validates that the algorithm prioritizes dynamic constraints over spatial brevity, accurately reflecting the significant energetic penalties imposed by the inclination (i) and eccentricity (e) differences in Route 6.

Route 5 (EFF: 8.87) was identified as the premier trajectory; despite constituting the longest physical path (26.29 AU), it necessitated the minimum energy expenditure per unit distance. Route 3 (EFF: 9.76) ranked second in efficiency. The superior performance of these specific routes validates the algorithm optimized not only a, e, i parameters but also Tisserand Invariant (Tj) compatibility.

The most significant finding of this study is the validation of the candidate gate objects—initially identified via the k=1 filter during the data mining phase—as the most efficient targets through physical route simulations. Notably, Route 5 (originating from Ukalegon) and Route 3 (originating from Jean-Claude), which occupy the apex of the efficiency ranking, explicitly incorporate the specific objects classified as the k=1 / Top 8 Anomaly during the preliminary stage of the research.

This result indicates that the Top k=1 anomalies found by the algorithm are not accidental; the formula detected hidden energy tunnels (dynamical passageways) within the chaotic asteroid belt that require low Δv (velocity change). The fact that Route 5 is 28% more advantageous in terms of efficiency (cost per AU) despite being 46% physically longer than Route 6 shows that the proposed mathematical model provides a superior strategic insight in space mission planning compared to classical distance-based approaches.

A key achievement of the algorithm was its ability to forecast energy efficiency purely through dataset matching density (k value) prior to detailed physical modeling. This characterizes the tool as an advanced 'orbit detector' capable of decoding hidden structures in chaotic data, a capability supported by a 100% alignment with subsequent physical simulations. The fact that the most efficient routes emerged directly from the k=1 group (the first 8 matches) strongly validates the intuitive premise of this study: if the 'Wow!' signal encodes a message, it likely utilizes a mathematical logic rooted in orbital mechanics. These findings suggest that the signal not only highlights a specific target for observation but also delineates the precise trajectory humanity must follow to intercept it.

It is concluded that Hypothesis 3 has been substantiated: if the 'Wow!' signal contains a cryptographic cipher, this encryption encodes a dual set of instructions—specifying not only the terminal destination ('Where to go' – 32532 Thereus) but also the optimal navigational vectors ('How to get there most efficiently' – Ukalegon and Jean-Claude Routes).

It has been proven that the main gate 55701 Ukalegon and intermediate gate 84011 Jean-Claude objects, selected from the k=1 matching group with the lowest percentage deviations proposed in Hypothesis 2, are the unique transfer points in physical space that minimize the energy cost efficiency required to reach 32532 Thereus, proposed in Hypothesis 1. Two routes with the most EFF efficiency to reach 32532 Thereus have been shown: 1) Earth → 3763 Qianxuesen → 15025 Uwontario → 55701 Ukalegon → 32532 Thereus and 2) Earth → 755 Quintilla → 84011 Jean-Claude → 32532 Thereus. The methodology utilized in this study has deciphered an obscured energetic topology, revealing that geometric proximity in space can be deceptive regarding orbital transfer costs.

It is substantiated that the Main Gate (55701 Ukalegon) and Intermediate Gate (84011 Jean-Claude)—selected from the k=1 group based on the lowest percentage deviations (Hypothesis 2)—function as unique spatial transfer nodes that minimize the energetic cost to the target 32532 Thereus (Hypothesis 1). Two primary trajectories yielding the highest efficiency (EFF) have been delineated: 1) Earth → 3763 Qianxuesen → 15025 Uwontario → 55701 Ukalegon → 32532 Thereus and 2) Earth → 755 Quintilla → 84011 Jean-Claude → 32532 Thereus. Collectively, these results attest that the algorithm has resolved a concealed 'energetic manifold,' establishing a functional high-efficiency transport grid within the solar system.

Section 5. Conclusions

The "Wow!" signal, recorded on August 15, 1977, by Ohio State University's Big Ear radio telescope, remains one of modern astronomy's most compelling enigmas. The famous "6EQUJ5" sequence—circled by Dr. Jerry Ehman—represents a technical transcript documenting the signal's intensity profile over time: a nearly perfect Gaussian curve that peaked sharply above background noise at 30 sigma [2]. This mathematical precision, coupled with the signal's distinctive characteristics, has long suggested that it may represent more than a random astrophysical event, it could be an intentionally transmitted signal.

Drawing upon the principles of serendipity and parsimony (Occam's Razor) that have guided scientific discovery throughout history, this study proposed that the "6EQUJ5" character sequence itself might function as a cipher. Through systematic application of the decoded formula "6EQ=5UJ," a deterministic, reproducible, and physically meaningful orbital architecture was extracted from what initially appeared to be chaotic astronomical data.

The comprehensive analyses conducted throughout this investigation have yielded three principal findings, which are synthesized below:

5.1. Rejection of Statistical Randomness and Temporal Consistency

Hypothesis 1, detailed in Section 1, posited that when the cipher formula was applied to the signal reception date of August 15, 1977, the solution corresponding to the 'Q' character would reveal the primary target encoded within the signal. Application of this procedure isolated 32532 Thereus as a unique match among 26,576 celestial objects. To assess the significance of this critical identification, extensive analyses were conducted across different temporal windows (1, 10, 100, and 111 days). These analyses conclusively demonstrated that the formula does not select objects randomly; rather, it systematically filters for bodies belonging to specific spatial classes.

While statistical analyses confirmed the formula's systematic selection bias toward certain object types, they did not establish 32532 Thereus as an isolated statistical anomaly in its own right. Nevertheless, its status as the sole match on the signal date provided sufficient grounds for designating it as the primary target. The formula's pronounced selectivity for objects with particular spatial characteristics necessitated that the investigation extends beyond this singular target. Specifically, it became imperative to examine the highest-concordance match group (Top 8) to understand the physical mechanism underlying this selectivity. This statistical requirement laid the groundwork for interrogating the physical nature of the selected objects in the study's second phase, thereby establishing the theoretical foundation for Hypothesis 2.

5.1.2. The Top 8 Anomaly and Physical Classification

Hypothesis 2 proposed that when the formula was solved exclusively for the Earth (E = Earth) variable, the resulting solutions would reveal strategic gateway candidates—critical nodes within a hidden Interplanetary Transport Network (ITN). These gateway candidates were predicted to reside within the specific Top 8 region identified by the Earth-only solution.

Within this theoretical framework, a multi-layered filtering process was implemented through data mining, combining Log-Log Deviation Histograms, Kernel Density Estimation (KDE), and Elbow Analysis methods. These analyses unequivocally established that the first eight matches (Top 8) constitute a unique anomaly cluster, separated from the remaining 20,582 matches not merely by numerical difference but by a statistical chasm. Detailed orbital parameter analyses (semi-major axis a, eccentricity e, inclination i, Tisserand parameter T_j) of this isolated "Top 8" cluster—employing violin plots and density comparisons—demonstrated that these objects do not represent a random collection. On the contrary, the data revealed that this group belongs to a specific dynamical class (associated with Jupiter Trojans and the Hilda Group), characterized by clustering within a narrow band around Tisserand Parameter $T_j \approx 3.0$, strong resonance interactions with Jupiter, exceptionally low orbital eccentricities, and inclinations optimized within a particular range. The remarkable concordance between statistical separation and physical properties established the foundation for Hypothesis 3.

5.1.3. Energetic Efficiency and Discovery of the "Invisible Highway"

Hypothesis 3 synthesized the preceding propositions by predicting that decipherment of the "6EQUJ5" sequence would not only reveal strategic gateway candidates (Hypothesis 2) and the ultimate destination target (Hypothesis 1) but would also map the topological structure of an energetically optimal hidden Interplanetary Transport Network (ITN).

Analyses conducted during this mapping process—employing a weighted cost formula and network resilience stress tests using Dijkstra's algorithm—resolved the relationship between physical distance and energetic cost, thereby uncovering the study's most striking and paradoxical result. Efficiency (EFF) analyses dramatically confirmed the existence of the topological map predicted by Hypothesis 3:

Distance-Energy Paradox: Route 6, which spans 18.00 AU and represents the shortest physical distance according to classical geometry, proved inefficient with an EFF score of 12.29, owing to substantial differences in orbital inclination and eccentricity. Conversely, Route 5 (via Ukalegon), despite traversing the longest physical distance at 26.29 AU, emerged as the most efficient trajectory with an EFF score of 8.87, by virtue of favorable Tisserand parameters and orbital phasing. This finding demonstrates that the formula identifies hidden energy tunnels (dynamical transit pathways) within the chaotic asteroid belt—pathways that require low Δv and provide fluid transitions through alignment of orbital phases and Tisserand parameters, independent of physical distance.

Validation of Strategic Gateways: Objects 55701 Ukalegon and 84011 Jean-Claude, identified during the data mining phase under Hypothesis 2 as "k=1" anomalies (the first eight matches), were shown in physical route simulations to serve as transfer nodes in Route 5 (Primary Route) and Route 3 (Intermediate Route)—the trajectories ranking highest in efficiency. This confirmed that these objects are not arbitrary waypoints, but rather strategic gateway candidates positioned on the Interplanetary Transport Network (ITN). The developed algorithm successfully predicted the most efficient energy manifolds based solely on data density, prior to any physical modeling.

Dual-Layer Encryption: In summary, Hypothesis 3 validates that the cryptographic structure of the "Wow!" signal contains a dual-layer instruction set: the signal encodes not only the ultimate destination (Where: 32532 Thereus) but also the optimal navigation vectors required to reach it (How: The Ukalegon and Jean-Claude Routes). The findings substantiate that the routes emerging from decipherment of the 6EQUJ5 sequence—specifically, (1) Earth → 3763 Qianxuesen → 15025 Uwontario → 55701 Ukalegon → 32532 Thereus and (2) Earth → 755 Quintilla → 84011 Jean-Claude → 32532 Thereus—map a functional, high-efficiency "Interplanetary Transport Network" infrastructure within the Solar System.

5.1.4. Theoretical Implications: Passive Infrastructure and Resource Utilization

While the primary focus of this study was the statistical and orbital validation of the decoded coordinates, the physical nature of the identified targets invites a broader theoretical interpretation regarding the search for extraterrestrial intelligence (SETI).

Persistence Over Transmission: The selection of Main Belt asteroids and Jupiter Trojans—bodies that have remained dynamically stable for billions of years—suggests a communication strategy prioritizing temporal durability overactive signaling. Unlike active probes which suffer from energy degradation, stable asteroids serve as passive, geological scale "markers," functioning effectively as gravitational milestones that outlast the operational lifespan of any artificial transmitter.

Logistic Hierarchy: The distinct roles of the targets implies a functional hierarchy. The Jupiter Trojans (e.g., Ukalegon), situated in the natural gravitational wells of Lagrange points, likely represent strategic "hubs" for low-energy orbital transfers. Conversely, the identified Main Belt objects may delineate specific transit corridors connecting these hubs.

The Objective: Finally, the designation of the Centaur 32532 Thereus as the terminal destination offers a compelling motive: resource acquisition. Centaurs, being rich in volatiles (water ice, methane, ammonia) compared to the dry inner solar system, are prime targets for *In-Situ Resource Utilization (ISRU)*. The deciphered network, therefore, may not merely be a map, but a blueprint for an energy-efficient supply chain designed to harvest volatiles from the chaotic Centaur region and transport them via the stable "invisible highway" of the ITN.

Section 6. Future Works

In this study, the hypothesis that the "Wow!" signal sequence "6EQUJ5" represents a deterministic cipher pointing to specific orbital dynamics and spatial coordinates has been supported through statistical analyses and orbital mechanics simulations. The findings indicate that 32532 Thereus serves as the ultimate destination, while 55701 Ukalegon and 84011 Jean-Claude function as critical gateways within the Interplanetary Transport Network (ITN). To deepen this investigation and validate these findings across multiple disciplines, the following research directions are recommended:

1. Multi-Wavelength Observational Campaigns of Candidate Celestial Bodies

Focused observational efforts should be directed toward 55701 Ukalegon (identified as the "Main Gate"), 84011 Jean-Claude (designated as the "Intermediate Gate"), and the ultimate target 32532 Thereus. To characterize the spectral properties, surface compositions, and rotational dynamics of these objects with greater precision, coordinated observation programs spanning radio, optical, and infrared wavelengths should be designed within both SETI and planetary science frameworks. Particular attention should be given to detecting potential techno signatures or anomalous thermal emissions on the surfaces or within the orbital environments of these bodies.

2. Machine Learning-Assisted Full Dataset Exploration

In the current study, route optimization using Dijkstra's algorithm was constrained to k=7 matching groups due to computational costs and statistical saturation observed at the k=4 level. Future investigations should employ unsupervised machine learning techniques and deep neural network architectures to enable comprehensive scanning of the entire database containing N=20,590 matches. This approach would reduce the computational burden from the $O(N^2)$ complexity inherent in brute-force methods to near-linear scaling, thereby facilitating the identification of additional "strategic intermediate nodes" that may have been overlooked in the current analysis.

3. Route Validation Through High-Fidelity N-Body Simulations

The cost function employed in this study (Eq. (13) and Eq. (14)) presents a theoretical energy efficiency model based on the similarity of orbital elements (a, e, i, T_j). In subsequent phases, this theoretical framework should be rigorously tested using high-precision N-body integrators that account for the gravitational influences of all major Solar System bodies. To demonstrate the practical viability of the proposed routes (particularly Route 5 and Route 3) for actual space mission design, detailed physical simulations should be conducted to calculate gravity assist maneuvers and the associated delta-v (Δv) requirements.

4. Investigation of Temporal Spatial Drift (Epoch Dependency)

Results from DBSCAN clustering analysis (Fig. 7) and Kernel Density Estimation (Fig. 6) reveal that objects identified through the formula exhibit a systematic migration from the inner Solar System (Jupiter-family comets) toward the outer Solar System (trans-Neptunian/Centaur regions) around the year 1990. The physical mechanisms underlying this bimodal distribution and the temporally expanding search window warrant thorough investigation. Theoretical astrophysical models should be developed to determine whether this drift reflects a time-dependent dynamic map encoded within the cipher algorithm or potentially indicates an as-yet-undiscovered long-period orbital resonance cycle.

5. Adaptation of Methodology to Other Historical Anomalous Signals

To assess the universality of the developed "orbital-based decryption" methodology and the "Invisible Highway" detection algorithm, these techniques should be applied to other unexplained radio signals in astronomical history or various Fast Radio Burst (FRB) events. This comparative approach would clarify whether the proposed methodology represents a coincidence specific to the "Wow!" signal or constitutes a generalizable framework for an advanced communication protocol.

These future investigations are expected to establish the foundation for a new paradigm in SETI research—one that transcends traditional signal processing techniques by integrating astrodynamics and data science methodologies.

References

[1] Race, M. S. (2008). Communicating about the discovery of extraterrestrial life: different searches, different issues. Acta Astronautica, 62(1), 71-78.

[2] Wright, J. T. (2021). Strategies and advice for the Search for Extraterrestrial Intelligence. Acta Astronautica, 188, 203-214.

[3] Kantha, S. S. (1996). Pros and cons in the search for extraterrestrial intelligence. Medical hypotheses, 46(3), 183-187.

[4] Impey, C. (2022). Life beyond Earth: How will it first be detected?. Acta Astronautica, 197, 387-398.

[5] Billingham, J. (1998). Cultural aspects of the search for extraterrestrial intelligence. Acta astronautica, 42(10-12), 711-719.

[6] Van Loon, A. J. (2005). The needless search for extraterrestrial fossils on Earth. Earth-Science Reviews, 68(3-4), 335-346.

[7] Ehman, J. R. (2010). “Wow!”-A Tantalizing Candidate. In Searching for Extraterrestrial Intelligence: SETI Past, Present, and Future (pp. 47-63). Berlin, Heidelberg: Springer Berlin Heidelberg.

[8] Ehman, J. R. (2010). The Big Ear Wow! Signal (30th Anniversary Report). Big Ear Radio Observatory. Available online at http://www. bigear. org/Wow30th/wow30th. htm.

[9] Gray, R. H., & Marvel, K. B. (2001). A VLA search for the Ohio state “Wow”. The Astrophysical Journal, 546(2), 1171.

[10] Paris, A., & Davies, E. (2015). Hydrogen clouds from comets 266/P christensen and P/2008 Y2 (Gibbs) are candidates for the source of the 1977 “WOW” signal. Journal of the Washington Academy of Sciences, 101(4), 25-32.

[11] Harp, G. R., Gray, R. H., Richards, J., Shostak, G. S., & Tarter, J. C. (2020). An ATA search for a repetition of the wow signal. The Astronomical Journal, 160(4), 162.

[12] Gray, R. H. (1994). A Search of the" Wow" locale for intermittent radio signals. Icarus, 112(2), 485-489.

[13] Caballero, A. (2022). An approximation to determine the source of the WOW! Signal. International Journal of Astrobiology, 21(3), 129-136.

[14] Makukov, M. A. (2013). The “Wow! signal” of the terrestrial genetic code. Icarus, 224(1), 228-242.

[15] Gray, R. H., & Ellingsen, S. (2002). A search for periodic emissions at the Wow Locale. The Astrophysical Journal, 578(2), 967.

[16] Paris, A. (2017). Hydrogen Line observations of cometary spectra at 1420 MHz. Journal of the Washington Academy of Sciences, 103(2), 1-20.

[17] Potochnik, A. (2017). Idealization and the Aims of Science. In Idealization and the Aims of Science. University of Chicago Press.

[18] Sio, U. N., & Ormerod, T. C. (2009). Does incubation enhance problem solving? A meta-analytic review. Psychological bulletin, 135(1), 94.

[**19**] Weisberg, R. W. (2019). Toward an integrated theory of insight in problem solving. In Insight and creativity in problem solving (pp. 5-39). Routledge.

[20] Sandkühler, S., & Bhattacharya, J. (2008). Deconstructing insight: EEG correlates of insightful problem solving. PLoS one, 3(1), e1459.

[21] Metcalfe, J., & Wiebe, D. (1987). Intuition in insight and noninsight problem solving. Memory & cognition, 15(3), 238-246.

[22] Danek, Amory H., and Jennifer Wiley. "What about false insights? Deconstructing the Aha! experience along its multiple dimensions for correct and incorrect solutions separately." Frontiers in psychology 7 (2017): 2077.

[23] Kounios, J., & Beeman, M. (2014). The cognitive neuroscience of insight. Annual review of psychology, 65, 71-93.

[24] Baker, A. (2022). Simplicity.'Stanford Encyclopedia of Philosophy.

[25] Sober, E. (2015). Ockham's razors: a user's manual. Cambridge University Press.

[26] Gauch, H. G. (2003). Scientific method in practice. Cambridge University Press.

[27] Chater, N., & Vitányi, P. (2003). Simplicity: a unifying principle in cognitive science?. Trends in cognitive sciences, 7(1), 19-22.

[28] Lombrozo, T. (2016). Explanatory preferences shape learning and inference. Trends in cognitive sciences, 20(10), 748-759.

[29] Copeland, S. (2019). On serendipity in science: discovery at the intersection of chance and wisdom. Synthese, 196(6), 2385-2406.

[30] Yaqub, O. (2018). Serendipity: Towards a taxonomy and a theory. Research Policy, 47(1), 169-179.

[31] Roberts, R. M. (1989). Serendipity: Accidental discoveries in science.[A description of important scientific facts discovered by accident]. Wiley (John) and Sons, New York, NY (USA).

[32] Brogren, C. H. (2024). Louis Pasteur—The life of a controversial scientist with a prepared mind, driven by curiosity, motivation, and competition. APMIS, 132(1), 7-30.

[33] Fleming, A. (1929). On the antibacterial action of cultures of a penicillium, with special reference to their use in the isolation of B. influenzae. British journal of experimental pathology, 10(3), 226.

[34] Corneli, J., Jordanous, A., Guckelsberger, C., Pease, A., & Colton, S. (2014). Modelling serendipity in a computational context. arXiv preprint arXiv:1411.0440.

[**35**] Glasser, O. (1993). Wilhelm Conrad Röntgen and the early history of the Roentgen rays (No. 1). Norman Publishing.

[36] Osepchuk, J. M. (2003). A history of microwave heating applications. IEEE Transactions on Microwave theory and Techniques, 32(9), 1200-1224.

[37] Ghofrani, H. A., Osterloh, I. H., & Grimminger, F. (2006). Sildenafil: from angina to erectile dysfunction to pulmonary hypertension and beyond. Nature reviews Drug discovery, 5(8), 689-702.

[**38**] Fahlberg, C., & Remsen, I. (1879). Ueber die oxydation des orthotoluolsulfamids. Berichte der deutschen chemischen Gesellschaft, 12(1), 469-473.

[39] Plunkett, R. J. (1986, April). The history of polytetrafluoroethylene: discovery and development. In High Performance Polymers: Their Origin and Development: Proceedings of the Symposium on the History of High Performance Polymers at the American Chemical Society Meeting held in New York, April 15–18, 1986 (pp. 261-266). Dordrecht: Springer Netherlands.

[40] Novoselov, K. S., Geim, A. K., Morozov, S. V., Jiang, D. E., Zhang, Y., Dubonos, S. V., ... & Firsov, A. A. (2004). Electric field effect in atomically thin carbon films. science, 306(5696), 666-669.

[41] Bhushan, B. (2009). Biomimetics: lessons from nature–an overview. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 367(1893), 1445-1486.

[42] Loadman, J. (2005). A short history of natural rubber research. Rubber Chemistry and Technology, 78(5), 754-768.